

Aggregate Risk in Dynamic Spatial Equilibrium: Assessing Climate Impacts Across the United States*

Preliminary and incomplete
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February, 2026

Abstract

We evaluate how climate change risk shapes the aggregate and local costs of climate change. We develop a dynamic spatial model of the U.S. economy and its 3,143 counties that features costly forward-looking migration, capital investment decisions, and risk premia. We incorporate aggregate uncertainty in warming trajectories and local uncertainty in exposure by leveraging the ‘Master Equation’ representation of the economy. We estimate structural elasticities and damage functions by matching reduced-form estimates linking population, income, and investment to coastal storms and heat waves. Our counterfactuals show that local risk raises capital through precautionary savings but generates sizable welfare losses for capitalists in exposed coastal areas. Aggregate risk leads to modest capital depletion in some southern coastal counties.

*We thank Bin Cheng for outstanding research assistance.

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1 Introduction

Climate change is one of the fundamental challenges that societies will have to grapple with in the coming decades and centuries. A defining feature of climate change over such long time scales is that it is uncertain. While there is broad scientific agreement that the Earth is warming rapidly due to anthropogenic greenhouse gas emissions, the extent of that warming by the end of the century is not known with precision. Geoscientific uncertainties—for instance, surrounding the equilibrium climate sensitivity—and economic and policy uncertainties—for instance, the extent of decarbonization policy—compound to widen the dispersion of plausible temperature paths.

Faced with this uncertainty, households and businesses will not wait passively for it to resolve, but instead take preemptive actions to adapt as best they can. The ability of economic actors to adapt depends critically on a second defining feature of climate change: its impacts are far from uniform across locations. By moving out of high-risk locations and redeploying capital in low-risk locations, workers and investors may limit their exposure to climate change risk more than if they were confined to staying in place. How does climate change risk reshape economic activity when it can reallocate across space?

In this paper, we answer this question by developing a dynamic spatial model of the United States economy with aggregate and local risk. We use a fine spatial resolution at the level of 3,143 counties to capture climate change impacts and adaptation. Our framework features costly forward-looking migration, capital investment decisions, and risk premia. The analysis remains computationally tractable despite aggregate risk in warming trajectories and the high-dimensional state space by leveraging the ‘Master Equation’ representation of the economy. We estimate structural elasticities and damage functions by matching reduced-form estimates linking population, income, and investment to coastal storms and heat waves. Our counterfactuals show, first, that risk premia associated with uncertain warming depress investment in coastal and southern locations. Second, workers respond to risk by migrating to safe locations, offsetting part of the losses from climate change. Third, uncertainty substantially magnifies welfare losses from climate change.

Our starting point is a framework with forward-looking dynamic migration and investment decisions at a high level of spatial resolution as in Bilal and Rossi-Hansberg (2023). As in Caliendo et al. (2019), forward-looking risk-averse workers decide where to live and work. They earn an equilibrium wage in their location, which they use to consume and lease housing at equilibrium rents. They receive random preference shocks for locations that motivate them to migrate subject to bilateral migration costs. We enrich this environment along several dimensions. As in Kleinman et al. (2023), risk-averse local capitalists own the capital stock in the location where they live and face a dynamic consumption-investment decision subject to adjustment costs that accounts for the future expected path of the economy and its climate. Capitalists rent out their capital stock to developers who use it to produce housing and commercial structures. Local firms produce consumption goods using labor and commercial structures with productivity that is heterogeneous across locations and varies with global temperatures. The evolution of the local paths of capital depreciation rates, productivities, and amenities depends on climate change with heterogeneous local exposures: as global mean temperatures rise, the frequency of storms and heat waves changes over time.

We introduce aggregate and local risk in this environment. The path of global mean temperature is not known with certainty. Instead, it is subject to aggregate shocks. Thus, local capital depreciation rates, productivities, and amenities fluctuate in tandem with aggregate shocks. In addition, we allow local exposures of capital depreciation rates, productivities, and amenities themselves to be risky. This local risk reflects idiosyncratic realizations of rare weather extremes. Local risk in capital depreciation generates local risk premia relative to an economy in which capital is a safe asset. We plan to incorporate two additional components of local risk: scientific uncertainty in how storms and heat waves change, location by location, and estimation uncertainty in climate impacts.

The resulting environment is a heterogeneous agent framework with a high-dimensional state space—population and capital in every county—and rich interdependencies between decisions and prices in all locations, and aggregate risk. To make progress, we leverage the ‘Master Equation’ methodology from Bilal (2023). Specifically, we use the ‘Second-order Approximation to the Master Equation’ (SAME), which consists of an analytic second-order perturbation around a long-run deterministic steady-state. It allows us

to solve efficiently for quantitative general equilibrium counterfactuals in heterogeneous agent models and capture the effects of aggregate risk, by solving a simple set of matrix and three-dimensional tensor Sylvester equations.

Equipped with our solution method, we estimate the central parameters of our framework: the migration elasticity, capitalist risk-aversion, which controls the investment elasticity, and the damage functions that map global temperature changes into local changes in capital depreciation rates, productivity, and amenities. We obtain data on storm and heat wave probabilities for each county from Bilal and Rossi-Hansberg (2023). Specifically, we measure how their frequency changes with global mean temperature in each county, and record local slopes and associated standard errors. We also obtain reduced-form estimates of these weather extremes on population, income, and investment from Bilal and Rossi-Hansberg (2023), with the associated standard errors. We obtain expert elicitations of the distribution of global mean temperature paths for the next century.

We target the reduced-form impulse response functions to quantify our framework. We estimate the migration elasticity and capitalist risk aversion jointly with the magnitude of shocks that correspond to a single extreme event realization. Given a guess for the migration elasticity and capitalist risk-aversion, and a standard calibration of the rest of the model parameters, we can invert the model in steady-state and obtain all the local characteristics, including local productivities, amenities, and the level of local capital investment costs.

We then leverage a key property of the SAME: impulse response functions are quadratic in the magnitude of shocks. Therefore, conditional on the two elasticities, it suffices to evaluate two impulse response functions for the target outcomes per extreme event—investment for storms and income per capita for heat waves—and compare them with the data. The magnitude of the underlying capital depreciation or productivity shock then solves a standard scalar quadratic equation. This observation sidesteps an additional numerical search over the magnitude of the shocks.

Next, we estimate the migration elasticity and capital risk-aversion with a method of simulated moments in an outer loop. We target the relative impulse response functions of employment and income to heat waves, and the relative impulse response function of investment to income to storms. We leverage the computational efficiency of our solution

method to propose global identification guarantees.

Storms and heat waves have large effects on capital depreciation, productivity, and amenities. Through the lens of our model, storms generate a 12% capital depreciation shock, while heat waves generate a 2% productivity shock together with a 0.6% amenity shock.¹ The resulting structural damage functions imply that the southeastern Atlantic coast should expect a 1 to 3 p.p. increase in the annual capital depreciation rate for every 1°C increase in global mean temperatures because of rising storm activity. Southern Florida should expect a 5% reduction in productivity, and a 1.6% reduction in amenities, for every 1°C increase in global mean temperature.

We are left with converting our estimates of the impact of a single extreme event into a damage function that maps global mean temperature to productivity, amenity, and capital depreciation losses. We interact damages from a given event with changes in the empirical probability of that event for each location in response to changes in temperature to obtain the average effect of rising temperature. We combine the same objects to construct how local capital depreciation, productivity, and amenity volatilities depend on global mean temperature.

We use the quantified model to evaluate the social cost of climate change risk in the U.S. economy. Our first exercise isolates the role of local and aggregate risk associated with climate change, without any realization of warming. We start with changes in local capital depreciation risk. Capital depreciation risk due to coastal storms leads to a precautionary savings motive: capital stocks rise on the southeastern Atlantic coast of the United States. Of course, this additional accumulation is a response to falling capital values because of higher local risk. On net, capitalists lose from rising local risk, up to 25% in consumption-equivalent welfare terms for changes in local risk corresponding to two degree Celsius of long-run warming. Workers benefit by up to 0.3% from changes in local capital depreciation risk because of the precautionary effect and rising capital stocks.

Aggregate capital depreciation risk due to uncertainty in long-run warming has the opposite effect. To understand this result, recall that local risk in idiosyncratic extreme weather realizations surprises capitalists. Thus, they cannot react to it on impact. By con-

¹We obtain a lower capital depreciation effect than Bilal and Rossi-Hansberg (2023) because capitalists are risk-averse instead of risk-neutral but do not face adjustment costs.

trast, aggregate risk due to long-run warming uncertainty leads to persistent deviations in capital returns around their mean. Capitalists can then intertemporally arbitrage these return differentials to their advantage. Thus, they consume more, accumulate less capital, and benefit comparably to what they lose from local risk.

Workers do not benefit similarly from aggregate warming risk for two reasons. First, they dislike lower capital stocks. Second, they cannot arbitrage intertemporally as easily as capitalists. Migration decisions are infrequent and subject to bilateral migration frictions. Thus, workers lose by up to 0.6% from aggregate capital depreciation risk on the southeastern Atlantic coast. These losses grow moderately larger once we incorporate aggregate productivity and amenity risk.

As opposed to capital, which cannot move, migration allows workers to escape risky locations. Migration targets safer locations away from the Atlantic coast and the Midwest, which are generally less exposed to climate impacts. We plan to unpack the direction of migration flows in more detail.

Our second exercise compares the magnitude of welfare losses from a median climate change scenario—a gradual increase of global mean temperature from 1°C to 4°C above pre-industrial levels by 2100, consistent with the IPCC’s business-as-usual scenario—with the associated cost of risk. We find that risk substantially magnifies welfare losses of workers. It also magnifies losses for capitalists, but to a lesser extent because of offsetting effects of local and aggregate risk. We plan to describe the spatial and aggregate patterns of amplification due to risk in more detail.

The rest of this document is organized as follows and primarily describes our methodology and preliminary results, rather than presenting a finished draft. Section 2 outlines the framework. Section 3 describes our solution method, defining the Master Equation, the “First-Order Approximation to the Master Equation” (FAME, Bilal, 2023; Bilal and Rossi-Hansberg, 2023), and the “Second-Order Approximation to the Master Equation” (SAME, Bilal, 2023). Section 4 details our quantification. Section 5 presents our preliminary results.

2 Framework with risky location and investment choices

This section presents the setup of our dynamic spatial model with aggregate and local risk. We model an economy with many locations and two types of agents: workers and capitalists. Workers can move subject to mobility costs and earn a wage, which they consume each period. Capitalists are fixed in a location, earn returns on their risky capital investments, and face a dynamic consumption-savings decision.

2.1 Agents and preferences

There is a unit mass of infinitely-lived workers who choose in which location $i \in \{1, \dots, \mathcal{I}\}$ to live. Workers living in location i have CRRA preferences $u(\tilde{C}_{it}) = \frac{\tilde{C}_{it}^{1-\gamma}-1}{1-\gamma}$, with $\tilde{C}_{it} = A_{it}C$, and A_{it} denoting amenities in the location where they live at time t and C denoting a Cobb-Douglas aggregator of a freely traded final good, used as the numeraire, and housing. Namely, $C = \left(\frac{c}{1-\beta}\right)^{1-\beta} \left(\frac{h}{\beta}\right)^{\beta}$, where β denotes the share of housing in expenditure. Workers discount the future at rate ρ . Workers are allowed to move at rate μ , in which case they draw extreme-value distributed idiosyncratic preference shocks for potential destinations, with dispersion parameter ν . If they move, they pay a bilateral moving cost τ_{ij} . We define the long-run (A_i) and time-varying (a_{it}) components of amenities in location i as $A_{it} = A_i e^{a_{it}}$.

Each location is populated by a unit measure of capitalists. They are immobile and have CRRA preferences $u^{\text{cap}}(C) = \frac{C^{1-\zeta}-1}{1-\zeta}$ over the final consumption good. They do not consume housing.

2.2 Technology

Workers in location i are employed by a representative firm that produces the final good using commercial structures and labor and faces constant returns to scale. Locations are endowed with a local productivity Z_{it} and a local stock of land L_i . A representative firm is endowed with a production function for the final good given by $Y_{it} = Z_{it} S_{it}^{\alpha} (N_{it}^P)^{1-\alpha}$, where S_{it} denotes local commercial structures and N_{it}^P the local number of workers employed in final good production. We denote by N_{it} the total number of workers in location

i at time t . We define the long-run (Z_i) and time-varying (χ_{it}) components of productivity as $Z_{it} \equiv Z_i e^{\chi_{it}}$.

In every location, there is an equilibrium capital stock K_{it} . Capital can be combined with land L_i and construction labor N_{it}^B to produce buildings B_{it} according to the production function $B_{it} = L_i^\omega (N_{it}^B)^\omega K_{it}^{1-\omega-\omega}$. Buildings B_{it} can be costlessly repurposed between residential housing and commercial structures, so that $B_{it} = S_{it} + H_{it}$. Denote by r_{it} the common rental rate of buildings to firms and workers.

Capitalists decide how much to invest in each location. Their only financial asset is their own capital stock, which is not mobile across locations. We denote by $R_{K,it}$ the return on a unit of capital. The local capital stock depreciates by $d\bar{\Delta}_{it}$ over a time interval of length dt , where

$$d\bar{\Delta}_{it} = \Delta_{it}dt + \Sigma_{it}dW_{it},$$

and dW_{it} is a standard Brownian motion that is i.i.d. across locations and time. The scaling factor Σ_{it} is “small” in the sense of the perturbation method described later. We define the long-run (Δ_i) and time-varying (δ_{it}) components of the mean capital depreciation rate by $\Delta_{it} \equiv \Delta_i + \delta_{it}$.

For tractability, we assume for now that land rents are consumed by absentee owners. We plan to relax this assumption in future work.

2.3 Climate change

The path of global mean temperatures, $\bar{T}_t$, is exogenous. Going forward, the vast majority of carbon emissions is expected to stem from developing economies and is thus largely exogenous to U.S. economic activity. Global mean temperatures, measured relative to pre-industrial levels, T^P , satisfy: $\bar{T}_t - T^P = \epsilon(\sigma_t z_t + T_t)$. T_t is a deterministic path corresponding to a central climate scenario, and z_t is a mean-zero shock capturing natural climate variability. The function σ_t scales the shock and encapsulates uncertainty in future carbon emissions. Two examples illustrate the role of σ_t clearly. First, when $\sigma_t = 1$, the shock z_t is additive and does not alter the variance of global temperature around its deterministic path. Second, when σ_t scales with T_t , the variance of global temperature

increases proportionally with the level of temperature itself. Finally, ϵ serves as a scaling factor introduced to simplify notation in our solution method.

Local productivity χ_{it} , amenities a_{it} , and capital depreciation δ_{it} depend on time only through the path of global mean temperatures. We parameterize these shifters through the following functions:

$$\chi_{it} = \chi_i (\bar{T}_t - T^P), \quad a_{it} = a_i (\bar{T}_t - T^P), \quad \delta_{it} = \delta_i (\bar{T}_t - T^P), \quad (1)$$

where $\chi_i(0) = a_i(0) = \delta_i(0) = 0$. Furthermore, we specify these damage functions as quadratic functions (which is immaterial since we will only consider up to second-order perturbations):

$$\chi_i(T) = \chi_{i1}T + \frac{\chi_{i2}T^2}{2}, \quad a_i(T) = a_{i1}T + \frac{a_{i2}T^2}{2}, \quad \delta_i(T) = \delta_{i1}T + \frac{\delta_{i2}T^2}{2}. \quad (2)$$

2.4 Static equilibrium

The maximization problem of firms implies that the wage w_{it} in location i satisfies the firm's first-order condition $w_{it} = (1 - \alpha)Z_{it}(S_{it}/N_{it}^P)^\alpha$. Similarly, the rental rate for structures satisfies $r_{it} = \alpha Z_{it}(S_{it}/N_{it}^P)^{-(1-\alpha)}$, since depreciation is not covered by the users of buildings. The optimal choice of labor in building production implies $\omega r_{it}B_{it}/N_{it}^B = w_{it}$. Furthermore, the local demand for housing by workers implies that an equilibrium in the housing market satisfies $\beta w_{it}N_{it} = r_{it}H_{it}$.

We determine equilibrium prices and quantities as functions of the local capital stock K_{it} and local number of workers N_{it} . We show that labor is allocated to goods and buildings production in constant shares: $N_{it}^P = xN_{it}$ and $H_{it} = yB_{it}$, where x, y are combinations of parameters described in equation (21), Appendix A.1. Using these shares and the production function of buildings, the wage and the rental rate of buildings become: $w_{it} = w_{i0}Z_{it}L_i^{\omega\alpha}(K_{it}^{1-\omega-\omega}N_{it}^{\omega-1})^\alpha$ and $r_{it} = r_{i0}Z_{it}L_i^{-\omega(1-\alpha)}(K_{it}^{1-\omega-\omega}N_{it}^{\omega-1})^{-(1-\alpha)}$, where w_{i0}, r_{i0} are combinations of parameters and local fundamentals expressed in Appendix A.1.

The rental rate of capital depends on the rental rate of buildings. Profit maximization by developers implies that it satisfies: $R_{K,it} = R_{0i}e^{\chi_{it}}K_{it}^{-\phi}N_{it}^\psi \equiv R_i(\chi_{it}, K_{it}, N_{it})$, where R_{0i}

is a location-specific constant that depends on the permanent component of productivity and available land, and ϕ and ψ are combinations of parameters detailed in Appendix A.1.

We can now solve for consumption given prices. Consumption by workers in location i is equal to the real wage. It satisfies $C_{it} = w_{it}/r_{it}^\beta = C_{0i}e^{(1-\beta)\chi_{it}} \left(K_{it}^{1-\omega-\omega}/N_{it}^{1-\omega} \right)^\xi$, where C_{0i} is a location-specific constant, and ξ a combination of parameters detailed in Appendix A.1. Finally, flow utility of workers in location i at time t is: $u \left(A_{it}w_{it}/r_{it}^\beta \right) = u \left(\tilde{C}_{0i}e^{a_{it}+(1-\beta)\chi_{it}} \left(K_{it}^{1-\omega-\omega}/N_{it}^{1-\omega} \right)^\xi \right) \equiv U_i(a_{it}, \chi_{it}, K_{it}, N_{it})$, where $\tilde{C}_{0i} \equiv A_i C_{0i}$.

2.5 Migration decisions

Workers solve a forward-looking dynamic migration decision problem. We denote by V_{it} the value of being located in i at time t . V_{it} satisfies the Hamilton-Jacobi-Bellman (HJB) equation given by:

$$\rho V_{it} = U_i(a_{it}, \chi_{it}, K_{it}, N_{it}) + \mathcal{M}_i[V] + \frac{\mathbb{E}[d_t V_{it}]}{dt}, \quad \mathcal{M}_i[V] \equiv \mu \left\{ \frac{1}{\nu} \log \left(\sum_j e^{\nu(V_{jt} - \tau_{ij})} \right) - V_{it} \right\}, \quad (3)$$

where we used standard discrete choice results and denoted $\mathcal{M}_i[V]$ the continuation value from migration. The last term in the worker's HJB equation represents the continuation value from changes in the aggregate state, where the notation d_t denotes the time increment due to changes in aggregates only and not due to individual migration. Standard discrete choice results then imply migration shares as a function of the vector of values V given by:

$$m_{ij}(V) = \frac{e^{\nu(V_j - \tau_{ij})}}{\sum_k e^{\nu(V_k - \tau_{ik})}}. \quad (4)$$

The population distribution evolves according to the Kolmogorov Forward (KF) equation:

$$\frac{dN_{it}}{dt} = \mu \left(\sum_k m_{ki}(V_t) N_{kt} - N_{it} \right) \equiv \mu \left((m^*(V_t) - \text{Id}) N_t \right)_i. \quad (5)$$

In equation (5), m denotes the matrix of migration shares $m_{ij}(V_t)$. m^* denotes the matrix transpose of the matrix m . We also denote by $M^*(V) = \mu(m^*(V) - \text{Id})$ a matrix that collects migration shares and the identity matrix, so that (5) becomes, in matrix notation, $\frac{dN_t}{dt} = M^*(V)N_t$. We denote $M(V) = \mu(m(V) - \text{Id})$ the transpose of $M^*(V)$ for future reference.

2.6 Investment decisions

Capitalists solve a forward-looking dynamic investment decision problem. We denote $\mathcal{P}_{it}(K)$ the value of being located in i at time t and holding K units of local capital. The level of capital evolves according to the law of motion:

$$dK_{it} = (I_{it} - \Delta_{it}K_{it})dt - \Sigma_{it}KdW_{it}, \quad (6)$$

where I_{it} denotes local investment, and $\Sigma_{it} = \Sigma_i \times \epsilon \sqrt{\sigma_t z_t + T_t}$, which scales with ϵ , the common scale parameter with respect to which we take the perturbation. We show how to micro found a diffusion process from infrequent weather extremes in Appendix C, and how this micro foundation delivers the definition of Σ_{it} that we choose. The capitalist budget constraint is $C_{it} + I_{it} = R_{it}K_{it}$. The capitalist HJB is then:

$$\rho \mathcal{P}_{it}(K) = \max_C \frac{C^{1-\zeta} - 1}{1-\zeta} + [(R_{it} - \Delta_{it})K - C]\mathcal{P}_{it,K} + \frac{\Sigma_{it}^2}{2} K^2 \mathcal{P}_{it,KK} + \mathbb{E}_t \left[\frac{d\mathcal{P}_{it}}{dt} \right],$$

We show in Appendix A.2 that $\mathcal{P}_{it}(K) = \frac{Q_{it}K^{1-\zeta} - 1/\rho}{1-\zeta}$, where Q_{it} satisfies:

$$\rho Q_{it} = Q_{it}^{1-1/\zeta} + (1-\zeta) \left[R_{it} - \Delta_{it} - Q_{it}^{-1/\zeta} - \frac{\zeta \Sigma_{it}^2}{2} \right] Q_{it} + \mathbb{E}_t \left[\frac{dQ_{it}}{dt} \right] \quad (7)$$

Furthermore, the investment policy satisfies $I_{it} = (R_{it} - Q_{it}^{-1/\zeta}) K_{it}$.

2.7 Steady-state

In the deterministic steady-state at baseline temperatures, $\epsilon = 0$. All time derivatives are equal to zero. The HJB equation of workers (3) and capitalist capitalists (7) become:

$$\rho V_i^{ss} = U_i(0, 0, K_i^{ss}, N_i^{ss}) + \mathcal{M}_i[V^{ss}], \quad \rho Q_i^{ss} = (Q_i^{ss})^{1-1/\zeta}. \quad (8)$$

The steady-state population distribution $N^{ss} = \{N_i^{ss}\}_i$ and capital distribution $K^{ss} = \{K_i^{ss}\}_i$ satisfy:

$$0 = M^*(V^{ss})N^{ss}, \quad \Delta_i = R_i^{ss} - (Q_i^{ss})^{-1/\zeta}. \quad (9)$$

Appendix B describes a simple algorithm to solve for the steady-state that leverages equations (8)-(9).

3 Solution Method

3.1 Strategy

Our economy is a dynamic general equilibrium economy in which the distributions of workers $N_t = \{N_{it}\}_i$ and capital $K_t = \{K_{it}\}_i$ are aggregate state variables that determine local prices. The distribution of workers and capital evolves slowly according to the laws of motions (5) and (6). These laws of motion in turn depend on the values of workers given by the HJB equations (3) and (7). We solve this dynamic fixed point problem by leveraging the ‘Master Equation’ representation of the economy.

The Master Equation approach is structured in two steps. In the first step, we merge the individual decision problems with the laws of motion of the population and capital distributions into a single Bellman equation: the Master Equation. The Master Equation is a state-space representation of the equilibrium. The second step consists in taking a local perturbation of the Master Equation in the scale parameter ϵ around a steady-state of the economy.

3.2 The Master Equation

We start with the Master Equation for workers. We combine the HJB equation (3) with the laws of motion (5) and (6) with a simple change of variables. We index value functions by the population and capital distributions instead of calendar time alone. The core

idea is that the population and capital distributions are no more than large-dimensional state variables from the perspective of any given worker. Of course, we need to keep calendar time as a separate index because of climate change embedded in global mean temperatures T_t . We denote log capital by $\mathcal{K}_t = \log K_t$.

Specifically, we change variables by writing $V_{it} = V_{it}(z_t, N_t, \mathcal{K}_t)$ and $Q_{it} = Q_{it}(z_t, N_t, \mathcal{K}_t)$, where now the time subscript t only captures dependence on deterministic temperature. Recall that $N_t = \{N_{it}\}_i$ denotes the vector of population across locations at time t , and $\mathcal{K}_t = \{\mathcal{K}_{it}\}_i$ similarly denotes the log capital distribution. Using this change of variables, we use the chain rule to express the continuation value from changes in aggregate states as:

$$\mathbb{E}_{it} \left[\frac{d_t V_{it}}{dt} \right] = \underbrace{\frac{\partial V_{it}}{\partial t}}_{\text{change in } T_t} + \underbrace{\mathcal{A}_\epsilon(z)[V_{it}]}_{\text{change in } z_t: \text{ continuation value}} + \underbrace{\sum_j \frac{\partial V_{it}}{\partial N_{jt}} \frac{dN_j}{dt}}_{\text{change in } N_t: \text{ chain rule}} + \underbrace{\sum_j \frac{\partial V_{it}}{\partial \mathcal{K}_j} \frac{d\mathcal{K}_{jt}}{dt}}_{\text{change in } \mathcal{K}_t: \text{ chain rule}} + \underbrace{\frac{1}{2} \sum_j \frac{\partial^2 V_{it}}{\partial \mathcal{K}_j^2} \Sigma_{jt}^2}_{\text{change in } \mathcal{K}_t: \text{ chain rule + Ito correction}}, \quad (10)$$

where $\mathcal{A}_\epsilon(z)$ is an operator that embeds the continuation value arising from natural climate variability. We later assume that z follows a continuous-time AR(1) process $dz_t = -\theta z_t + dW_t$. In this case, the operator $\mathcal{A}_\epsilon(z)$ is given by:

$$\mathcal{A}_\epsilon(z)[V] = -\theta z \epsilon \frac{\partial V}{\partial z} + \frac{\epsilon^2}{2} \frac{\partial^2 V}{\partial z^2}. \quad (11)$$

The last two components of (10) encode how workers' values change with the population and capital distributions across locations. When workers reallocate across locations in equilibrium or when the distribution of capital shifts in equilibrium, the distribution of local wages, building rental rates, and capital rental rates changes. These changes in these three price distributions affect residents of location i .

We then use the laws of motion of the population and capital distributions (5) and (6) to express the time change of population and capital in (10). After this substitution into the continuation value (10) and into the HJB (3), we obtain the Master Equation for workers. We omit the full dependence of the value function and its derivatives on $(\mathcal{K}, N, t, \epsilon)$ on the right-hand-side for brevity, and we also omit the direct dependence of $a_{it}, \chi_{it}, \Delta_{it}$

on $\varepsilon(\sigma_t z + T_t)$:

$$\begin{aligned}
\rho V_{it}(z, N, \mathcal{K}) &= \underbrace{U_i(a_{it}, \chi_{it}, N_i, \mathcal{K}_i)}_{\text{flow payoff}} + \underbrace{\mathcal{M}_i[V]}_{\text{continuation value from own migration}} \\
&+ \underbrace{\sum_j \frac{\partial V_{it}}{\partial N_j} (M^*(V_t) N_t)_j + \mathcal{A}_\varepsilon(z)[V_{it}] + \frac{\partial V_i}{\partial t}}_{\text{continuation values from aggregate changes in population, } z \text{ and } t} \\
&+ \underbrace{\sum_j \frac{\partial V_{it}}{\partial \mathcal{K}_j} \left\{ R_i(\chi_{it}, N_i, \mathcal{K}_i) - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right\} + \frac{1}{2} \sum_j \frac{\partial^2 V_{it}}{\partial \mathcal{K}_j^2} \Sigma_{jt}^2}_{\text{continuation values from aggregate changes in capital distribution}}.
\end{aligned} \tag{12}$$

Note that the Master Equation for workers depends on the value of capitalists Q because the latter affects investment, capital accumulation, prices, and, ultimately, worker welfare.

A similar logic delivers the Master Equation for capitalists, which is given by

$$\begin{aligned}
\rho Q_{it}(z, N, \mathcal{K}) &= \underbrace{Q_{it}^{1-1/\zeta} + (1-\zeta) \left[R_i(\chi_{it}, N_i, \mathcal{K}_i) - \Delta_{it} - Q_{it}^{-1/\zeta} - \frac{\zeta \Sigma_{it}^2}{2} \right]}_{\text{flow payoff + continuation values from own investment}} Q_{it} \\
&+ \underbrace{\sum_j \frac{\partial Q_{it}}{\partial N_j} (M^*(V) N_t)_j + \mathcal{A}_\varepsilon(z)[Q_{it}] + \frac{\partial Q_{it}}{\partial t}}_{\text{continuation values from aggregate changes in population, } z \text{ and } t} \\
&+ \underbrace{\sum_j \frac{\partial Q_{it}}{\partial \mathcal{K}_j} \left\{ R_i(\chi_{it}, N_i, \mathcal{K}_i) - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right\} + \frac{1}{2} \sum_j \frac{\partial^2 Q_{it}}{\partial \mathcal{K}_j^2} \Sigma_{jt}^2}_{\text{continuation values from aggregate changes in capital distribution}}.
\end{aligned} \tag{13}$$

We omit the equilibrium definition, since it is standard. The main advantage of the Master Equations (12)–(13) is that they deliver a parsimonious characterization of the economy's general equilibrium. The derivations above establish our first result.

Proposition 1. (Equilibrium)

A solution (V, Q) to the Master Equations (12)–(13) together with a solution to the population and

capital distributions $\{N_t, K_t\}_t$ to the KF equations (5) and (6) is an equilibrium of our economy.

3.3 The FAME

Although the Master Equations (12)–(13) provide a parsimonious characterization of equilibrium, they are difficult to solve numerically because they depend on high-dimensional distributions. As a result, standard global methods quickly run into the curse of dimensionality. We instead adopt an analytic approach that both clarifies the economic mechanisms and substantially reduces the computational burden. In particular, we look for a local solution to (12)–(13) when the scale parameter ϵ is sufficiently small. We begin with a first-order approximation, solving the First-order Approximation to the Master Equations (FAME), and later use a second-order approximation in the same spirit.

The FAME is a first-order Taylor expansion of the Master Equations (12)–(13) in ϵ around the initial steady-state. We consider distributions and shocks that are close enough to steady-state. Moreover, given the nature of the evolution of capital, (6), we make the approximation considering log capital, instead of capital in levels. Namely, denote:

$$N = N^{ss} + \epsilon n, \quad \text{and} \quad \mathcal{K} = \mathcal{K}^{ss} + \epsilon k,$$

where the vector n denotes scaled deviations from steady-state for the population distribution and k denotes scaled deviations from steady-state for the log capital distribution. We then seek a solution of the form:

$$\begin{aligned} V_{it}(z, N^{ss} + \epsilon n, \mathcal{K}^{ss} e^{\epsilon k}) &= V_i^{ss} + \epsilon \left\{ \sum_j V_{ij}^N n_j + \sum_j V_{ij}^K k_j + v_{it}^Z z + v_{it}^T \right\} + \mathcal{O}(\epsilon^2), \\ Q_{it}(z, N^{ss} + \epsilon n, \mathcal{K}^{ss} e^{\epsilon k}) &= Q_i^{ss} + \epsilon \left\{ \sum_j Q_{ij}^N n_j + \sum_j Q_{ij}^K k_j + q_{it}^Z z + q_{it}^T \right\} + \mathcal{O}(\epsilon^2). \end{aligned} \quad (14)$$

The additive separability follows from the first-order Taylor expansion in ϵ .

Rather than solving for the full equilibrium objects V and Q in the Master Equations, the first-order method reduces the problem to a finite set of coefficients: the directional derivatives v_{ij}^N , v_{ij}^K , v_{it}^Z , and v_{it}^T of the value function with respect to the population and capital distributions, temperature shocks, and temperature trends. We refer to these coefficients as “impulse values.” Plugging the first-order Taylor expansions (14) into the

Master Equations (12)–(13) yields a system of restrictions that pins down these impulse values, which we call the FAME. Similar to Bilal and Rossi-Hansberg (2023), we summarize the FAME results in the following proposition. All proofs are in Appendix D.

Proposition 2. (*Deterministic and trend FAME*)

The matrices $v^N, v^K, q^N, q^K, v_t^T, q_t^T$ satisfy the generalized Sylvester matrix equation in $v^d = \begin{pmatrix} v^N & v^K \\ q^N & q^K \end{pmatrix}$ and the Ordinary Differential Equation systems in $v_t^T = \begin{pmatrix} v_t^T \\ q_t^T \end{pmatrix}$ and $p_t = \begin{pmatrix} n_t \\ k_t \end{pmatrix}$:

$$\begin{aligned}\rho v^d &= D^d + Mv^d + v^d M^* + v^d P v^d, \\ \rho v_t^T &= D_t^T + (M + v^d P)v_t^T + \frac{\partial v_t^T}{\partial t}, \\ \frac{dp_t}{dt} &= D_t^p + P(v_t^T + z_t v^Z) + (M^* + P v^d)p_t,\end{aligned}$$

where the matrices D^d, M, P and vectors D_t^T, D_t^p only depend on parameters, steady-state objects, and T_t^D . They are specified in Appendix D together with the stochastic FAME for v^Z and q^Z .

The deterministic FAME characterizes the deterministic impulse values v^N, v^K, q^N, q^K as the solution to a Bellman equation. In Proposition 2, this Bellman equation can be written as a nonlinear Sylvester matrix equation for the impulse-value matrix v^d . The key advantage is interpretability: the deterministic FAME summarizes how the value of workers and capitalists in location i changes in equilibrium when an additional worker enters location j or an additional unit of capital is added in location j .

The right-hand side of the deterministic FAME decomposes into four terms, each mapping to a distinct economic force. First, the direct price-impact matrix D^d captures how marginal changes in the population and capital distributions shift equilibrium prices and therefore flow payoffs. For workers, changes in real wages are summarized by the parameter combination $\xi(1 - \omega)$ and are then converted into utility changes via the diagonal marginal-utility term. For capitalists, the lower blocks of D^d capture how the same distributional shifts affect the return on capital.

Second, the term Mv^d encodes the partial-equilibrium option value of migration. The FAME shows that, at first order, workers can evaluate this option value using the steady-state migration matrix M embedded in M . This simplification follows from an envelope argument: migration decisions are already optimal at the steady state, so the first-order

valuation of marginal changes in location fundamentals can be computed holding the migration policy at its steady-state form.

Third, the term $\mathbf{v}^d \mathbf{M}^*$ captures a first general-equilibrium force. When assessing the effect of an additional worker in location j , agents in location i must account for the fact that this marginal entrant will migrate over time, which affects future population distributions and thus future payoffs. The FAME reveals that these expectations are summarized by the same steady-state migration matrix. The corresponding forward-looking effect for capital does not appear because there is no gross capital growth in steady state.

Fourth, the term $\mathbf{v}^d \mathbf{P} \mathbf{v}^d$ captures a second general-equilibrium force: it reflects how the marginal entrant changes the law of motion for the population and capital distributions and how agents value those induced changes. The trend FAME mirrors the deterministic structure. It adds a direct climate-impact component, D_t^T , which depends linearly on global mean temperature T_t^D and the linearized damage coefficients $a_i \equiv a'_i(0)$, $\chi_i \equiv \chi'_i(0)$, and $\delta_i \equiv \delta'_i(0)$, and then propagates this impact through analogous partial- and general-equilibrium channels.

These first-order results imply three practical properties that come directly from the local perturbation. First, the deterministic FAME is finite-dimensional: the impulse-value object $\mathbf{v}^d$ has dimension $(2\mathcal{I})^2$, rather than being infinite-dimensional like the original nonlinear Master Equations (12)–(13). Second, all inputs to the FAME can be constructed from steady-state objects, many of which admit closed-form expressions. Third, the system is block-recursive: the deterministic FAME can be solved first, the trend and stochastic impulse values can then be obtained conditional on $\mathbf{v}^d$, and finally the equilibrium path of allocations p_t can be computed by forward iteration given the impulse values.

This structure delivers a straightforward computational algorithm. Given steady-state objects, one constructs the $2\mathcal{I} \times 2\mathcal{I}$ matrices D_d , $\mathbf{M}$, and $\mathbf{P}$ and then solves for the deterministic impulse value $\mathbf{v}^d$ that satisfies the nonlinear Sylvester equation $0 = D_d + (\mathbf{M} - \rho \text{Id}) \mathbf{v}^d + \mathbf{v}^d \mathbf{M}^* + \mathbf{v}^d \mathbf{P} \mathbf{v}^d$. With $\mathbf{v}^d$ in hand, one computes the trend impulse value using a backward iteration implied by Proposition 2. Finally, given the deterministic and trend impulse values, impulse responses to shocks or initial conditions, and the resulting allocation paths, follow from a simple forward iteration.

Although tractable, the FAME has limitations. Importantly for our purposes, it fea-

tures certainty equivalence. As a result, risk has no first-order effects on workers' and capitalists' values or on the equilibrium distributions. To incorporate both aggregate and local risk into our analysis, we therefore proceed to a second-order approximation.

3.4 The SAME

The SAME is a second-order Taylor expansion of the (12)-(13) in ϵ around the initial steady-state. For the second-order approximation, we look for solutions:

$$V_{it} = V_i^{ss} + \epsilon v_{1i} + \frac{\epsilon^2}{2} v_{2i} + \mathcal{O}(\epsilon^3), \quad Q_i = Q_i^{ss} + \epsilon q_{1i} + \frac{\epsilon^2}{2} q_{2i} + \mathcal{O}(\epsilon^3), \quad (15)$$

where v_{1i} and q_{1i} come from the FAME:

$$v_{1i} = \sum_j V_{ij}^N n_j + \sum_j V_{ij}^K k_j + v_{it}^Z z + v_{it}^T \quad (16)$$

$$q_{1i} = \sum_j Q_{ij}^N n_j + \sum_j Q_{ij}^K k_j + q_{it}^Z z + q_{it}^T. \quad (17)$$

The second-order term for workers, v_{2i} , is given by

$$\begin{aligned} v_{2i} = & \sum_j \sum_s \mathcal{V}_{ijs}^{N,N} n_j n_s + \sum_j \sum_s \mathcal{V}_{ijs}^{K,K} k_j k_s + \sum_j \sum_s \mathcal{V}_{ijs}^{N,K} n_j k_s + \sum_j \sum_s \mathcal{V}_{ijs}^{K,N} k_j n_s \\ & + 2 \sum_j V_{ijt}^{N,Z} n_j z + 2 \sum_j V_{ijt}^{K,Z} k_j z + 2 \sum_j V_{ijt}^{N,T} n_j + 2 \sum_j V_{ijt}^{K,T} k_j \\ & + \bar{v}_{it} + v_{it}^{Z,Z} z^2 + v_{it}^{T,T} + 2v_{it}^{Z,T} z. \end{aligned} \quad (18)$$

The three-dimensional matrices, tensors, $\mathcal{V}_{ijs}^{N,N}$, $\mathcal{V}_{ijs}^{K,K}$, $\mathcal{V}_{ijs}^{N,K}$, and $\mathcal{V}_{ijs}^{K,N}$ correspond to the second-order directional derivatives of the worker's value function with respect to the population and capital distributions. The matrices $V_{ijt}^{N,Z}$, $V_{ijt}^{K,Z}$, $V_{ijt}^{N,T}$, and $V_{ijt}^{K,T}$ represent second-order cross directional derivatives of the value function with respect to the population distribution, capital distribution, temperature shocks, and temperature trends. Similarly, $v_{it}^{Z,Z}$ and $v_{it}^{T,T}$ are the second-order derivatives of the worker's value with respect to temperature shocks and temperature trends, with $v_{it}^{Z,T}$ denoting the cross derivative. Finally, $\bar{v}_{it}$ represents the ex-ante valuation of risk arising from the second-order

continuation value of aggregate temperature shocks.

Similarly, the second-order term q_{2i} takes the form:

$$\begin{aligned} q_{2i} = & \sum_j \sum_s \mathcal{Q}_{ijs}^{N,N} n_j n_s + \sum_j \sum_s \mathcal{Q}_{ijs}^{K,K} k_j k_s + \sum_j \sum_s \mathcal{Q}_{ijs}^{N,K} n_j k_s + \sum_j \sum_s \mathcal{Q}_{ijs}^{K,N} k_j n_s \\ & + 2 \sum_j \mathcal{Q}_{ijt}^{N,Z} n_j z + 2 \sum_j \mathcal{Q}_{ijt}^{K,Z} k_j z + 2 \sum_j \mathcal{Q}_{ijt}^{N,T} n_j + 2 \sum_j \mathcal{Q}_{ijt}^{K,T} k_j \\ & + \bar{q}_{it} + q_{it}^{Z,Z} z^2 + q_{it}^{T,T} + 2q_{it}^{Z,T} z, \end{aligned} \quad (19)$$

with the analogous interpretation that each object corresponds to a second-order directional derivative of the capitalist's value function with respect to the population and capital distributions, temperature shocks, and temperature trends.

We provide a methodological contribution by characterizing the SAME. Similar to the FAME, we summarize the SAME results in the following proposition. All proofs are in Appendix E.

Proposition 3. (*Deterministic and trend SAME*)

The tensors $\mathcal{V}^{N,N}, \mathcal{V}^{N,K}, \mathcal{V}^{K,N}, \mathcal{V}^{K,K}, \mathcal{Q}^{N,N}, \mathcal{Q}^{N,K}, \mathcal{Q}^{K,N}, \mathcal{Q}^{K,K}$, the matrices $V^{N,T}, V^{K,T}, Q^{N,T}, Q^{K,T}$ and the vectors $v_t^{T,T}, q_t^{T,T}, \bar{v}_t, \bar{q}_t$ satisfy the generalized Sylvester matrix equation in $\mathcal{T} = (\mathcal{V}, \mathcal{Q})$ and the Ordinary Differential Equation systems in $V_t^T = \begin{pmatrix} V^{N,T} & V^{K,T} \\ Q^{N,T} & Q^{K,T} \end{pmatrix}$, $v_t^{T,T} = \begin{pmatrix} v_t^{T,T} \\ q_t^{T,T} \end{pmatrix}$ and $\hat{v}_t = \begin{pmatrix} \bar{v}_t \\ \bar{q}_t \end{pmatrix}$:

$$\begin{aligned} \rho \mathcal{T} &= \mathbb{T} + \mathcal{T} \times_1 \Phi + \mathcal{T} \times_2 \Theta + \mathcal{T} \times_3 \Theta, \\ \rho V_t^T &= \tilde{D}_t^T + \Phi V_t^T + V_t^T \Theta + \frac{\partial V_t^T}{\partial t}, \\ \rho v_t^{T,T} &= \tilde{D}_t^{T,T} + \Phi v_t^{T,T} + v_t^{T,T} \Theta + \frac{\partial v_t^{T,T}}{\partial t}, \\ \rho \hat{v}_t &= \hat{d}_t + \Phi \hat{v}_t + \hat{v}_t \Theta + \frac{\partial \hat{v}_t}{\partial t} \end{aligned}$$

where the tensor $\mathbb{T}$, the matrices $\tilde{D}_t^T, \tilde{D}_t^{T,T}$ and the vector $\hat{d}_t$ only depend on parameters, steady-state objects, and T_t and σ_t . They are specified in Appendix E together with the stochastic SAME and the law of motion for population and capital. Moreover, $\times_k$ represents the tensor multiplication operation with respect to the k component.

4 Model Quantification

4.1 Data

We assemble two types of data. First, we compile economic information at the county-year level from multiple sources. Employment, wages, and related statistics are drawn from the Bureau of Economic Analysis, while investment data at five-year intervals come from the Census of Manufactures. Together, these datasets cover the period from 1960 to 2019.

Second, we construct a detailed weather dataset. We use daily data on surface temperature, wind speed, and precipitation on a 0.5° by 0.5° grid beginning in 1901 from the Inter-Sectoral Impact Model Intercomparison Project (ISMIP). These reanalysis data combine historical station-level weather observations with climate model outputs to generate high-resolution weather series. We aggregate this information to build a dataset of annual weather extremes for each U.S. county and year.

4.2 Baseline parameters

We start by determining baseline preferences parameters. Importantly, we estimate worker's migration elasticity and capitalist RRA parameter in Section 4.4 below. Throughout the rest of the paper, we interpret an interval $[t, t + 1)$ as one year.

Preference parameters are determined as follows. We use a pure rate of time preference $\rho = 0.02$, a value that is standard in the climate change literature (Rennert et al., 2022). We set the housing expenditure share for workers to the common value of $\beta = 0.3$ (Davis and Ortalo-Magne, 2011). We momentarily set the worker's risk-aversion parameter to $\gamma = 1$. In current work we relax this assumptions.

The next step is to determine the parameters governing migration decisions. We set $\mu = 2.30$ so that 90% of workers have the option to migrate within a year. We estimate bilateral migration costs τ_{ij} consistent with the empirical migration rate at baseline, as described in more detail below. Local amenities are estimated using the model inversion as described in the next section.

The parameters in the production function of goods and buildings are set to standard

values. The share of commercial structures in goods production in the model includes both capital and physical structures in the data. Typical estimates put the first share at 0.3, and the second at 0.1, so $\alpha = 0.4$. To obtain the share of labor ϖ in the production function of buildings, we measure the fraction of labor in the structures production sector x from the BEA input-output data. We obtain $x = 0.95$. Thus, we recover $\varpi = \frac{(1-\alpha)(1-x)}{\alpha x + (1-\alpha)\beta} = 0.05$. To obtain the share of land in the building production function ω , we use common estimates from the real estate literature. For a typical urban property, the value of land represents around 35% of the property value, while the building value represents the other 65% (Davis et al., 2021) Hence, we set $\omega = 0.35$. Local productivities are the result of the model inversion, as described below.

The final set of parameters is related to investment choices. We impose a common capital depreciation rate of $\Delta_i = \Delta = 0.08$ for all counties at baseline. Of course, climate change affects the depreciation rate heterogeneously across locations. We estimate the investment elasticity and the local investment costs as described below.

4.3 Inversion of fundamentals

Our second step is to estimate time-invariant fundamentals for every location. Suppose for now that we have estimates of the migration elasticity ν and the investment elasticity ζ .

We use 2012 data for wages w_i , employment N_i , investment I_i , and migration flows m_{ij} for every pair of counties i, j . We interpret 2012 as the initial steady-state of the economy. We use a recursive scheme to recover the time-invariant fundamentals in every location, Z_i, A_i, c_{i0} , and τ_{ij} . The following proposition, proven in Appendix ??, shows that given the observed data we can uniquely recover these fundamentals.

Proposition 4. *(Inversion of fundamentals)*

Given data $w_i, N_i, \tilde{I}_i, m_{ij}$, there exists a unique set of vectors of fundamentals, (Z_i, A_i, c_{i0}) , and a unique symmetric migration cost matrix, τ_{ij} .

The proof of Proposition 4 is constructive and thus provides an algorithm to recover fundamentals. The proof extends standard static inversion arguments to our fully forward-looking dynamic setting. We can only recover symmetric migration costs because we

invert them non-parametrically.²

4.4 Migration Elasticity and Capitalist Preferences

Our inversion argument relies on knowledge of the migration elasticity ν and capitalist RRA parameter ζ . We estimate those internally using the reduced-form estimates of the impact of extreme events on economic activity of Bilal and Rossi-Hansberg (2023). Thus, we nest our inversion procedure in an outer loop in which we search over candidate values for ν and ζ .

Consistently with Bilal and Rossi-Hansberg (2023), we assume that storms only generate a capital depreciation shock in coastal counties. We denote its scalar magnitude by $\Gamma^{\text{cap dep}}$. Heat waves generate a productivity shock together with an amenity shock in warm counties. We denote by Γ^{prod} the scalar value of the productivity shock, and by $\eta \times \Gamma^{\text{prod}}$ the magnitude of the amenity shock.³

We leverage a key property of the FAME to estimate the migration and investment elasticities without having to specify the magnitude of shocks. The response of the economy to shocks is linear in the magnitude of the shock by construction. Hence, relative impulse responses are independent from the magnitude of a common shock. For instance, in the FAME, the impulse response of employment relative to the response of income per capita following a productivity shock is independent from the magnitude of this productivity shock. If we target relative impulse responses, we can thus use a shock of an arbitrary magnitude. This property allows us to estimate ν , ζ , and η independently of the magnitude of the underlying shocks $\Gamma^{\text{cap dep}}$, Γ^{prod} .

We then proceed by indirect inference. Our approach is similar to a standard grid search, although the details of our implementation differ for efficiency. In a first step, we populate our parameter space using parameter vectors $\theta \equiv (\nu, \eta, \zeta)$ from a Sobol sequence. This approach covers the multivariate space more efficiently than with a standard tensor product grid. We chose the range of this parameter space to be wide enough to include a broad range of possible values and of course to include our point estimates.

²Without symmetry, they are not separately identified from local amenities. If we imposed instead that migration costs depend on some distance metric, we could, of course, estimate this dependence flexibly.

³We set the persistence of the productivity shock ψ^{prod} to match a half-life of 5 years. As we show below, this persistence aligns with the mean-reversion patterns in the data.

In a second step, we invert fundamentals as in Section C.1 and solve for the steady-state following Section 2.7 and Appendix B. Next, we solve the steady-state and the FAME of our economy. We further simulate the dynamic response of the entire economy to a 1% transitory productivity shock, together with a $\eta\%$ amenity shock. We also simulate the dynamic response of the entire economy to a 1% capital depreciation shock. It suffices to consider a 1% shock for our purposes because of the the FAME is linear and we consider relative responses to estimate elasticities, a point to which we return below. From these counterfactual responses, we construct model analogs of the event study estimates from Bilal and Rossi-Hansberg (2023). We account for the empirical spatial correlation of storms and heat waves by treating each location proportionally to the empirical number of extreme events. We then use these model-based event study estimates to construct target moments that inform (ν, η, ζ) .

In a third step, we find the parameter vector (ν, η, ζ) among our set that best matches the data. Then, starting from this parameter vector, we use a standard minimum distance solver to adjust the parameters (ν, η, ζ) and close any remaining discrepancy relative to the data.

What moments are best fit for our purposes? Our strategy estimates the three parameters $\theta = (\nu, \eta, \zeta)$ jointly, but we discuss how we select each moment to provide intuition for identification. We provide numerical confirmation that these parameters determine the relevant moments below.

The gravity structure of migration indicates that the migration elasticity ν is related to the response of local employment relative to the response of income per capita, holding amenities fixed. Thus, we use the response of employment relative to income per capita following a capital depreciation shock, which leaves amenity unchanged.

To inform the amenity-productivity ratio η , we use the cumulative impulse response of employment relative to the cumulative impulse response of income per capita at 10 years after a productivity-amenity shock, $\frac{\text{cir}_{10, \text{emp}, \text{prod-am}, \theta}}{\text{cir}_{10, \text{ipc}, \text{prod-am}, \theta}}$. We target the corresponding moment in the data following a heat wave in warm counties. Intuitively, if employment responds strongly relative to income per capita holding the migration elasticity ν fixed, amenities must be a large component of the effect of heat waves.

The cumulative impulse response of investment relative to employment 10 years after

a capital depreciation shock, $\frac{cir_{10,inv,cap\ dep,\theta}}{cir_{10,emp,cap\ dep,\theta}}$ informs the capitalists preference parameter ζ . Intuitively, the size of the employment response encapsulates the overall magnitude of the shock. If investment responds strongly relative to employment, the capitalists RRA parameter ζ must be large. We target its counterpart in the data following storms in coastal counties.

We invert fundamentals, solve for the steady-state of the model, the FAME, and impulse responses for 2,000 triplets $\theta = (\nu, \eta, \zeta)$. We construct our target moments and contrast them with their data counterparts. Once we identify the triplet among the 2,000 that provides the closest fit, we use a standard Newton search to close the remaining gap between model and data moments. The figure confirms our global identification argument numerically.⁴

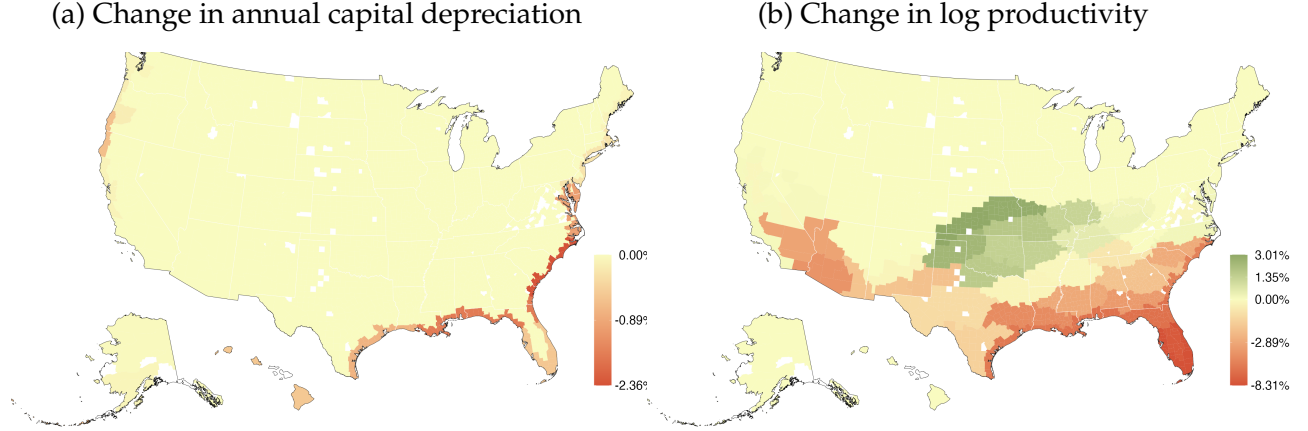
We estimate the migration elasticity to be $\nu = 0.24$. Consistently with our county-level specification, our value is just slightly above the value of 0.2 in the state-level specification in Caliendo et al. (2019). We estimate that amenities respond $\eta = 0.17$ times as much as productivity following heat waves. This relative response is somewhat larger than the average one in Cruz and Rossi-Hansberg (2023) for the entire world. The larger relative role of amenities in the context of the U.S. is consistent with production being better shielded from extreme heat through adaptation, such as air conditioning, than in developing nations. Our estimate of the capitalists RRA parameter $\zeta = 3.41$ is in range with the values reported in the literature.

4.5 Damage functions

With the migration and investment elasticities in hand, we turn to estimating the damage functions for productivity, amenities, and capital depreciation χ_i, a_i, δ_i . We do so in two steps. In the first step, we estimate common economic damages from a given realization of storms $\Gamma^{cap\ dep}$ and heat waves Γ^{prod} using the cumulative impulse responses from Bilal and Rossi-Hansberg (2023). In the second step, we interact these damages from single events with local changes in the probability that these events occur as global mean

⁴In order to be able to cover the parameter space efficiently, we aggregate counties into 100 clusters for this exercise. We choose these clusters with a k-means algorithm to group clusters based on latitude, longitude, population, income, and coastal status. Our results vary only slightly when we increase the number of clusters.

Figure 1: Damage functions from heat and storms in the United States.



Note: Effect of a 1°C increase in global temperature on productivity and annual capital depreciation rates across counties in the U.S. Constructed according to equation (??). Panel (a): coefficients χ_i such that $d \log Z_{it} = \chi_i dT_t$. Panel (b): negative of coefficients δ_i such that $d\Delta_{it} = \delta_i dT_t$.

temperatures rise. As the estimation procedure follows closely Bilal and Rossi-Hansberg (2023), we omit its full exposition and proceed to describe the results.

Figure 1 maps our estimated damage functions. Panel 1(a) shows the impact of a 1°C increase in global mean temperatures on capital depreciation for every county in the U.S. The impact is highly heterogeneous across space and only imperfectly correlated with the one for heat waves. Storms increase capital depreciation rates substantially in the South-Eastern Atlantic coast. In Louisiana and North and South Carolina, a 1°C increase in global temperature increases the capital depreciation rate by more than 2 p.p. This effect amounts to more than a 6 p.p. increase for a warming scenario of 3°C, more than doubling the baseline capital depreciation rate. These capital depreciation effects are substantial, but affect a much smaller fraction of counties than heat waves. Through the lens of our structural model, we can compare their relative importance.

Panel 1(b) shows the impact of a 1°C increase in global mean temperatures on productivity for every county in the U.S. The impact is highly heterogeneous across space. We find that southern Florida will experience productivity reductions in excess of 5% due to the rapidly rising occurrence of heat waves. For a 3°C warming scenario, this reduction compounds to more than a 15% reduction in productivity. Productivity losses of comparable magnitude are pervasive throughout the South of the U.S. Northern Texas and Oklahoma gain moderately, as our historical data shows that they experience a reduction

in the frequency of heat waves. By construction, the change in amenities is proportional to the change in productivity and thus we omit the corresponding map.

5 The impact of climate risk in the US

We conduct counterfactual exercises to assess the effects of aggregate and local risk on the U.S. spatial economic configuration. We start the economy in steady state. We take 2025 as the year in which agents learn the warming path T_t and the paths of aggregate and local risk, σ_t and Σ_{it} . We consider a warming scenario in which T_t rises gradually by 2°C by 2100 and remains constant thereafter. We set $\sigma_t = T_t/2$, capturing the idea that as global temperature increases, the variance of temperature increases as well. We choose the scaling factor corresponding to the IPCC's 95% confidence interval on global temperature.

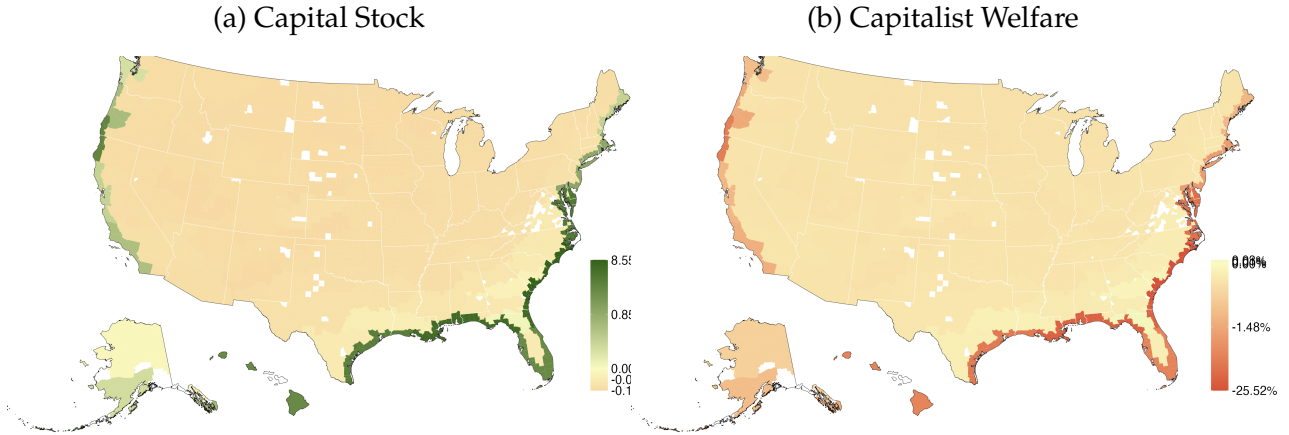
5.1 Local and aggregate risk in the long-run

We first study the long-run effects of local and aggregate risk. To do so, we take as our baseline the steady-state economy under a 2°C temperature increase with no local or aggregate risk, that is, $\sigma_t = 0$ and $\Sigma_{it} = 0$ for all i and t . We then compute new stochastic steady states in which we first introduce local climate risk by turning on Σ_{it} , and then introduce aggregate risk by turning on σ_t .

Figure 2 displays the effects of local risk on capitalists. Figure 2(a) illustrates the heterogeneous impact on the local capital stock. The idiosyncratic nature of local risk leads capitalists to increase their capital stock in almost all counties. The reason is that idiosyncratic risk creates a precautionary savings motive for capitalists. Faced with greater uncertainty about future shocks, capitalists respond by accumulating capital, which is their only financial asset. Figure 2(b) shows that local risk leads to substantial welfare losses for capitalists in southern coastal locations. Despite holding more capital, in places like coastal South Carolina, capitalists lose up to 25% in consumption-equivalent terms from the increase in local risk.

Aggregate risk, on the other hand, has a moderate positive impact on capitalists. Fig-

Figure 2: Effect of local risk on capitalists.



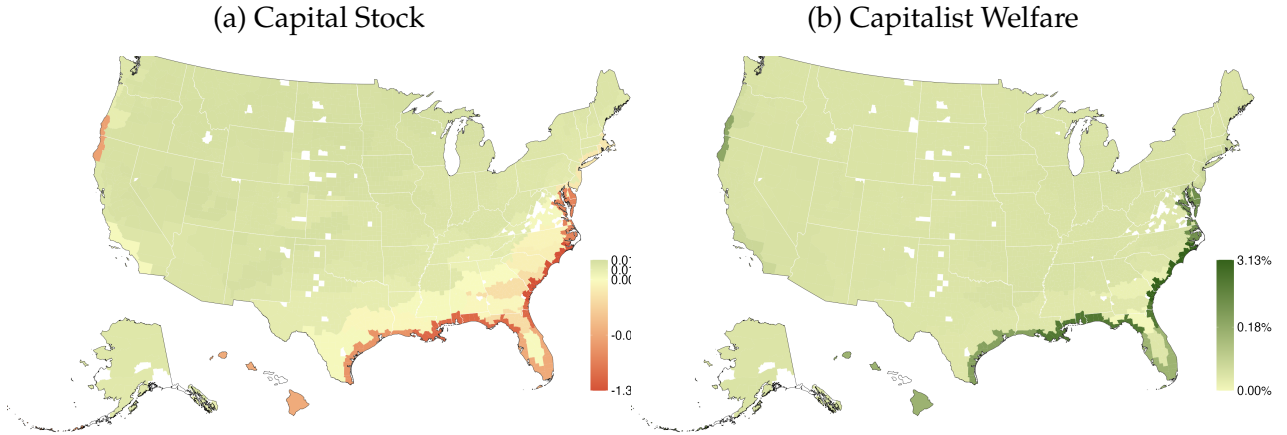
Note: Effect of local risk on capitalist under a 2°C increase in global temperature. Panel (a): change in capital stock relative to the deterministic steady state. Panel (b): change in capitalist welfare relative to the deterministic steady state.

Figure 3 illustrates the effect of aggregate risk on the capital stock and capitalist welfare across U.S. counties. To isolate aggregate risk, we set $\Sigma_{it} = 0$ and turn on the path of σ_t . Figure 3(a) shows that the capital stock slightly decreases in coastal counties in the southern region. The magnitude of this decline is about five times smaller than the increase in capital induced by local risk. The reason capitalists decumulate capital under aggregate risk is that aggregate risk acts as a state variable on which they can condition their decisions. In particular, long-run uncertainty about warming generates persistent deviations in capital returns around their mean. Capitalists can intertemporally arbitrage these return differentials to their advantage. As a result, they consume more and accumulate less capital.

Figure 3(b) shows that capitalists gain from aggregate risk throughout the country. By reacting to aggregate shocks and arbitraging return differentials, capitalists who are more exposed to climate risk experience welfare gains above 3% in consumption-equivalent terms.

Figure 6 in Appendix I shows the effects of local and aggregate risk on workers. These effects mirror those for capitalists. Local risk generates modest welfare gains for workers, who benefit from a higher capital stock. Coastal locations that experience larger capital gains also experience population gains. In contrast, aggregate risk generates modest costs for workers. Because the capital stock is lower, wages decline, leading to lower worker

Figure 3: Effect of aggregate risk on capitalists.



Note: Effect of aggregate risk on capitalist under a 2°C increase in global temperature. Panel (a): change in capital stock relative to the deterministic steady state. Panel (b): change in capitalist welfare relative to the deterministic steady state.

welfare. Migration responses to aggregate risk are modest.

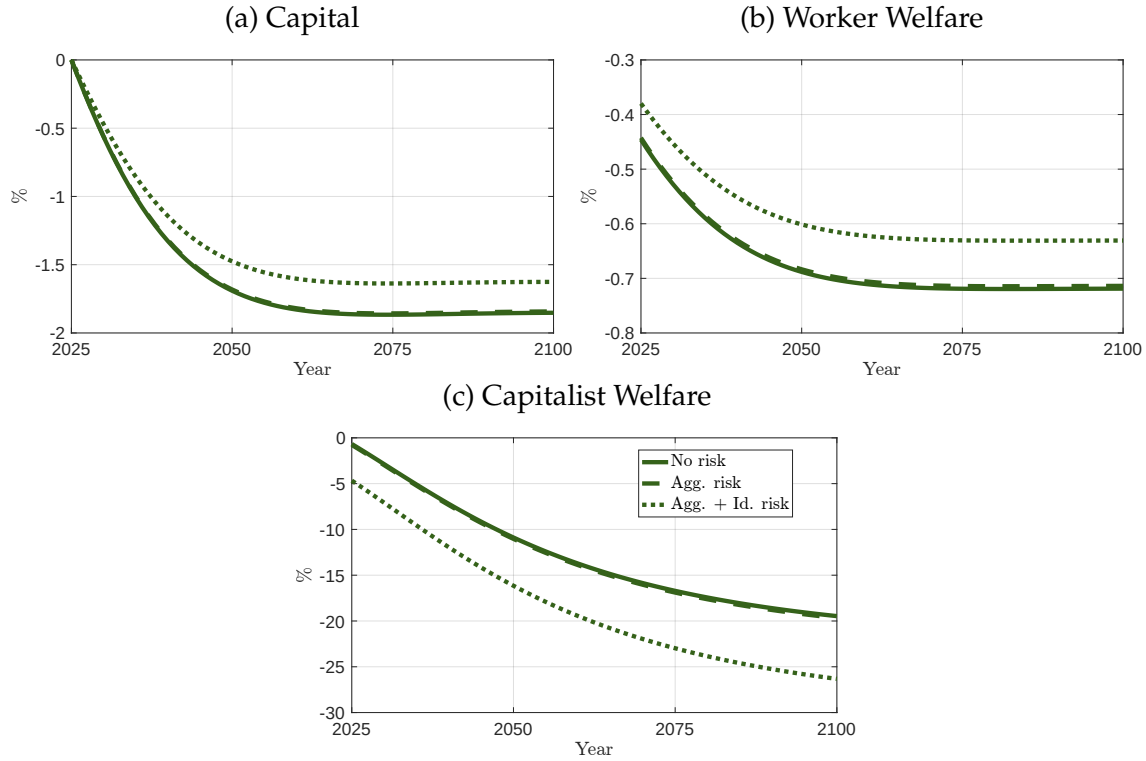
5.2 Local and aggregate risk during the transition

Now we analyze the impact of local and aggregate risk during the transition to a warmer world. We take the 2025 steady state as the baseline scenario and study the transition to a new steady state with a 2 °C temperature increase by 2100. We activate local and aggregate risk during the transition.

Figure 4 shows aggregate trends during the transition under three scenarios. The first scenario (solid line) considers an increase in temperature but no risk. The second scenario (dashed line) considers temperature increases with aggregate risk. The third scenario (dotted line) considers temperature increases with both aggregate and local risk. Figure 4(a) shows that higher global temperatures directly reduce the capital stock by increasing capital depreciation. By 2100, the U.S. economy experiences a decline in aggregate capital of slightly less than 2%. Aggregate risk has a negligible effect on the capital stock, consistent with Figure 3. In contrast, idiosyncratic (local) risk has a permanent effect, raising the capital stock by about 3 pp in the long run. The effects of both types of risk on capital accumulation scale with the increase in temperature.

Figures 4(b) and 4(c) display workers' and capitalists' welfare during the transition. Workers experience welfare losses of 0.7% in consumption-equivalent terms following a

Figure 4: Effect of local and aggregate risk during transition.



Note: Effect of aggregate local and aggregate risk during the transition towards a 2°C increase in global temperature by 2100. Panel (a): change in capital stock relative to the deterministic steady state. Panel (b): change in aggregate workers welfare relative to the deterministic steady state. Panel (c): change in aggregate capitalist welfare relative to the deterministic steady state.

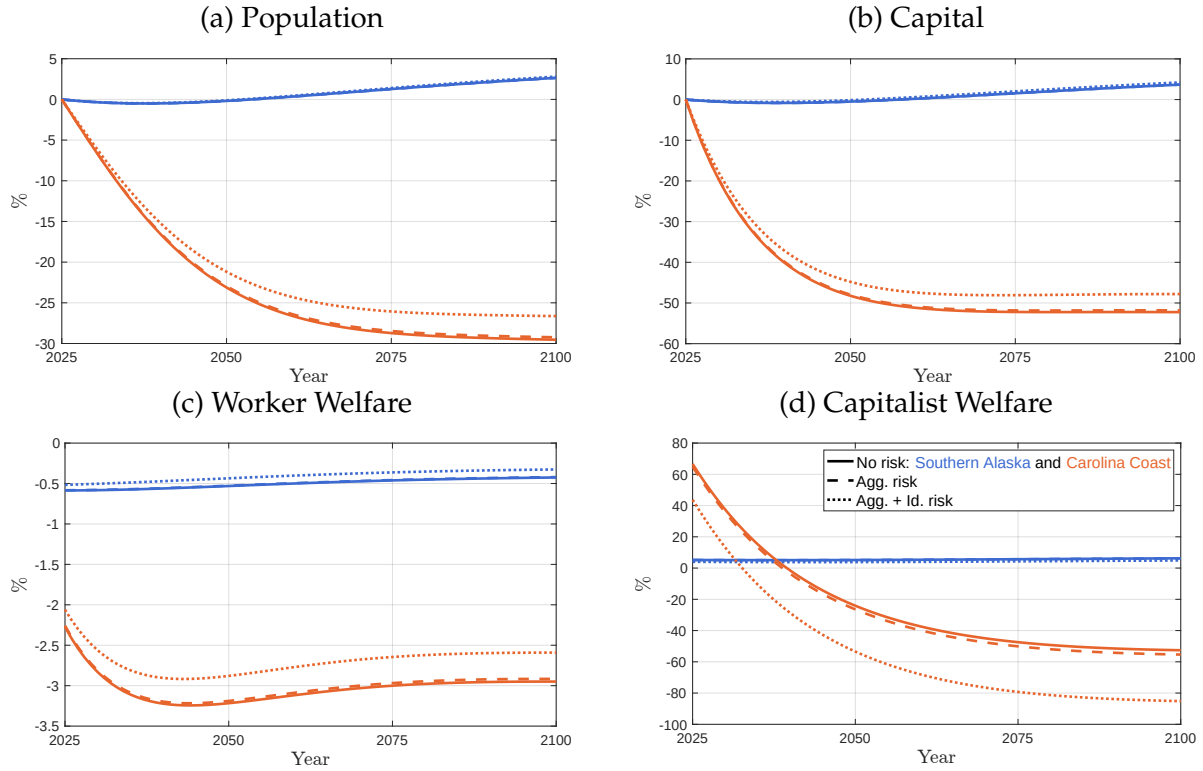
2 °C increase in temperature. Introducing aggregate risk leaves the welfare effects largely unchanged. Nevertheless, local risk mitigates these welfare losses by 1 pp in the long run.

In contrast, capitalists suffer a direct welfare loss of 20% in consumption-equivalent terms from higher temperatures. Similarly to workers, aggregate risk has a negligible effect on capitalists in both the short and long run. Idiosyncratic risk magnifies capitalists' welfare losses by 5% in the short run and by 7% in the long run.

Finally, Figure 5 shows the transitional dynamics for Southern Alaska and the South Carolina coast. Counties in southern Alaska experience the largest worker inflows due to the increase in global temperatures, around 3% relative to the 2025 steady state. The capital stock in this area increases modestly. Overall, workers experience a relatively stable welfare reduction of about 0.5% over time, while capitalists gain around 5%.

In contrast, coastal South Carolina is the region in the US that experiences the largest

Figure 5: Transitional dynamics for Alaska and South Carolina Coast.



Note: Effect of aggregate local and aggregate risk for selected locations during the transition towards a 2°C increase in global temperature by 2100. Blue line corresponds to Southern Alaska counties and orange line to coastal South Carolina counties. Panel (a): change in population relative to the deterministic steady state. Panel (b): change in capital relative to the deterministic steady state. Panel (c): change workers welfare relative to the deterministic steady state. Panel (d): change in capitalist welfare relative to the deterministic steady state.

population decline. The increase in global temperatures, together with aggregate and local risk, leads to a population decrease of around 30% and a reduction in the local capital stock of about 50%. Local risk modestly mitigates the losses in both population and capital. Both workers and capitalists are significantly affected by global warming, experiencing long-run welfare losses of 3% and 50%, respectively, in consumption-equivalent terms. Local risk reduces welfare losses for workers by 0.5% but magnifies the losses for capitalists by almost 30%. Interestingly, the dynamics of worker welfare are highly non-linear and non-monotonic, illustrating the advantages of using SAME as a solution method.

APPENDIX

A Setup

A.1 Static Equilibrium

Final good production. The wage w_{it} in location i satisfies the firm's first-order condition $w_{it} = (1 - \alpha)Z_{it}(S_{it}/N_{it}^P)^\alpha$. Neither renters nor firms cover depreciation, so that the rental rate for structures satisfies $r_{it} = \alpha Z_{it}(S_{it}/N_{it}^P)^{-(1-\alpha)}$.

Buildings production. The optimal choice of labor in building production implies $\omega r_{it} B_{it} / N_{it}^B = w_{it}$.

Housing demand. Local demand for housing clears the housing market $\beta w_{it} N_{it} = r_{it} H_{it}$.

Equilibrium prices. We now solve for equilibrium prices and quantities as functions of the local capital stock K_{it} and local number of workers N_{it} . Combining labor demand in final good and building production, we obtain $\omega \alpha \frac{B_{it}}{N_{it}^B} = (1 - \alpha) \frac{S_{it}}{N_{it}^P}$. Housing demand rewrites $\beta(1 - \alpha) N_{it} \frac{S_{it}}{N_{it}^P} = \alpha H_{it}$.

We look for shares x, y such that

$$N_{it}^P = x N_{it} \text{ , } H_{it} = y B_{it} \tag{20}$$

and so $N_{it}^B = (1 - x) N_{it}$ and $S_{it} = (1 - y) B_{it}$. Substituting into the previous equations, we obtain

$$\frac{\omega \alpha}{1 - \alpha} = \frac{(1 - y)(1 - x)}{x} \text{ , } \frac{\beta(1 - \alpha)}{\alpha} = \frac{xy}{1 - y}.$$

These two equations imply⁵

$$x = \frac{(1-\alpha)(1-\omega\beta)}{\alpha\omega + (1-\alpha)} , \quad y = \beta \frac{(1-\alpha) + \alpha\omega}{\alpha + (1-\alpha)\beta} , \quad 1-y = \alpha \frac{1-\beta\omega}{\alpha + (1-\alpha)\beta}. \quad (21)$$

Using these shares, we express the wage and the rental rate as

$$\begin{aligned} w_{it} &= (1-\alpha)\alpha^\alpha \Xi^{-\alpha} Z_{it} (B_{it}/N_{it})^\alpha \\ r_{it} &= \alpha^\alpha \Xi^{1-\alpha} Z_{it} (B_{it}/N_{it})^{-(1-\alpha)} \end{aligned} \quad (22)$$

where

$$\Xi = (1-\alpha) \frac{\alpha + (1-\alpha)\beta}{\alpha\omega + (1-\alpha)}$$

Before proceeding, it is useful to substitute out buildings and express prices in terms of capital:

$$\begin{aligned} w_{it} &= (1-\alpha)\alpha^\alpha \Xi^{-\alpha} (1-x)^{\omega\alpha} Z_{it} L_i^{\omega\alpha} (K_{it}^{1-\omega-\omega} N_{it}^{\omega-1})^\alpha \\ r_{it} &= \alpha^\alpha \Xi^{1-\alpha} (1-x)^{-(1-\alpha)\omega} Z_{it} L_i^{-\omega(1-\alpha)} (K_{it}^{1-\omega-\omega} N_{it}^{\omega-1})^{-(1-\alpha)} \end{aligned} \quad (23)$$

Rental rate of capital. We also need the rental rate of capital:

$$\begin{aligned} R_{K,it} &= (1-\omega-\omega)(1-x)^\omega K_{it}^{-(\omega+\omega)} L_i^\omega N_{it}^\omega r_{it} \\ &= (1-\omega-\omega)\alpha^\alpha \Xi^{1-\alpha} (1-x)^{\alpha\omega} Z_{it} L_i^{\omega\alpha} K_{it}^{-\phi} N_{it}^{1-\alpha+\alpha\omega} \end{aligned} \quad (24)$$

We obtain:

$$\begin{aligned} R_{K,it} &= \underbrace{(1-\omega-\omega)\alpha^\alpha \Xi^{1-\alpha} (1-x)^{\alpha\omega} Z_{it} L_i^{\omega\alpha}}_{\equiv R_{0i}} e^{\chi_{it}} K_{it}^{-\phi} N_{it}^\psi \\ &= R_{0i} e^{\chi_{it}} K_{it}^{-\phi} N_{it}^\psi \\ &\equiv R_i(\chi_{it}, K_{it}, N_{it}) \end{aligned} \quad (25)$$

⁵Multiplying both equations leads to $y(1-x) = \omega\beta$ which can then be substituted into either of the equations.

where

$$\begin{aligned}\phi &= \omega + \varpi + (1 - \omega - \varpi)(1 - \alpha) \\ \psi &= 1 - \alpha + \alpha\varpi\end{aligned}$$

Consumption and flow utility. We also express total consumption (real wage) in location i as

$$\begin{aligned}C_{it} &= \frac{w_{it}}{r_{it}^\beta} \\ &= \frac{(1 - \alpha)\alpha^\alpha \Xi^{-\alpha} (1 - x)^{\varpi\alpha} Z_{it} L_i^{\omega\alpha} (K_{it}^{1-\omega-\varpi} N_{it}^{\varpi-1})^\alpha}{\left[\alpha^\alpha \Xi^{1-\alpha} (1 - x)^{-(1-\alpha)\varpi} Z_{it} L_i^{-\omega(1-\alpha)} (K_{it}^{1-\omega-\varpi} N_{it}^{\varpi-1})^{-(1-\alpha)} \right]^\beta} \\ &= (1 - \alpha)\alpha^{\alpha(1-\beta)} \Xi^{-\xi} (1 - x)^{\varpi\xi} Z_{it}^{1-\beta} L_i^{\omega\xi} K_{it}^{(1-\omega-\varpi)\xi} N_{it}^{-\xi(1-\varpi)} \\ &= \underbrace{(1 - \alpha)\alpha^{\alpha(1-\beta)} \Xi^{-\xi} (1 - x)^{\varpi\xi} Z_{it}^{1-\beta} L_i^{\omega\xi}}_{\equiv C_{0i}} \cdot e^{(1-\beta)\chi_{it}} \left(\frac{K_{it}^{1-\omega-\varpi}}{N_{it}^{1-\varpi}} \right)^\xi \\ &\equiv C_{0i} e^{(1-\beta)\chi_{it}} \left(\frac{K_{it}^{1-\omega-\varpi}}{N_{it}^{1-\varpi}} \right)^\xi\end{aligned}\tag{26}$$

where

$$\xi = \alpha + \beta(1 - \alpha)$$

With consumption at hand, we express the flow utility in location i at time t :

$$\begin{aligned}u\left(A_{it} \frac{w_{it}}{r_{it}^\beta}\right) &= u\left(\tilde{C}_{0i} e^{a_{it} + (1-\beta)\chi_{it}} \left(\frac{K_{it}^{1-\omega-\varpi}}{N_{it}^{1-\varpi}}\right)^\xi\right) \\ &\equiv U_i(a_{it}, \chi_{it}, K_{it}, N_{it}),\end{aligned}\tag{27}$$

where $\tilde{C}_{0i} \equiv A_i C_{0i}$.

A.2 Capitalists

We guess and verify that:

$$\mathcal{Q}_{it}(K) = \frac{Q_{it}K^{1-\zeta}}{1-\zeta} + P_{it} \quad C = c_{it}K \quad (28)$$

Under this guess: $\mathcal{Q}_{it,K} = Q_{it}K^{-\zeta}$ and $\mathcal{Q}_{it,KK} = -\zeta Q_{it}K^{-\zeta-1}$. Substituting the guess into the HJB:

$$\rho \frac{Q_{it}K^{1-\zeta}}{1-\zeta} + \rho P_{it} = \max_c \frac{c^{1-\zeta}K^{1-\zeta} - 1}{1-\zeta} + [(R_{it} - \Delta_{it}) - c]Q_{it}K^{1-\zeta} - \frac{\zeta \Sigma_{it}^2}{2} Q_{it}K^{1-\zeta} + \frac{K^{1-\zeta}}{1-\zeta} \mathbb{E}_t \left[\frac{dQ_{it}}{dt} \right] + \mathbb{E}_t$$

Identifying components:

$$\begin{aligned} \rho Q_{it} &= \max_c c^{1-\zeta} + (1-\zeta) [(R_{it} - \Delta_{it}) - c] Q_{it} - \frac{\zeta(1-\zeta)\Sigma_{it}^2}{2} Q_{it} + \mathbb{E}_t \left[\frac{dQ_{it}}{dt} \right] \\ \rho P_{it} &= -\frac{1}{1-\zeta} + \mathbb{E}_t \left[\frac{dP_{it}}{dt} \right] \Rightarrow P_{it} = -\frac{1}{\rho(1-\zeta)} \end{aligned}$$

So we get the FOC:

$$c^{-\zeta} = Q_{it} \Rightarrow c = Q_{it}^{-1/\zeta} \Rightarrow \frac{I}{K} = R_{it} - c = R_{it} - Q_{it}^{-1/\zeta}$$

where we have used that $I = R_{it}K - C_{it}$.

Substitute back into the HJB:

$$\begin{aligned} \rho Q_{it} &= Q_{it}^{1-1/\zeta} + (1-\zeta) \left[R_{it} - \Delta_{it} - Q_{it}^{-1/\zeta} - \frac{\zeta \Sigma_{it}^2}{2} \right] Q_{it} + \mathbb{E}_t \left[\frac{dQ_{it}}{dt} \right] \\ &= \zeta Q_{it}^{1-1/\zeta} + (1-\zeta) \left[R_{it} - \Delta_{it} - \frac{\zeta \Sigma_{it}^2}{2} \right] Q_{it} + \mathbb{E}_t \left[\frac{dQ_{it}}{dt} \right]. \end{aligned}$$

With log utility $\zeta = 1$, we obtain:

$$\rho Q_{it} = 1 \quad \mathcal{Q}_{it}(K) = \frac{1}{\rho} \log K,$$

and so proportional risk Σ_{it} does not matter as income and substitution effects cancel out.

B Steady State Computation

Solving for steady-state values is relatively straightforward:

1. Start by using (8) to solve for Q_i^{ss} and the consumption rate:

$$(Q_i^{ss})^{-1/\zeta} = \rho = c^{ss}$$

2. Given Q_i^{ss} , then use (9) to solve for the equilibrium capital as a function of population:

$$K_i^{ss} = \left(\frac{R_{0i}(N_i^{ss})^\psi}{\rho + \Delta_i} \right)^{1/\phi} \equiv \bar{K}_i(N_i^{ss}) \quad (29)$$

3. Substitute (29) into (8) to obtain

$$\rho V_i^{ss} = U_i(K_i(N_i^{ss}), N_i^{ss}) + \mathcal{M}_i[V^{ss}] \quad (30)$$

4. Alternate on (30) and (9) to jointly solve the equilibrium values V_i^{ss} and population N_i^{ss} :

- (a) Start from a uniform guess for N_i^{ss} (or any other guess)
- (b) Solve (30) for V_i^{ss}
- (c) Update N_i^{ss} by solving (9) given V_i^{ss}
- (d) Keep iterating on (b-c) until both N_i^{ss} and V_i^{ss} have converged.

C Microfoundation of the capital depreciation process

To estimate the capital depreciation process and scale Σ_{it} accordingly, we lay out a simple microfoundation starting from Poisson capital depreciation shocks.

Suppose momentarily that there are infrequent, large depreciation shocks. These depreciation shocks of magnitude $1 - D < 1$ arise with some Poisson rate that depends on

the global mean temperature relative to the pre-industrial level. D is the amount of capital that is destroyed by the shock. We denote the relative temperature by T_t . We omit i subscripts for now. The law of motion of local capital follows a drift-compound Poisson process: $\frac{dK_t}{K_t} = \frac{I_t}{K_t}dt + d\bar{\Delta}_t$, where $d\bar{\Delta}_t = \Delta_t dt + DdJ_t$, where $J_t \in \{0, 1\}$ indicates whether a Poisson shock hits at all. $\lambda(T)$ denotes the Poisson intensity of these Poisson shocks and depends on global mean temperature. D is a single scalar that denotes the size of the shock when it hits. This law of motion implies that when a Poisson shock of magnitude D hits, capital jumps from K_t to $DK_t < K_t$. We are interested in the structure of the generator associated with the Poisson destruction part:

$$\mathcal{P}[V](K) = \lambda(T)(V((1-D)K) - V(K)) \quad (31)$$

We assume that D is small enough in the definition of the generator.⁶ Using a second-order expansion in D : $V((1-D)K) - V(K) = -V'(K)KD + \frac{1}{2}V''(K)K^2D^2$. Substituting into (31):

$$\mathcal{D}[V](K) = -KV'(K)\lambda(T)D + \frac{K^2V''(K)}{2}\lambda(T)D^2$$

So, holding D fixed, the generator indeed has the same functional form than that of a Brownian motion. However, the implied variance still scales linearly with $\lambda(T)$. Therefore, specifying $\lambda(T) = \lambda_0 \epsilon T$ as we did for the FAME only does not let us use the classic structure of “small noise” expansion in which the volatility scales with ϵ and the variance scales with ϵ^2 .⁷

However, we can posit that depreciation shocks are also small. In fact, that assumption is necessary to justify the Brownian motion generator in the first place. We operationalize

⁶We observe that, to obtain a Brownian motion as a limit, the *only* possibility is to take a limit when the support of D collapses to 1. Otherwise, derivatives cannot appear. We work with the generator, which is the dual of the law of motion. It is easier to take an expansion in the space of functions rather than in a space of stochastic processes.

⁷The usual way to get risk to be second order is to have shocks scale with T_t . But that is not what we are measuring. We are measuring probabilities holding the size of shocks fixed.

One possibility is to include volatility components in the FAME. That is likely a major lift, because second order derivatives then enter the FAME and so the FAME and SAME become interlinked and jointly determined. If we can avoid it, we would rather avoid it.

this idea as follows. We posit:

$$D = \sqrt{\epsilon \bar{D}},$$

where ϵ is the same scale parameter as the one that we use for the FAME/SAME. We identify terms between our Poisson approximation and our Brownian motion law of motion:

$$\delta_{it} = D\lambda_i(\epsilon T_t) \quad \Sigma_{it} = \sqrt{D^2\lambda_i(\epsilon T_t)} = D\sqrt{\lambda_i(\epsilon T_t)}.$$

In the data, we estimate that the probability of storms is linear in global mean temperature,⁸ and so:

$$\lambda_i(\epsilon T_t) = \bar{\lambda}_i \times \epsilon T_t.$$

Substituting into the drift and volatility equations, we obtain:

$$\delta_{it} = \epsilon^{3/2} \cdot \bar{D}\bar{\lambda}_i T_t \equiv \epsilon^{3/2} \cdot \delta_{i1} T_t \quad \Sigma_{it} = \epsilon \cdot \bar{D}\sqrt{\bar{\lambda}_i T_t} \equiv \epsilon \Sigma_i \sqrt{T_t},$$

where we defined:

$$\delta_{i1} = \bar{D}\bar{\lambda}_i \quad \Sigma_i = \bar{D}\sqrt{\bar{\lambda}_i}$$

The first equation (almost) coincides with what we used in the ACC/the FAME only. The second equation lets us define volatility. The variance is then *linear*, not quadratic, in T_t , but it is *quadratic* in ϵ :

$$\Sigma_{it}^2 = \epsilon^2 \Sigma_i^2 T_t. \tag{32}$$

At first sight, we face a tension: although the FAME is supposed to be linear in ϵ , the leading capital depreciation terms scale in $\epsilon^{3/2} \ll \epsilon$. In fact, this tension is not a mathematical consistency issue.

⁸In fact, it looks slightly convex. We might include this convexity in the SAME later on as a δ_{i2} term.

The key observation is that the FAME is additive. Therefore, it is entirely internally consistent to structure the perturbation as follows:

1. Solve the deterministic FAME, which is independent from δ_i
2. Solve the trend FAME for productivity/amenities, which only depends on χ_i, a_i
3. Then only solve for the trend FAME for capital depreciation, which depends only on δ_i and scales with $\epsilon^{3/2}$. Because it is additive, no calculation changes relative to what we are currently doing in the FAME
4. IRFs are also additive in the trend FAMES. We can thus immediately sum the ϵ and the $\epsilon^{3/2}$ components into a “FAME+”.

Therefore, in practice, we do not need to change anything in the FAME.⁹

C.1 Steady-state inversion

We start from (??). The objective is to determine Δ_i and K_i from data on I_i, N_i, w_i . We have the basic identity:

$$\frac{I_i}{K_i} = R_i - \rho = \Delta_i.$$

We will also use the identity coming from expenditure shares: $R_i K_i = (1 - \omega - \varpi)r_i B_i$ and $\varpi r_i B_i = w_i(1 - x)N_i$. Hence:

$$R_i K_i = \frac{1 - \omega - \varpi}{\varpi}(1 - x)w_i N_i.$$

Hence:

$$I_i = \frac{1 - \omega - \varpi}{\varpi}(1 - x)w_i N_i - \rho K_i.$$

⁹We only need to acknowledge that, to be consistent with the definition of Σ_{it} , the capital depreciation terms are, mathematically, $\epsilon^{3/2}$.

We can thus express the local capital stock as:

$$K_i = \frac{1}{\rho} \left[\frac{(1 - \omega - \varpi)(1 - x)}{\varpi} w_i N_i - I_i \right].$$

We will have to a bit careful: payments to labor and investment have to be in the same units. And capital will have to remain positive. We then obtain the local capital depreciation rate as:

$$\Delta_i = \frac{I_i}{K_i}.$$

We recover an estimate of the stock of buildings B_i and housing H_i from the constant shares rule (20)-(21) and the buildings production function as before:

$$B_i = L_i^\omega \left((1 - x) N_i \right)^\omega K_i^{1-\omega-\varpi} , \quad H_i = y B_i$$

The wage equation (22) then delivers an estimate of fundamental productivity in county i as before:

$$Z_i = \frac{w_i}{(1 - \alpha) \alpha^\alpha \Xi^{-\alpha} (B_i / N_i)^\alpha}.$$

From there we construct $C_{0i} = (1 - \alpha) \alpha^{\alpha(1-\beta)} \Xi^{-\xi} (1 - x)^{\omega\xi} Z_i^{1-\beta} L_i^{\omega\xi}$ and $R_{0i} = (1 - \omega - \varpi) \alpha^\alpha \Xi^{1-\alpha} (1 - x)^{\alpha\varpi} Z_i L_i^{\omega\alpha}$ as before.

We recover migration costs as before. We construct migration shares as $m_{ij} = \frac{\mathfrak{m}_{ij}}{\sum_k \mathfrak{m}_{ik}}$. The equation for migration shares (4) implies that $X_{ij} \equiv \log m_{ij} - \log m_{ii} = \nu(V_j - V_i) - \nu\tau_{ij} \equiv Y_j - Y_i + \varepsilon_{ij}$ after imposing the normalization $\tau_{ii} = 0$. We exploit that $X_{ij} + X_{ji} = \varepsilon_{ij} + \varepsilon_{ji}$. We assume symmetric migration costs $\tau_{ij} = \tau_{ji}$, which we then recover as

$$\tau_{ij} = -\frac{X_{ij} + X_{ji}}{2\nu}.$$

We are left with estimating local amenities, which is now slightly different from before. We proceed in two steps. The first step consists in recovering local valuations from the steady-state population distribution. Defining $X_i = e^{\nu V_i}$ and $T_{ki} = e^{-\nu\tau_{ki}}$, equation (9)

implies, together with a normalization of values:

$$X_i = \frac{N_i}{\sum_k \frac{N_k T_{ki}}{\sum_j T_{kj} X_j}} , \quad \sum_i X_i = 1. \quad (33)$$

The system of equations (33) satisfies the gross substitutes property, and so has at most one solution. A simple iterative method delivers the solution to equation (33), and we recover $V_i = \frac{1}{\nu} \log(X_i)$.

With an estimate of local valuations V_i at hand, the second step consists in inverting the HJB (3) to recover amenities. In particular, we obtain flow utility as:

$$U_i = \rho V_i - \mathcal{M}_i[V].$$

From there, we recover amenities as:

$$A_i = \frac{u^{-1}(U_i)}{C_{0i} K_i^{(1-\omega-\varpi)\xi} N_i^{-(1-\varpi)\xi}}.$$

D FAME

D.1 Flow payoffs

D.1.1 Workers

The flow payoffs become, to first order,

$$\frac{U_{it} - U_i^{ss}}{\epsilon} \approx u'(\tilde{C}_i^{ss}) \tilde{C}_i^{ss} \cdot \left((a_{i1} + (1 - \beta)\chi_{i1})(\sigma_t z + T_t) + \xi(1 - \omega - \varpi)k_i - \xi(1 - \varpi)\frac{n_i}{N_i^{ss}} \right)$$

with $u'(C) = C^{-\gamma}$. In vector notation,

$$\begin{aligned} \frac{U_t - U^{ss}}{\epsilon} &= \mathbf{diag}\left(u'(\tilde{C}_i^{ss})\tilde{C}_i^{ss}\right) \left[\mathbf{vec}(a_{i1} + (1 - \beta)\chi_{i1})(\sigma_t z + T_t) \right. \\ &\quad \left. + \zeta(1 - \omega - \varpi)k - \zeta(1 - \omega)\mathbf{diag}\left(1/N_i^{ss}\right)n \right] \\ &\equiv D_{UK}k + D_{UN}n + d_{UT}(\sigma_t z + T_t), \end{aligned} \quad (34)$$

where we have defined:

$$\begin{aligned} d_{UT} &= \mathbf{vec}\left(u'(\tilde{C}_i^{ss})\tilde{C}_i^{ss}(a_{i1} + (1 - \beta)\chi_{i1})\right), \\ D_{UK} &= \zeta(1 - \omega - \varpi)\mathbf{diag}\left(u'(\tilde{C}_i^{ss})\tilde{C}_i^{ss}\right), \quad D_{UN} = -\zeta(1 - \omega)\mathbf{diag}\left(u'(\tilde{C}_i^{ss})\tilde{C}_i^{ss}/N_i^{ss}\right). \end{aligned} \quad (35)$$

Alternatively, we could have written $\epsilon^{-1}(U_t - U^{ss}) = d_{UZ}\sigma_t z + d_{UT}T_t + D_{UK}k - D_{UN}n$, with $d_{UZ} = d_{UT}$ for completion. Nevertheless, to avoid coding mistakes, we just keep the vector d_{UT} for the derivations.

D.1.2 Capitalists

We include the flow payoff from the return to capital and the continuation value from investment. To first order and in vector notation, we obtain

$$\begin{aligned} &\epsilon^{-1} \left[\left(\zeta Q_{it}^{1-1/\zeta} + (1 - \zeta) \left\{ R_{it} - \Delta_{it} - \frac{\zeta \Sigma_{it}^2}{2} \right\} Q_{it} \right) - \left(\zeta (Q_i^{ss})^{1-1/\gamma} + (1 - \zeta) \left\{ (R_i^{ss} - \Delta_i - 0) \right\} Q_{it} \right) \right] \\ &= D_{QK}k + D_{QN}n + d_{QT}(\sigma_t z + T_t) + D_{QQ}[Q^N n + Q^K k + q_t^Z z + q_t^T] \end{aligned}$$

where

$$\begin{aligned} D_{QK} &= -(1 - \zeta)\phi\mathbf{diag}\left(R_i^{ss}Q_i^{ss}\right), \quad D_{QN} = (1 - \zeta)\psi\mathbf{diag}\left(R_i^{ss}Q_i^{ss}/N_i^{ss}\right), \\ D_{QQ} &= 0, \quad d_{QT} = (1 - \zeta)\mathbf{vec}\left((R_i^{ss}\chi_{i1} - \delta_{i1})Q_i^{ss}\right). \end{aligned} \quad (36)$$

where once again we omit the vector d_{QZ} , which is equal to d_{RT} , to simplify the code.

Also note that now the matrices D_{QK}, D_{QN} do not correspond to the derivatives of the interest rate anymore. Note also that the term corresponding to the steady-state capital drift times the change in Q drops out because the capital drift is zero in steady-state.

Finally, there might have been a sign typo in how D_{RN} entered the FAME, and a typo in the sign of D_{RK} . We need to check the code (it was present in the ACC code too).

We will also change the sign convention for how d_{RT} enters the FAME to avoid excess minus signs.

Other than these changes (and correcting the signs!), we can simply plug in these new equations into our existing code.

For use below, we also define the derivative of the interest rate alone:

$$D_{RK} = -\phi \mathbf{diag}(R_i^{ss}), \quad D_{RN} = \psi \mathbf{diag}(R_i^{ss}/N_i^{ss}) \quad d_{RT} = \mathbf{vec}(R_i^{ss} \chi_{i1}).$$

D.2 Continuation value from migration

The continuation value from migration becomes

$$\frac{\mathcal{M}[V] - \mathcal{M}[V^{ss}]}{\epsilon} \approx MV^N n + MV^K k + Mv_t^Z z + Mv_t^T, \quad (37)$$

where henceforth M denotes the steady-state matrix $M(V^{ss})$.

Continuation value from aggregate shocks. We impose z to follow an AR(1) process with mean-reversion parameter θ and unit variance. Therefore, the continuation values become

$$\begin{aligned} \mathcal{A}_\epsilon(z)[V] &= \mathcal{A}_\epsilon(z)[v_t^Z z] = -\epsilon \theta z v_t^Z, \\ \mathcal{A}_\epsilon(z)[Q] &= \mathcal{A}_\epsilon(z)[q_t^Z z] = -\epsilon \theta z q_t^Z. \end{aligned}$$

D.3 Population distribution

To linearize the law of motion of population, first note that

$$\frac{\partial m_{ji}}{\partial V_k} = \begin{cases} -vm_{ji}m_{jk} & \text{if } i \neq k \\ vm_{ji}(1 - m_{ji}) & \text{if } i = k \end{cases}$$

and so

$$\begin{aligned} M_{ij}^*(V + dV) &= M_{ij}^*(V) + \mu \sum_k \frac{\partial m_{ji}}{\partial V_k} dV_k \\ &= M_{ij}^*(V) + \mu v m_{ji} \left\{ dV_i - [m \cdot dV]_j \right\} \end{aligned}$$

Then

$$\begin{aligned} M_i^*(V + dV)[N^{ss}] &= M_i^*(V)[N^{ss}] + \mu v \sum_j m_{ji} \left\{ dV_i - [m \cdot dV]_j \right\} N_j^{ss} \\ &= M_i^*(V)[N^{ss}] + \nu \mu \left[\left(\mathbf{diag}(m^* N^{ss}) - m^* \mathbf{diag}(N^{ss}) m \right) \cdot dV \right]_i \\ &\equiv M_i^*(V)[N^{ss}] + (G \cdot dV)_i \end{aligned}$$

where

$$G = \nu \mu \left(\mathbf{diag}(m^* N^{ss}) - m^* \mathbf{diag}(N^{ss}) m \right) \quad (38)$$

D.4 Continuation value from changes in population distribution

We can then move on to the continuation value from changes in the population distribution. In vector notation,

$$\epsilon^{-1} \left(\sum_j \frac{\partial V_i}{\partial N_j} (M^*(V)N)_j \right)_{i=1}^I \approx V^N M^* n + V^N G (V^N n + V^K k + v_t^Z z + v_t^T) \quad (39)$$

Similarly,

$$\epsilon^{-1} \left(\sum_j \frac{\partial Q_i}{\partial N_j} (M^*(V)N)_j \right)_{i=1}^I \approx Q^N M^* n + Q^N G(V^N n + V^K k + v_t^Z z + v_t^T) \quad (40)$$

D.5 Capital distribution

The log capital LoM becomes to first order:

$$\frac{dk_t}{dt} = D_{RK} k_t + D_{RN} n_t + d_{IT}(\sigma_t z + T_t) + D_{IQ} [Q^N n_t + Q^K k_t + q_t^Z z + q_t^T],$$

where the Ito correction drops out to first order, and where:

$$d_{IT} = \mathbf{vec} \left(R_i^{SS} \chi_{i1} - \delta_{i1} \right) \quad D_{IQ} = \mathbf{diag} \left(\frac{(Q_i^{ss})^{-1-1/\zeta}}{\zeta} \right)$$

There are two changes relative to what we had before: (1) the definition of D_{IQ}, d_{IT} has changed (also note the change of sign in the definition/how they enter the LoM), and (2) two new terms enter, $D_{RK} k + D_{RN} n$, that we will need to keep track of in the continuation values.

D.6 Continuation value from changes in capital distribution

All the Ito correction terms are second order, so we can ignore them in the FAME. We obtain

$$\begin{aligned} & \epsilon^{-1} \sum_j \frac{\partial V_i}{\partial \mathcal{K}_j} \left\{ R_i(\chi_{it}, N_i, \mathcal{K}_i) - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right\} \\ & \approx \sum_j V_{ij}^K \left\{ D_{RK} k_t + D_{RN} n_t + d_{IT}(\sigma_t z + T_t) + D_{IQ} [Q^N n_t + Q^K k_t + q_t^Z z + q_t^T] \right\} \\ & = V^K \left\{ D_{IQ} [Q^N n + Q^K k + q_t^Z z + q_t^T] + D_{RK} k + D_{RN} n + (\sigma_t z + T_t) d_{IT} \right\} \quad (41) \end{aligned}$$

Similarly,

$$\begin{aligned} & \epsilon^{-1} \sum_j \frac{\partial Q_i}{\partial \mathcal{K}_j} \left\{ R_i(\chi_{it}, N_i, \mathcal{K}_i) - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right\} \\ & \approx Q^K \left\{ D_{IQ} \left[Q^N n + Q^K k + q_t^Z z + q_t^T \right] + D_{RK} k + D_{RN} n + (\sigma_t z + T_t) d_{IT} \right\} \quad (42) \end{aligned}$$

D.7 Expansion of prices and investment

For estimation, it is also useful to collect the expansion of prices along any transition. We have, in level changes:

$$\begin{aligned} \hat{w}_t &= D_{wN} n_t + D_{wK} k_t + d_{wT} (\sigma_t z_t + T_t) \\ \hat{R}_t &= D_{RN} n_t + D_{RK} k_t + d_{RT} (\sigma_t z_t + T_t) \\ \hat{I}_t &= D_{IN} n_t + D_{IK} k_t + d_{IQ} \hat{Q}_t + d_{IT} (\sigma_t z_t + T_t) \end{aligned}$$

where:

$$\begin{aligned} D_{wN} &= -\alpha(1 - \omega) \mathbf{diag}(w_i^{ss} / N_i^{ss}) & D_{wK} &= \alpha(1 - \omega - \varpi) \mathbf{diag}(w_i^{ss}) & d_{wT} &= \mathbf{vec}(\chi_i w_i^{ss}) \\ D_{IN} &= \mathbf{diag}(K_i^{ss}) D_{RN} & D_{IK} &= \mathbf{diag}(K_i^{ss}) D_{RK} + \mathbf{diag}(I_i^{ss}) & D_{IQ} &= \mathbf{diag}(K_i^{ss}) D_{IQ} \\ d_{IT} &= \mathbf{diag}(K_i^{ss}) d_{RT}, \end{aligned}$$

and D_{RN}, D_{RK}, d_{RT} are defined in Section D.1.2.

D.8 FAME in vector notation

D.8.1 Workers

Thus, the linearized master equation for the worker value becomes, in vector notation,

$$\begin{aligned}
& \rho(V^N n + V^K k + v_t^Z z + v_t^T) - \frac{\partial v_t^Z}{\partial t} z - \frac{\partial v_t^T}{\partial t} \\
& \quad \text{flow payoff} \\
& = \overbrace{D_{UK} k + D_{UN} n + (\sigma_t z + T_t) d_{UT}}^{\text{continuation value from migration}} \\
& + \overbrace{M(V^N n + V^K k + v_t^Z z + v_t^T)}^{\text{c.v. from climate shocks}} \\
& - \overbrace{\theta z v_t^Z}^{\text{c.v. from climate shocks}} \\
& + \overbrace{V^N M^* n + V^N G(V^N n + V^K k + v_t^Z z + v_t^T)}^{\text{continuation value from changes in the population distribution}} \\
& + \underbrace{V^K \left\{ D_{IQ} \left[Q^N n + Q^K k + q_t^Z z + q_t^T \right] + D_{RK} k + D_{RN} n + (\sigma_t z + T_t) d_{IT} \right\}}_{\text{continuation value from changes in the capital distribution}}
\end{aligned}$$

Identifying coefficients and unknowns in bold notation,

$$\begin{aligned}
\rho \mathbf{V}^N &= D_{UN} + M \mathbf{V}^N + \mathbf{V}^N M^* + \mathbf{V}^K D_{RN} + \mathbf{V}^N G \mathbf{V}^N + \mathbf{V}^K D_{IQ} \mathbf{Q}^N \\
\rho \mathbf{V}^K &= D_{UK} + M \mathbf{V}^K + \mathbf{V}^K D_{RK} + \mathbf{V}^N G \mathbf{V}^K + \mathbf{V}^K D_{IQ} \mathbf{Q}^K \\
(\rho + \theta) \mathbf{v}_t^Z - \frac{\partial \mathbf{v}_t^Z}{\partial t} &= d_{UT} \sigma_t + M \mathbf{v}_t^Z + \mathbf{V}^N G \mathbf{v}_t^Z + \mathbf{V}^K D_{IQ} \mathbf{q}_t^Z + \mathbf{V}^K d_{IT} \sigma_t \\
\rho \mathbf{v}_t^T - \frac{\partial \mathbf{v}_t^T}{\partial t} &= d_{UT} T_t + M \mathbf{v}_t^T + \mathbf{V}^N G \mathbf{v}_t^T + \mathbf{V}^K D_{IQ} \mathbf{q}_t^T + \mathbf{V}^K d_{IT} T_t
\end{aligned}$$

D.8.2 Capitalists

Similarly, the linearized master equation for capitalists becomes, in vector notation,

$$\begin{aligned}
& \rho(Q^N n + Q^K k + q_t^Z z + q_t^T) - \frac{\partial q_t^Z}{\partial t} - \frac{\partial q_t^T}{\partial t} \\
& \quad \text{flow payoff + own continuation value} \\
& = \underbrace{D_{QK}k + D_{QN}n + d_{QT}(\sigma_t z + T_t) + D_{QQ}[Q^N n + Q^K k + q_t^Z z + q_t^T]}_{\text{c.v. from climate shocks}} \\
& - \underbrace{\theta z q_t^Z}_{\text{c.v. from climate shocks}} \\
& + \underbrace{Q^N M^* n + Q^N G(V^N n + V^K k + v_t^Z z + v_t^T)}_{\text{continuation value from changes in the population distribution}} \\
& + \underbrace{Q^K \left\{ D_{IQ} [Q^N n + Q^K k + q_t^Z z + q_t^T] + D_{RK}k + D_{RN}n + (\sigma_t z + T_t)d_{IT} \right\}}_{\text{continuation value from changes in the capital distribution}}
\end{aligned}$$

Identifying coefficients and with unknowns in bold notation,

$$\begin{aligned}
\rho Q^N &= D_{QN} + \mathbf{Q}^N M^* + \mathbf{Q}^K D_{RN} + D_{QQ} \mathbf{Q}^N + \mathbf{Q}^N G \mathbf{V}^N + \mathbf{Q}^K D_{IQ} \mathbf{Q}^N \\
\rho \mathbf{Q}^K &= D_{QK} + \mathbf{Q}^K D_{RK} + D_{QQ} \mathbf{Q}^K + \mathbf{Q}^N G \mathbf{V}^K + \mathbf{Q}^K D_{IQ} \mathbf{Q}^K \\
(\rho + \theta) q_t^Z - \frac{\partial q_t^Z}{\partial t} &= d_{QT} \sigma_t + \mathbf{Q}^N G v_t^Z + \mathbf{Q}^K D_{IQ} q_t^Z + \mathbf{Q}^K d_{IT} \sigma_t \\
\rho q_t^T - \frac{\partial q_t^T}{\partial t} &= d_{QT} T_t + \mathbf{Q}^N G v_t^T + \mathbf{Q}^K D_{IQ} q_t^T + \mathbf{Q}^K d_{IT} T_t
\end{aligned}$$

We can collect terms and solve for a stacked system step by step.

D.9 Deterministic FAME

$$\begin{aligned}
\rho \begin{pmatrix} \mathbf{V}^N & \mathbf{V}^K \\ \mathbf{Q}^N & \mathbf{Q}^K \end{pmatrix} &= \begin{pmatrix} D_{UN} & D_{UK} \\ D_{QN} & D_{QK} \end{pmatrix} + \begin{pmatrix} M & 0 \\ 0 & D_{QQ} \end{pmatrix} \begin{pmatrix} \mathbf{V}^N & \mathbf{V}^K \\ \mathbf{Q}^N & \mathbf{Q}^K \end{pmatrix} + \begin{pmatrix} \mathbf{V}^N & \mathbf{V}^K \\ \mathbf{Q}^N & \mathbf{Q}^K \end{pmatrix} \begin{pmatrix} M^* & 0 \\ D_{RN} & D_{RK} \end{pmatrix} \\
&+ \begin{pmatrix} \mathbf{V}^N & \mathbf{V}^K \\ \mathbf{Q}^N & \mathbf{Q}^K \end{pmatrix} \begin{pmatrix} G & 0 \\ 0 & D_{IQ} \end{pmatrix} \begin{pmatrix} \mathbf{V}^N & \mathbf{V}^K \\ \mathbf{Q}^N & \mathbf{Q}^K \end{pmatrix} \tag{43}
\end{aligned}$$

We recognize a nonlinear Sylvester equation which we can easily solve by iterating on it.

D.10 Stochastic FAME: z

Once we have the solution V^N, V^K, Q^N, Q^K , we can solve for v_t^Z and q_t^Z . Define the vector Ψ and the matrix Φ as:

$$\Psi = \begin{pmatrix} d_{UT} + V^K d_{IT} \\ d_{QT} + Q^K d_{IT} \end{pmatrix}, \quad \Phi = \begin{pmatrix} M + V^N G & V^K D_{IQ} \\ Q^N G & Q^K D_{IQ} \end{pmatrix} \quad (44)$$

Then, v_t^Z and q_t^Z satisfy a linear Sylvester equation:

$$(\rho + \theta) \begin{pmatrix} v_t^Z \\ q_t^Z \end{pmatrix} = \Psi \sigma_t + \Phi \begin{pmatrix} v_t^Z \\ q_t^Z \end{pmatrix} + \begin{pmatrix} v_t^Z \\ q_t^Z \end{pmatrix} \mathcal{D}_T^*, \quad (45)$$

where $\mathcal{D}_T$ is a matrix that discretizes the time derivative. We solve for $p = (v_t^Z; q_t^Z)$ by iterating back in time:

$$\begin{aligned} p_\infty &= ((\rho + \theta)\text{Id} - \Phi)^{-1} \Psi \sigma_\infty \\ p_t &= \left((1 + (\rho + \theta)dt) - dt \Phi \right)^{-1} \left(p_{t+dt} + \Psi \sigma_t dt \right) \end{aligned}$$

D.11 Trend FAME: T_t

Finally, we solve for the time-dependent piece. v_t^T and q_t^T satisfy:

$$\rho \begin{pmatrix} v_t^T \\ q_t^T \end{pmatrix} = \Psi T_t + \Phi \begin{pmatrix} v_t^T \\ q_t^T \end{pmatrix} + \begin{pmatrix} v_t^T \\ q_t^T \end{pmatrix} \mathcal{D}_T^* \quad (46)$$

This is, again, a linear Sylvester equation that we solve iteratively for $p \equiv (v_t^T; q_t^T)$ by iterating backward in time:

$$\begin{aligned} p_\infty &= (\rho \text{Id} - \Phi)^{-1} \Psi T_\infty^D \\ p_t &= \left((1 + \rho dt) - dt \Phi \right)^{-1} \left(p_{t+dt} + \Psi T_t dt \right) \end{aligned}$$

D.12 Linearized KFEs

To obtain the linearized laws of motions given T_t and a sequence z_t , we simply use the same terms as those in the “continuation value from changes in the population/capital distribution” from the FAMEs, since those correspond precisely to the linearization of the KFE. Let k_t be the vector of capital stocks across locations at time t , and n_t be the vector of populations across locations at time t . Then, in vector notation, the linearized KFE’s are:

$$\begin{aligned}\frac{dn_t}{dt} &= M^* n_t + G(V^N n_t + V^K k_t + v_t^Z z_t + v_t^T) \\ \frac{dk_t}{dt} &= D_{IQ} [Q^N n_t + Q^K k_t + q_t^Z z_t + q_t^T] + D_{RN} n_t + D_{RK} k_t + (\sigma_t z_t + T_t) d_{IT} \quad (47)\end{aligned}$$

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Written in matrix form:

$$\begin{aligned}v_1 &= V^N n + V^K k + v_t^Z z + v_t^T, \\ v_2 &= \mathcal{V}^{N,N} \times_3 n \times_2 n + \mathcal{V}^{N,K} \times_3 k \times_2 n + \mathcal{V}^{K,N} \times_3 n \times_2 k + \mathcal{V}^{K,K} \times_3 k \times_2 k \\ &\quad + 2V_t^{N,Z} n z + 2V_t^{K,Z} k z + 2V_t^{N,T} n + 2V_t^{K,T} k \\ &\quad + \bar{v}_t + v_t^{Z,Z} z^2 + v_t^{T,T} + 2v_t^{Z,T} z + v_t^\Sigma,\end{aligned} \quad (48)$$

and

$$\begin{aligned}q_1 &= Q^N n + Q^K k + q_t^Z z + q_t^T \\ q_2 &= \mathcal{Q}^{N,N} \times_3 n \times_2 n + \mathcal{Q}^{N,K} \times_3 k \times_2 n + \mathcal{Q}^{K,N} \times_3 n \times_2 k + \mathcal{Q}^{K,K} \times_3 k \times_2 k \\ &\quad + 2Q_t^{N,Z} n z + 2Q_t^{K,Z} k z + 2Q_t^{N,T} n + 2Q_t^{K,T} k \\ &\quad + \bar{q}_t + q_t^{Z,Z} z^2 + q_t^{T,T} + 2q_t^{Z,T} z + q_t^\Sigma,\end{aligned} \quad (49)$$

where $\times_3$ and $\times_2$ are the 3-node and 2-node tensor products, respectively, which we defined earlier.

Importantly, from (18), we notice that the tensors $\mathcal{V}^{N,K}$ and $\mathcal{V}^{K,N}$ are symmetric in their last two coordinates. Similarly, the expression (19) implies that $\mathcal{Q}^{N,K}$ and $\mathcal{Q}^{K,N}$ are also

symmetric in their last two coordinates.

E.1 Flow payoffs

E.1.1 Workers

We start with the utility function. For the second-order effect of the distribution, we have:

$$\begin{aligned}\left. \frac{\partial^2 U_{it}}{\partial N_{it}^2} \right|_{SS} &= \frac{\xi(1-\omega)\tilde{C}_i^{ss} \left[\xi(1-\omega)u''(\tilde{C}_i^{ss})\tilde{C}_i^{ss} + (1+\xi(1-\omega))u'(\tilde{C}_i^{ss}) \right]}{(N_i^{ss})^2}, \\ \left. \frac{\partial^2 U_{it}}{\partial \mathcal{K}_{it} \partial N_{it}} \right|_{SS} &= -\frac{\xi^2(1-\omega)(1-\omega-\omega)\tilde{C}_i^{ss} \left[u''(\tilde{C}_i^{ss})\tilde{C}_i^{ss} + u'(\tilde{C}_i^{ss}) \right]}{N_i^{ss}}, \\ \left. \frac{\partial^2 U_{it}}{\partial \mathcal{K}_{it}^2} \right|_{SS} &= [\xi(1-\omega-\omega)]^2 \tilde{C}_i^{ss} \left[u''(\tilde{C}_i^{ss})\tilde{C}_i^{ss} + u'(\tilde{C}_i^{ss}) \right].\end{aligned}$$

For the cross effect of the distribution and the aggregate shocks, we have:

$$\begin{aligned}\left. \frac{\partial^2 U_{it}}{\partial(\sigma_t z_t) \partial N_{it}} \right|_{SS} &= \left. \frac{\partial^2 U_{it}}{\partial T_t \partial N_{it}} \right|_{SS} = -\frac{\epsilon \xi(1-\omega)(a_{i1} + (1-\beta)\chi_{i1})\tilde{C}_i^{ss} \left[u''(\tilde{C}_i^{ss})\tilde{C}_i^{ss} + u'(\tilde{C}_i^{ss}) \right]}{N_i^{ss}}, \\ \left. \frac{\partial^2 U_{it}}{\partial(\sigma_t z_t) \partial \mathcal{K}_{it}} \right|_{SS} &= \left. \frac{\partial^2 U_{it}}{\partial T_t \partial \mathcal{K}_{it}} \right|_{SS} = \epsilon \xi(1-\omega-\omega)(a_{i1} + (1-\beta)\chi_{i1})\tilde{C}_i^{ss} \left[u''(\tilde{C}_i^{ss})\tilde{C}_i^{ss} + u'(\tilde{C}_i^{ss}) \right].\end{aligned}$$

Finally, for the second-order effect of the aggregate shocks, we have:

$$\begin{aligned}\left. \frac{\partial^2 U_{it}}{\partial(\sigma_t z_t)^2} \right|_{SS} &= \left. \frac{\partial^2 U_{it}}{\partial(T_t)^2} \right|_{SS} = \left. \frac{\partial^2 U_{it}}{\partial(\sigma_t z_t) \partial T_t} \right|_{SS} \\ &= \epsilon^2 \left\{ u''(\tilde{C}_i^{ss}) \left(\tilde{C}_i^{ss} \right)^2 (a_{i1} + (1-\beta)\chi_{i1})^2 + u'(\tilde{C}_i^{ss}) \tilde{C}_i^{ss} \left((a_{i1} + (1-\beta)\chi_{i1})^2 + a_{i2} + (1-\beta) \right) \right\}\end{aligned}$$

Putting all the terms together, we have:

$$\begin{aligned}
U_t - U^{ss} & \approx \epsilon \{ \mathbb{D}_{UN} n + \mathbb{D}_{UK} k + \mathbf{d}_{UT} (\sigma_t z + T_t) \} \leftarrow \text{first-order term} \\
& + \frac{\epsilon^2}{2} \{ \mathbb{D}_{UNN} \times_3 n \times_2 n + \mathbb{D}_{UNK} \times_3 k \times_2 n + \mathbb{D}_{UKN} \times_3 n \times_2 k + \mathbb{D}_{UKK} \times_3 k \times_2 k \} \\
& + \epsilon^2 \mathbb{D}_{UKT} k (\sigma_t z + T_t) + \epsilon^2 \mathbb{D}_{UNT} n (\sigma_t z + T_t) \\
& + \frac{\epsilon^2}{2} \mathbf{d}_{UTT} \left[(\sigma_t z)^2 + (T_t)^2 + 2(\sigma_t z) T_t \right],
\end{aligned}$$

where $\mathbb{D}_{UNN}$, $\mathbb{D}_{UKN}$, $\mathbb{D}_{UNK}$, and $\mathbb{D}_{UKK}$ are diagonal tensors defined as:

$$\begin{aligned}
\mathbb{D}_{UNN} &= \xi(1 - \omega) \mathbf{diag}_3 \left(\frac{\tilde{C}_i^{ss} \left[\xi(1 - \omega) u'' \left(\tilde{C}_i^{ss} \right) \tilde{C}_i^{ss} + (1 + \xi(1 - \omega)) u' \left(\tilde{C}_i^{ss} \right) \right]}{(N_i^{ss})^2} \right), \\
\mathbb{D}_{UKN} &= \mathbb{D}_{UNK} = -\xi^2(1 - \omega)(1 - \omega - \omega) \mathbf{diag}_3 \left(\frac{\tilde{C}_i^{ss} \left[u'' \left(\tilde{C}_i^{ss} \right) \tilde{C}_i^{ss} + u' \left(\tilde{C}_i^{ss} \right) \right]}{N_i^{ss}} \right), \\
\mathbb{D}_{UKK} &= [\xi(1 - \omega - \omega)]^2 \mathbf{diag}_3 \left(\tilde{C}_i^{ss} \left[u'' \left(\tilde{C}_i^{ss} \right) \tilde{C}_i^{ss} + u' \left(\tilde{C}_i^{ss} \right) \right] \right),
\end{aligned}$$

where $\mathbf{diag}_3$ is the 3-dimensional diagonal operator; and $\mathbb{D}_{UN}$, $\mathbb{D}_{UK}$, $\mathbb{D}_{UZ}$, and $\mathbf{d}_{UT}$ are defined as in section 4. The remaining objects are defined as follows:

$$\begin{aligned}
\mathbb{D}_{UNT} &= -\xi(1 - \omega) \mathbf{diag} \left(\frac{(a_{i1} + (1 - \beta)\chi_{i1}) \tilde{C}_i^{ss} \left[u'' \left(\tilde{C}_i^{ss} \right) \tilde{C}_i^{ss} + u' \left(\tilde{C}_i^{ss} \right) \right]}{N_i^{ss}} \right), \\
\mathbb{D}_{UKT} &= \xi(1 - \omega - \omega) \mathbf{diag} \left((a_{i1} + (1 - \beta)\chi_{i1}) \tilde{C}_i^{ss} \left[u'' \left(\tilde{C}_i^{ss} \right) \tilde{C}_i^{ss} + u' \left(\tilde{C}_i^{ss} \right) \right] \right), \\
\mathbf{d}_{UTT} &= \mathbf{vec} \left(u'' \left(\tilde{C}_i^{ss} \right) \left(\tilde{C}_i^{ss} \right)^2 (a_{i1} + (1 - \beta)\chi_{i1})^2 \right) + \mathbf{vec} \left(u' \left(\tilde{C}_i^{ss} \right) \tilde{C}_i^{ss} \left((a_{i1} + (1 - \beta)\chi_{i1})^2 + a_{i2} + (1 - \beta)\chi_{i2} \right) \right)
\end{aligned}$$

E.1.2 Capitalists

We let $\bar{Q}_{it} \equiv \zeta Q_{it}^{1-1/\zeta} + (1-\zeta) \left\{ R_{it} - \Delta_{it} - \frac{\zeta \Sigma_{it}^2}{2} \right\} Q_{it}$, to be the flow pay-off for the capitalists. First, for the second-order effect of the distribution, we have:

$$\begin{aligned} \left. \frac{\partial^2 \bar{Q}_{it}}{\partial N_{it}^2} \right|_{SS} &= \frac{\psi(\psi-1)(1-\zeta) Q_i^{ss} R_i^{ss}}{(N_i^{ss})^2}, \\ \left. \frac{\partial^2 \bar{Q}_{it}}{\partial \mathcal{K}_{it} \partial N_{it}} \right|_{SS} &= -\frac{\psi\phi(1-\zeta) Q_i^{ss} R_i^{ss}}{N_i^{ss}}, \\ \left. \frac{\partial^2 \bar{Q}_{it}}{\partial \mathcal{K}_{it}^2} \right|_{SS} &= \phi^2(1-\zeta) Q_i^{ss} R_i^{ss}. \end{aligned}$$

Second, for the cross-effect of the distribution and the aggregate shocks, we have:

$$\begin{aligned} \left. \frac{\partial^2 \bar{Q}_{it}}{\partial (\sigma_t z) \partial N_{it}} \right|_{SS} &= \left. \frac{\partial^2 \bar{Q}_{it}}{\partial T_t \partial N_{it}} \right|_{SS} = \epsilon\psi(1-\zeta) Q_i^{ss} \chi_{i1} \frac{R_i^{ss}}{N_i^{ss}}, \\ \left. \frac{\partial^2 \bar{Q}_{it}}{\partial (\sigma_t z) \partial \mathcal{K}_{it}} \right|_{SS} &= \left. \frac{\partial^2 \bar{Q}_{it}}{\partial T_t \partial \mathcal{K}_{it}} \right|_{SS} = -\epsilon\phi(1-\zeta) Q_i^{ss} \chi_{i1} R_i^{ss}. \end{aligned}$$

Third, the coefficients of the second-order effect of the aggregate shocks are given by:

$$\left. \frac{\partial^2 \bar{Q}_{it}}{\partial (\sigma_t z)^2} \right|_{SS} = \left. \frac{\partial^2 \bar{Q}_{it}}{\partial (T_t)^2} \right|_{SS} = \left. \frac{\partial^2 \bar{Q}_{it}}{\partial (\sigma_t z) \partial T_t} \right|_{SS} = \epsilon^2(1-\zeta) Q_i^{ss} \left[R_i^{ss} (\chi_{i1}^2 + \chi_{i2}) - \delta_{i2} \right].$$

Forth, coefficients for the capital depreciation coming from the Brownian motion:

$$\left. \frac{\partial \bar{Q}_{it}}{\partial \Sigma_{it}^2} \right|_{SS} = -\epsilon^2 \frac{(1-\zeta)\zeta}{2} Q_i^{ss}$$

Finally, the terms involving Q_{it} :

$$\begin{aligned}
\left. \frac{\partial^2 \bar{Q}_{it}}{\partial Q_{it}^2} \right|_{SS} &= \frac{1-\zeta}{\zeta} (Q_i^{ss})^{-\frac{1+\zeta}{\zeta}} \\
\left. \frac{\partial^2 \bar{Q}_{it}}{\partial \mathcal{K}_{it} \partial Q_{it}} \right|_{SS} &= -\phi(1-\zeta) R_i^{ss} \\
\left. \frac{\partial^2 \bar{Q}_{it}}{\partial N_{it} \partial Q_{it}} \right|_{SS} &= \psi(1-\zeta) \frac{R_i^{ss}}{N_i^{ss}} \\
\left. \frac{\partial^2 \bar{Q}_{it}}{\partial (\sigma_t z) \partial Q_{it}} \right|_{SS} &= \left. \frac{\partial^2 \bar{Q}_{it}}{\partial T_t \partial Q_{it}} \right|_{SS} = \epsilon(1-\zeta) [R_i^{ss} \chi_{i1} - \delta_{i1}].
\end{aligned}$$

Putting all the terms together, we have that:

$$\begin{aligned}
\bar{Q}_{it} - \bar{Q}_i^{ss} &\approx \bar{Q}_{i,N} (\epsilon n_i) + \bar{Q}_{i,\mathcal{K}} (\epsilon k_i) + \bar{Q}_{i,Q} \left(\epsilon q_{1i} + \frac{\epsilon^2}{2} q_{2i} \right) + \bar{Q}_{i,Tz} + \bar{Q}_{i,T} T_t \\
&+ \frac{1}{2} \bar{Q}_{i,NN} (\epsilon n_i)^2 + \frac{1}{2} \bar{Q}_{i,\mathcal{K}\mathcal{K}} (\epsilon k_i)^2 + \bar{Q}_{i,N\mathcal{K}} (\epsilon n_i) (\epsilon k_i) \\
&+ \frac{1}{2} \bar{Q}_{i,QQ} \left(\epsilon q_{1i} + \frac{\epsilon^2}{2} q_{2i} \right)^2 \\
&+ \bar{Q}_{i,NT} (\epsilon n_i) \sigma_{tz} + \bar{Q}_{i,NT} (\epsilon n_i) T_t \\
&+ \bar{Q}_{i,\mathcal{K}T} (\epsilon k_i) \sigma_{tz} + \bar{Q}_{i,\mathcal{K}T} (\epsilon k_i) T_t \\
&+ \bar{Q}_{i,QN} \left(\epsilon q_{1i} + \frac{\epsilon^2}{2} q_{2i} \right) n_i + \bar{Q}_{i,Q\mathcal{K}} \left(\epsilon q_{1i} + \frac{\epsilon^2}{2} q_{2i} \right) k_i + \bar{Q}_{i,QT} \left(\epsilon q_{1i} + \frac{\epsilon^2}{2} q_{2i} \right) (\sigma_{tz} + T_t) \\
&+ \frac{1}{2} \bar{Q}_{i,TT} (\sigma_{tz})^2 + \frac{1}{2} \bar{Q}_{i,TT} (T_t)^2 + \bar{Q}_{i,TT} (\sigma_{tz}) T_t + \bar{Q}_{i,\Sigma} \Sigma_i^2 \zeta_t,
\end{aligned}$$

where $\bar{Q}_{i,XY}$ denote the cross partial derivative of $\bar{Q}_i$ with respect to X and Y , evaluated at steady state. Now, before writing all terms in matrix notation, we work on the quadratic term q_{1i}^2 :

$$\begin{aligned}
q_{1i}^2 = & \sum_j \sum_s Q_{ij}^N Q_{is}^N n_j n_s + \sum_j \sum_s Q_{ij}^K Q_{is}^K k_j k_s + \sum_j \sum_s Q_{ij}^N Q_{is}^K n_j k_s + \sum_j \sum_s Q_{ij}^K Q_{is}^N k_j n_s \\
& + \left(q_{it}^Z z \right)^2 + \left(q_{it}^T \right)^2 + 2 q_{it}^Z z q_{it}^T \\
& + 2 \sum_j Q_{ij}^N n_j q_{it}^Z z + 2 \sum_j Q_{ij}^N n_j q_{it}^T \\
& + 2 \sum_j Q_{ij}^K k_j q_{it}^Z z + 2 \sum_j Q_{ij}^K k_j q_{it}^T.
\end{aligned}$$

First, note that:

$$\begin{aligned}
\sum_j \sum_s Q_{ij}^N Q_{is}^N n_j n_s &= \left[\mathbb{Q}^{N,N} \times_3 n \times_2 n \right]_i, & \text{with } \mathbb{Q}_{ijs}^{N,N} &= Q_{ij}^N Q_{is}^N \\
\sum_j \sum_s Q_{ij}^N Q_{is}^K n_j k_s &= \left[\mathbb{Q}^{N,K} \times_3 k \times_2 n \right]_i, & \text{with } \mathbb{Q}_{ijs}^{N,K} &= Q_{ij}^N Q_{is}^K \\
\sum_j \sum_s Q_{ij}^K Q_{is}^N k_j n_s &= \left[\mathbb{Q}^{K,N} \times_3 n \times_2 k \right]_i, & \text{with } \mathbb{Q}_{ijs}^{K,N} &= Q_{ij}^K Q_{is}^N \\
\sum_j \sum_s Q_{ij}^K Q_{is}^K k_j k_s &= \left[\mathbb{Q}^{K,K} \times_3 k \times_2 k \right]_i, & \text{with } \mathbb{Q}_{ijs}^{K,K} &= Q_{ij}^K Q_{is}^K
\end{aligned}$$

On the other hand:

$$\begin{aligned}
\sum_j Q_{ij}^N q_{it}^Z z n_j &= \left[\mathbf{diag}(q_t^Z) \mathbb{Q}^N n z \right]_i, & \sum_j Q_{ij}^N q_{it}^T n_j &= \left[\mathbf{diag}(q_t^T) \mathbb{Q}^N n \right]_i, \\
\sum_j Q_{ij}^K q_{it}^Z z k_j &= \left[\mathbf{diag}(q_t^Z) \mathbb{Q}^K k z \right]_i, & \sum_j Q_{ij}^K q_{it}^T k_j &= \left[\mathbf{diag}(q_t^T) \mathbb{Q}^K k \right]_i,
\end{aligned}$$

where **diag** is the diagonal operator. Putting all terms together, we get:

$$\begin{aligned}
q_1 \cdot q_1 = & \mathbb{Q}^{N,N} \times_3 n \times_2 n + \mathbb{Q}^{N,K} \times_3 k \times_2 n + \mathbb{Q}^{K,N} \times_3 n \times_2 k + \mathbb{Q}^{K,K} \times_3 k \times_2 k \\
& + (q_t^Z \cdot q_t^Z) z^2 + q_t^T \cdot q_t^T + 2(q_t^Z \cdot q_t^T) z \\
& + 2\mathbf{diag}(q_t^Z) \mathbb{Q}^N n z + 2\mathbf{diag}(q_t^T) \mathbb{Q}^N n \\
& + 2\mathbf{diag}(q_t^Z) \mathbb{Q}^K k z + 2\mathbf{diag}(q_t^T) \mathbb{Q}^K k.
\end{aligned}$$

where the operator $\cdot$ denotes the inner product between two vectors. Then, to the leading order, we get:

$$\begin{aligned}
\bar{Q}_t - \bar{Q}^{ss} &\approx \epsilon \{ \mathbb{D}_{QN} n + \mathbb{D}_{QK} k + \mathbb{d}_{QT} (\sigma_t z + T_t) \} \\
&+ \frac{\epsilon^2}{2} \mathbb{D}_{QQ} \left\{ \mathcal{Q}^{N,N} \times_3 n \times_2 n + \mathcal{Q}^{N,K} \times_3 k \times_2 n + \mathcal{Q}^{K,N} \times_3 n \times_2 k + \mathcal{Q}^{K,K} \times_3 k \times_2 k \right. \\
&\quad + 2\mathcal{Q}_t^{N,Z} n z + 2\mathcal{Q}_t^{K,Z} k z + 2\mathcal{Q}_t^{N,T} n + 2\mathcal{Q}_t^{K,T} k \\
&\quad \left. + \bar{q}_t + q_t^{Z,Z} z^2 + q_t^{T,T} + 2q_t^{Z,T} z + q_t^\Sigma \right\} \\
&+ \frac{\epsilon^2}{2} \{ \mathbb{D}_{QNN} \times_3 n \times_2 n + \mathbb{D}_{QKK} \times_3 k \times_2 k + \mathbb{D}_{QNK} \times_3 n \times_2 k + \mathbb{D}_{QKN} \times_3 k \times_2 n \} \\
&+ \frac{\epsilon^2}{2} \mathbb{D}_{QQQ} \left\{ \mathbb{Q}^{N,N} \times_3 n \times_2 n + \mathbb{Q}^{N,K} \times_3 k \times_2 n + \mathbb{Q}^{K,N} \times_3 n \times_2 k + \mathbb{Q}^{K,K} \times_3 k \times_2 k \right. \\
&\quad + (q_t^Z \cdot q_t^Z) z^2 + q_t^T \cdot q_t^T + 2(q_t^Z \cdot q_t^T) z \\
&\quad + 2\mathbf{diag}(q_t^Z) \mathbb{Q}^N n z + 2\mathbf{diag}(q_t^T) \mathbb{Q}^N n \\
&\quad \left. + 2\mathbf{diag}(q_t^Z) \mathbb{Q}^K k z + 2\mathbf{diag}(q_t^T) \mathbb{Q}^K k \right\} \\
&+ \epsilon^2 \left\{ \mathbb{D}_{QNT} n [\sigma_t z + T_t] + \mathbb{D}_{QKT} k [\sigma_t z + T_t] \right\} \\
&+ \epsilon^2 \left\{ \left(\mathbb{D}_{QQN} \times_1 \mathbb{Q}^N \right) \times_2 n \times_3 n + \left(\mathbb{D}_{QQN} \times_1 \mathbb{Q}^K \right) \times_2 k \times_3 n + (\mathbb{D}_{QQN} \times_1 q_t^Z) n z + (\mathbb{D}_{QQN} \times_1 q_t^T) n \right. \\
&+ \epsilon^2 \left\{ \left(\mathbb{D}_{QQK} \times_1 \mathbb{Q}^N \right) \times_2 n \times_3 k + \left(\mathbb{D}_{QQK} \times_1 \mathbb{Q}^K \right) \times_2 k \times_3 k + (\mathbb{D}_{QQK} \times_1 q_t^Z) k z + (\mathbb{D}_{QQK} \times_1 q_t^T) k \right. \\
&+ \epsilon^2 \mathbb{D}_{QQT} [\mathbb{Q}^N n + \mathbb{Q}^K k + q_t^Z z + q_t^T] [\sigma_t z + T_t] \\
&\left. + \frac{\epsilon^2}{2} \mathbb{d}_{QTT} \left[(\sigma_t z)^2 + (T_t)^2 + 2(\sigma_t z) T_t \right] + \frac{\epsilon^2}{2} \Sigma^2 \mathbb{d}_{Q\sigma T \zeta t} \right\}
\end{aligned}$$

where $\mathbb{D}_{QNN}$, $\mathbb{D}_{QKN}$, $\mathbb{D}_{QNK}$, $\mathbb{D}_{QKK}$, $\mathbb{D}_{QQN}$ and $\mathbb{D}_{QQK}$ are diagonal tensors defined as:

$$\begin{aligned}
\mathbb{D}_{QNN} &= \psi(\psi - 1)(1 - \zeta) \mathbf{diag}_3 \left(\frac{Q_i^{ss} R_i^{ss}}{(N_i^{ss})^2} \right), & \mathbb{D}_{QKK} &= \phi^2(1 - \zeta) \mathbf{diag}_3 (Q_i^{ss} R_i^{ss}), \\
\mathbb{D}_{QNK} &= \mathbb{D}_{QKN} = -\psi\phi(1 - \zeta) \mathbf{diag}_3 \left(\frac{Q_i^{ss} R_i^{ss}}{N_i^{ss}} \right), \\
\mathbb{D}_{QQN} &= (1 - \zeta)\psi \mathbf{diag}_3 \left(\frac{R_i^{ss}}{N_i^{ss}} \right) & \mathbb{D}_{QQK} &= -(1 - \zeta)\phi \mathbf{diag}_3 (R_i^{ss})
\end{aligned}$$

Moreover, Σ^2 is a diagonal matrix with components Σ_i^2 , and $\mathbb{D}_{QK}$, $\mathbb{D}_{QN}$, $\mathbb{D}_{QQ}$, and $\mathbb{d}_{QT}$

are defined as in section 4. The remaining matrices are and vectors:

$$\begin{aligned}
\mathbb{D}_{QQQ} &= \frac{1-\zeta}{\zeta} \mathbf{diag} \left((Q_i^{ss})^{-1-\frac{1}{\zeta}} \right), & \mathbb{D}_{QQT} &= (1-\zeta) \mathbf{diag} (R_i^{ss} \chi_{i1} - \delta_{i1}) \\
\mathbb{D}_{QNT} &= \psi(1-\zeta) \mathbf{diag} \left(\chi_{i1} \frac{Q_i^{ss} R_i^{ss}}{N_i^{ss}} \right), & \mathbb{D}_{QKT} &= -\phi(1-\zeta) \mathbf{diag} (\chi_{i1} Q_i^{ss} R_i^{ss}), \\
\mathbf{d}_{QTT} &= (1-\zeta) \mathbf{vec} \left(Q_i^{ss} \left(R_i^{ss} (\chi_{i1}^2 + \chi_{i2}) - \delta_{i2} \right) \right), & \mathbf{d}_{QST} &= -\frac{(1-\zeta)\zeta}{2} \mathbf{vec} (Q_i^{ss}).
\end{aligned}$$

Grouping terms according to their order and using $\mathbb{D}_{QQ} = 0$ gives the final expression:

$$\begin{aligned}
\overline{Q}_t - \overline{Q}_i^{ss} &\approx \epsilon \left\{ \mathbb{D}_{QK}k + \mathbb{D}_{QN}n + \mathbf{d}_{QT}(\sigma_t z + T_t) \right\} \leftarrow \text{first-order term} \\
&+ \frac{\epsilon^2}{2} \left\{ \mathbb{D}_{QNN} + \mathbb{Q}^{N,N} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQN} \times_1 Q^N \right\} \times_3 n \times_2 n \\
&+ \frac{\epsilon^2}{2} \left\{ \mathbb{D}_{QKN} + \mathbb{Q}^{N,K} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQN} \times_1 Q^K \right\} \times_3 k \times_2 n \\
&+ \frac{\epsilon^2}{2} \left\{ \mathbb{D}_{QNK} + \mathbb{Q}^{K,N} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQK} \times_1 Q^N \right\} \times_3 n \times_2 k \\
&+ \frac{\epsilon^2}{2} \left\{ \mathbb{D}_{QKK} + \mathbb{Q}^{K,K} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQK} \times_1 Q^K \right\} \times_3 k \times_2 k \\
&+ \epsilon^2 \left[\mathbb{D}_{QQQ} \mathbf{diag}(q_t^Z) Q^N + (\mathbb{D}_{QNT} + \mathbb{D}_{QQT} Q^N) \sigma_t + \mathbb{D}_{QQN} \times_1 q_t^Z \right] nz \\
&+ \epsilon^2 \left[\mathbb{D}_{QQQ} \mathbf{diag}(q_t^T) Q^N + (\mathbb{D}_{QNT} + \mathbb{D}_{QQT} Q^N) T_t + \mathbb{D}_{QQN} \times_1 q_t^T \right] n \\
&+ \epsilon^2 \left[\mathbb{D}_{QQQ} \mathbf{diag}(q_t^Z) Q^K + (\mathbb{D}_{QKT} + \mathbb{D}_{QQT} Q^K) \sigma_t + \mathbb{D}_{QQK} \times_1 q_t^Z \right] kz \\
&+ \epsilon^2 \left[\mathbb{D}_{QQQ} \mathbf{diag}(q_t^T) Q^K + (\mathbb{D}_{QKT} + \mathbb{D}_{QQT} Q^K) T_t + \mathbb{D}_{QQK} \times_1 q_t^T \right] k \\
&+ \frac{\epsilon^2}{2} \left[\mathbb{D}_{QQQ} (q_t^Z \cdot q_t^Z) + 2\mathbb{D}_{QQT} q_t^Z \sigma_t + \mathbf{d}_{QTT} \sigma_t^2 \right] z^2 \\
&+ \frac{\epsilon^2}{2} \left[\mathbb{D}_{QQQ} (q_t^T \cdot q_t^T) + 2\mathbb{D}_{QQT} q_t^T T_t + \mathbf{d}_{QTT} (T_t)^2 \right] \\
&+ \epsilon^2 \left[\mathbb{D}_{QQQ} q_t^Z q_t^T + \mathbb{D}_{QQT} (q_t^Z T_t + q_t^T \sigma_t) + \mathbf{d}_{QTT} \sigma_t T_t \right] z. \\
&+ \frac{\epsilon^2}{2} \Sigma^2 \mathbf{d}_{QST} \zeta_t.
\end{aligned}$$

E.2 Continuation Value from Migration

We have that:

$$\mathcal{M}_i(V_t) - \mathcal{M}_i(V^{ss}) \approx \nabla[\mathcal{M}_i](V^{ss})(V - V^{ss}) + \frac{1}{2}(V - V^{ss})^T H[\mathcal{M}_i](V^{ss})(V - V^{ss}),$$

where $\nabla[\mathcal{M}_i](\cdot)$ and $H[\mathcal{M}_i](\cdot)$ are the gradient and the Hessian of $\mathcal{M}_i$, respectively. We have that, up to a second order, $V - V^{ss} = \epsilon v_1 + \frac{\epsilon^2}{2} v_2$, with v_1 and v_2 defined in (16) and (18). Therefore, omitting the dependence on V^{ss} , we can write the expression above as:

$$\mathcal{M}_i(V_t) - \mathcal{M}_i(V^{ss}) \approx \epsilon [Mv_1]_i + \frac{\epsilon^2}{2} [Mv_2]_i + \frac{1}{2}(\epsilon v_1 + \frac{\epsilon^2}{2} v_2)^T H[\mathcal{M}_i](V^{ss})(\epsilon v_1 + \frac{\epsilon^2}{2} v_2).$$

In matrix form:

$$\begin{aligned} Mv_1 &= MV^N n + MV^K k + Mv_t^Z z + Mv_t^T, \\ Mv_2 &= M \left[\mathcal{V}^{N,N} \times_3 n \times_2 n + \mathcal{V}^{N,K} \times_3 k \times_2 n + \mathcal{V}^{K,N} \times_3 n \times_2 k + \mathcal{V}^{K,K} \times_3 k \times_2 k \right. \\ &\quad \left. + 2V_t^{N,Z} n z + 2V_t^{K,Z} k z + 2V_t^{N,T} n + 2V_t^{K,T} k + \bar{v}_t + v_t^{Z,Z} z^2 + v_t^{T,T} + 2v_t^{Z,T} z + v_t^{\Sigma,Z} z + v_t^{\Sigma,T} \right] \\ &= \mathcal{V}^{N,N} \times_1 M \times_3 n \times_2 n + \mathcal{V}^{N,K} \times_1 M \times_3 k \times_2 n + \mathcal{V}^{K,N} \times_1 M \times_3 n \times_2 k + \mathcal{V}^{K,K} \times_1 M \times_3 k \times_2 k \\ &\quad + 2MV_t^{N,Z} n z + 2MV_t^{K,Z} k z + 2MV_t^{N,T} n + 2MV_t^{K,T} k + M\bar{v}_t + Mv_t^{Z,Z} z^2 + Mv_t^{T,T} + 2Mv_t^{Z,T} z \\ &\quad + Mv_t^{\Sigma,Z} z + Mv_t^{\Sigma,T}. \end{aligned}$$

On the other hand, the Hessian matrix is given by:

$$H[\mathcal{M}_i](V^{ss}) = \left[\frac{\partial m_{ik}(V^{ss})}{\partial V_h} \right]_{hk}.$$

To ease notation, we omit the dependence of the migration probabilities on the steady-state vector V^{ss} . Then, we have that:

$$(\epsilon v_1 + \frac{\epsilon^2}{2} v_2)^T H[\mathcal{M}_i](V^{ss})(\epsilon v_1 + \frac{\epsilon^2}{2} v_2) = \mu \nu \left[\sum_k m_{ik}(\epsilon v_{1k} + \frac{\epsilon^2}{2} v_{2k})^2 - \sum_h \sum_k m_{ih} m_{ik}(\epsilon v_{1h} + \frac{\epsilon^2}{2} v_{2h})(\epsilon v_{1k} + \frac{\epsilon^2}{2} v_{2k}) \right]$$

Note that the second-order terms in v_2 will drop out of the approximation when raised to

a higher power or if multiplied by v_1 . Thus, we have that:

$$(\epsilon v_1 + \frac{\epsilon^2}{2} v_2)^T H[\mathcal{M}_i] (V^{ss})(\epsilon v_1 + \frac{\epsilon^2}{2} v_2) = \epsilon^2 \mu \nu \left[\sum_k m_{ik} v_{1k}^2 - \sum_h \sum_k m_{ih} m_{ik} v_{1h} v_{1k} \right],$$

which in matrix notation is:

$$(\epsilon v_1 + \frac{\epsilon^2}{2} v_2)^T H[\mathcal{M}] (V^{ss})(\epsilon v_1 + \frac{\epsilon^2}{2} v_2) = \epsilon^2 \mu \nu m(v_1 \cdot v_1) - \epsilon^2 \mu \nu \mathbb{M} \times_3 (v_1) \times_2 (v_1),$$

where $\mathbb{M}$ is a tensor defined as $\mathbb{M}_{ikh} = -m_{ik} m_{ih}$. We can use the expression for $q_1 \cdot q_1$ in the previous section to expand the term $m(v_1 \cdot v_1)$:

$$\begin{aligned} m(v_1 \cdot v_1) = & \mathbb{V}^{N,N} \times_1 m \times_3 n \times_2 n + \mathbb{V}^{N,K} \times_1 m \times_3 k \times_2 n + \mathbb{V}^{K,N} \times_1 m \times_3 n \times_2 k + \mathbb{V}^{K,K} \times_1 m \times_3 k \times_2 k \\ & + m \left(v_t^Z \cdot v_t^Z \right) z^2 + m \left(v_t^T \cdot v_t^T \right) + 2m \left(v_t^Z \cdot v_t^T \right) z \\ & + 2m \mathbf{diag}(v_t^Z) V^N n z + 2m \mathbf{diag}(v_t^T) V^N n \\ & + 2m \mathbf{diag}(v_t^Z) V^K n z + 2m \mathbf{diag}(v_t^T) V^K n, \end{aligned}$$

where the tensors $\mathbb{V}^{N,N}$, $\mathbb{V}^{N,K}$, $\mathbb{V}^{K,N}$, and $\mathbb{V}^{K,K}$ are defined as:

$$\mathbb{V}_{ijs}^{N,N} = V_{ij}^N V_{is}^N \quad \mathbb{V}_{ijs}^{N,K} = V_{ij}^N V_{is}^K \quad \mathbb{V}_{ijs}^{K,N} = V_{ij}^K V_{is}^N \quad \mathbb{V}_{ijs}^{K,K} = V_{ij}^K V_{is}^K$$

We can use Lemmas 1 and 3 in the appendix jointly with the fact that $\mathbb{M}$ is symmetric in its last two coordinates to expand the term $\mathbb{M} \times_3 v_1 \times_2 v_1$ as:

$$\begin{aligned} \mathbb{M} \times_3 (v_1) \times_2 (v_1) = & \left(\mathbb{M} \times_3 V^N \times_2 V^N \right) \times_3 n \times_2 n + \left(\mathbb{M} \times_3 V^K \times_2 V^N \right) \times_3 k \times_2 n \\ & + \left(\mathbb{M} \times_3 V^N \times_2 V^K \right) \times_3 n \times_2 k + \left(\mathbb{M} \times_3 V^K \times_2 V^K \right) \times_3 k \times_2 k \\ & + 2 \left(\mathbb{M} \times_2 V^N \times_3 v_t^Z \right) n z + 2 \left(\mathbb{M} \times_2 V^N \times_3 v_t^T \right) n \\ & + 2 \left(\mathbb{M} \times_2 V^K \times_3 v_t^Z \right) k z + 2 \left(\mathbb{M} \times_2 V^K \times_3 v_t^T \right) k \\ & + \mathbb{M} \times_3 v_t^Z \times_2 v_t^Z z^2 + \mathbb{M} \times_3 v_t^T \times_2 v_t^T + 2 \mathbb{M} \times_3 v_t^Z \times_2 v_t^T z. \end{aligned}$$

Gathering all the terms gives:

$$\begin{aligned}
\mathcal{M}(V_t) - \mathcal{M}(V^{ss}) = & \epsilon \left(MV^N n + MV^K k + Mv_t^Z z + Mv_t^T + M \right) \\
& + \frac{\epsilon^2}{2} \left[\mathcal{V}^{N,N} \times_1 M \times_3 n \times_2 n + \mathcal{V}^{N,K} \times_1 M \times_3 k \times_2 n + \mathcal{V}^{K,N} \times_1 M \times_3 n \times_2 k + \mathcal{V}^{K,K} \times_1 M \times_3 k \times_2 k \right. \\
& \quad + 2MV_t^{N,Z} nz + 2MV_t^{K,Z} kz + 2MV_t^{N,T} n + 2MV_t^{K,T} k + M\bar{v}_t + Mv_t^{Z,Z} z^2 + Mv_t^{T,T} + 2Mv_t^{Z,T} z \\
& \quad \left. + Mv_t^{\Sigma,Z} z + Mv_t^{\Sigma,T} \right] \\
& + \frac{\epsilon^2 \mu v}{2} \left[\mathbb{V}^{N,N} \times_1 m \times_3 n \times_2 n + \mathbb{V}^{N,K} \times_1 m \times_3 k \times_2 n + \mathbb{V}^{K,N} \times_1 m \times_3 n \times_2 k + \mathbb{V}^{K,K} \times_1 m \times_3 k \times_2 k \right. \\
& \quad + m \left(v_t^Z \cdot v_t^Z \right) z^2 + m \left(v_t^T \cdot v_t^T \right) + 2m \left(v_t^Z \cdot v_t^T \right) z \\
& \quad \left. + 2m \mathbf{diag}(v_t^Z) V^N nz + 2m \mathbf{diag}(v_t^T) V^N n + 2m \mathbf{diag}(v_t^Z) V^K nz + 2m \mathbf{diag}(v_t^T) V^K n \right] \\
& + \frac{\epsilon^2 \mu v}{2} \left[\left(\mathbb{M} \times_3 V^N \times_2 V^N \right) \times_3 n \times_2 n + \left(\mathbb{M} \times_3 V^K \times_2 V^N \right) \times_3 k \times_2 n \right. \\
& \quad + \left(\mathbb{M} \times_3 V^N \times_2 V^K \right) \times_3 n \times_2 k + \left(\mathbb{M} \times_3 V^K \times_2 V^K \right) \times_3 k \times_2 k \\
& \quad + 2 \left(\mathbb{M} \times_2 V^N \times_3 v_t^Z \right) nz + 2 \left(\mathbb{M} \times_2 V^N \times_3 v_t^T \right) n + 2 \left(\mathbb{M} \times_2 V^K \times_3 v_t^Z \right) kz + 2 \left(\mathbb{M} \times_2 V^K \times_3 v_t^T \right) k \\
& \quad \left. + \mathbb{M} \times_3 v_t^Z \times_2 v_t^Z z^2 + \mathbb{M} \times_3 v_t^T \times_2 v_t^T + 2\mathbb{M} \times_3 v_t^Z \times_2 v_t^T z \right]
\end{aligned}$$

We obtain the final expression grouping terms according to their order:

$$\begin{aligned}
\mathcal{M}(V_t) - \mathcal{M}(V^{ss}) = & \epsilon \left(MV^N n + MV^K k + Mv_t^Z z + Mv_t^T \right) \leftarrow \text{first-order term} \\
& + \frac{\epsilon^2}{2} \left[\mathcal{V}^{N,N} \times_1 M + \mu\nu \left(\mathbb{V}^{N,N} \times_1 m + \mathbb{M} \times_3 V^N \times_2 V^N \right) \right] \times_3 n \times_2 n \\
& + \frac{\epsilon^2}{2} \left[\mathcal{V}^{N,K} \times_1 M + \mu\nu \left(\mathbb{V}^{N,K} \times_1 m + \mathbb{M} \times_3 V^K \times_2 V^N \right) \right] \times_3 k \times_2 n \\
& + \frac{\epsilon^2}{2} \left[\mathcal{V}^{K,N} \times_1 M + \mu\nu \left(\mathbb{V}^{K,N} \times_1 m + \mathbb{M} \times_3 V^N \times_2 V^K \right) \right] \times_3 n \times_2 k \\
& + \frac{\epsilon^2}{2} \left[\mathcal{V}^{K,K} \times_1 M + \mu\nu \left(\mathbb{V}^{K,K} \times_1 m + \mathbb{M} \times_3 V^K \times_2 V^K \right) \right] \times_3 k \times_2 k \\
& + \epsilon^2 \left[MV_t^{N,Z} + \mu\nu \left(m \mathbf{diag}(v_t^Z) V^N + \mathbb{M} \times_2 V^N \times_3 v_t^Z \right) \right] nz \\
& + \epsilon^2 \left[MV_t^{N,T} + \mu\nu \left(m \mathbf{diag}(v_t^T) V^N + \mathbb{M} \times_2 V^N \times_3 v_t^T \right) \right] n \\
& + \epsilon^2 \left[MV_t^{K,Z} + \mu\nu \left(m \mathbf{diag}(v_t^Z) V^K + \mathbb{M} \times_2 V^K \times_3 v_t^Z \right) \right] kz \\
& + \epsilon^2 \left[MV_t^{K,T} + \mu\nu \left(m \mathbf{diag}(v_t^T) V^K + \mathbb{M} \times_2 V^K \times_3 v_t^T \right) \right] k \\
& + \frac{\epsilon^2}{2} \left[Mv_t^{Z,Z} + \mu\nu \left(m(v_t^Z \cdot v_t^Z) + \mathbb{M} \times_3 v_t^Z \times_2 v_t^Z \right) \right] z^2 \\
& + \frac{\epsilon^2}{2} \left[Mv_t^{T,T} + \mu\nu \left(m(v_t^T \cdot v_t^T) + \mathbb{M} \times_3 v_t^T \times_2 v_t^T \right) \right] \\
& + \epsilon^2 \left[Mv_t^{Z,T} + \mu\nu \left(m(v_t^Z \cdot v_t^T) + \mathbb{M} \times_3 v_t^Z \times_2 v_t^T \right) \right] z \\
& + \frac{\epsilon^2}{2} Mv_t^{\Sigma,Z} z + \frac{\epsilon^2}{2} Mv_t^{\Sigma,T} \\
& + \frac{\epsilon^2}{2} M\bar{v}_t
\end{aligned}$$

E.3 Continuation Value from Changes in Population Distribution

E.3.1 Workers

The continuation value for workers from changes in the population distribution is given by:

$$\mu \sum_j \frac{\partial V_i}{\partial N_j} \left(\sum_k m_{kj}(V) N_k - N_j \right)$$

We will approximate individual terms and then take the product. First, applying the

chain rule to (15), (16) and (18), we get that:

$$\frac{\partial V_i}{\partial N_j} = V_{ij}^N + \frac{\epsilon}{2} \left(\sum_s \mathcal{V}_{ijs}^{N,N} n_s + \sum_j \mathcal{V}_{ijs}^{N,N} n_j + \sum_s \mathcal{V}_{ijs}^{N,K} k_s + \sum_j \mathcal{V}_{ijs}^{K,N} k_j + 2V_{ijt}^{N,Z} z + 2V_{ijt}^{N,T} \right).$$

Now, fixing j and k , we work on the term $m_{kj}N_k - N_j$. First, we start approximating m_{kj} :

$$M_{jk}^*(V_{SS} + dv) = M_{jk}(V_{SS})^* + \nabla [M_{jk}^*](V_{SS})dv + \frac{1}{2}dv^*H[M_{jk}^*](V_{SS})dv$$

Recall that we have already derived the gradient component in the previous section, which we now define as a tensor:

$$\frac{\partial m_{kj}}{\partial V_h} = \begin{cases} -vm_{kj}m_{kh} & \text{if } j \neq h \\ vm_{kj}(1 - m_{kj}) & \text{if } j = h \end{cases}$$

Therefore,

$$\frac{\partial M_{jk}^*}{\partial V_h} = \mu v \begin{cases} -m_{kj}m_{kh} & \text{if } j \neq h \\ m_{kj}(1 - m_{kj}) & \text{if } j = h \end{cases} \equiv \mathbb{G}_{jkh}$$

This implies that,

$$\left(\nabla [M_{jk}^*](V_{SS})dv \right)_{jk} = \mathbb{G} \times_3 dv$$

On the other hand, for the Hessian, we have that $H[M_{jk}^*] = \mu \frac{\partial^2 m_{kj}}{\partial V_h \partial V_s}$. Now,

$$\begin{aligned} \frac{\partial^2 m_{kj}}{\partial^2 V_h} &= v^2 m_{kj} m_{kh} (2m_{kh} - 1) & \text{for } h \neq j \\ \frac{\partial^2 m_{kj}}{\partial V_h \partial V_s} &= v^2 2m_{kj} m_{kh} m_{ks} & \text{for } h \neq s \text{ and } h, s \neq j \\ \frac{\partial^2 m_{kj}}{\partial V_h \partial V_j} &= v^2 m_{kj} m_{kh} (2m_{kj} - 1) & \text{for } h \neq j \\ \frac{\partial^2 m_{kj}}{\partial^2 V_j} &= v^2 m_{kj} (1 - 3m_{kj} + 2m_{kj}^2) \end{aligned}$$

Hence, define:

$$\mathbb{H}_{kjhs} = \mu \frac{\partial^2 m_{jk}}{\partial V_h \partial V_s},$$

a four-dimensional tensor. Using these expressions, one obtains:

$$\left(dv^* H \left[M_{jk}^* \right] dv \right)_{jk} = \mathbb{H} \times_4 dv \times_3 dv,$$

where the tensor product with respect to the fourth coordinate, $\times_4$, is defined as analogously to $\times_3$ in Section ?? . Then, to the leading order (where the 0th order term drops out in steady-state):

$$\begin{aligned} (M(V)^* N)_j &= \sum_k \left(M_{jk}^* + \mathbb{G}_{jk\bullet} \times_3 dv + \frac{1}{2} \mathbb{H}_{jk\bullet\bullet} \times_4 dv \times_3 dv \right) (N_k + \epsilon n_k), \\ &= (\mathbb{G}_{j\bullet\bullet} \times_2 N) dv + \epsilon (M^* n)_j \\ &\quad + \epsilon \mathbb{G}_{j\bullet\bullet} \times_3 dv \times_2 n + \frac{1}{2} (\mathbb{H}_{j\bullet\bullet\bullet} \times_2 N) \times_3 dv \times_2 dv, \\ &= \epsilon \left\{ (\mathbb{G}_{j\bullet\bullet} \times_2 N) v_1 + (M^* n)_j \right\} \\ &\quad + \epsilon^2 \left\{ \frac{1}{2} (\mathbb{G}_{j\bullet\bullet} \times_2 N) v_2 + \mathbb{G}_{j\bullet\bullet} \times_3 v_1 \times_2 n + \frac{1}{2} (\mathbb{H}_{j\bullet\bullet\bullet} \times_2 N) \times_3 v_1 \times_2 v_1 \right\} \\ &\equiv \epsilon \mathcal{M}_j + \epsilon^2 \mathcal{N}_j, \end{aligned} \tag{50}$$

where we drop 3rd-order terms to get to the second and third lines. Note that:

$$\begin{aligned} \mathcal{M} &= M^* n + (\mathbb{G} \times_2 N) v_1, \\ &= M^* n + (\mathbb{G} \times_2 N) \left(V^N n + V^K k + v_t^Z z + v_t^T \right), \\ &= F^N n + F^K k + f_t^Z z + f_t^T, \end{aligned}$$

with

$$\begin{aligned} F^N &= M^* + (\mathbb{G} \times_2 N) V^N, & f_t^Z &= (\mathbb{G} \times_2 N) v_t^Z, \\ F^K &= (\mathbb{G} \times_2 N) V^K, & f_t^T &= (\mathbb{G} \times_2 N) v_t^T. \end{aligned}$$

Thus, again, to the leading order:

$$\begin{aligned}
& \sum_j \frac{\partial V_i}{\partial N_j} (M^*(V)N)_j = \\
& \sum_j \left\{ V_{ij}^N + \frac{\epsilon}{2} \left(\sum_s \mathcal{V}_{ijs}^{N,N} n_s + \sum_s \mathcal{V}_{isj}^{N,N} n_s + \sum_s \mathcal{V}_{ijs}^{N,K} k_s + \sum_s \mathcal{V}_{isj}^{K,N} k_s + 2V_{ijt}^{N,Z} z + 2V_{ijt}^{N,T} \right) \right\} \\
& \quad \times \left\{ \epsilon \mathcal{M}_j + \epsilon^2 \mathcal{N}_j \right\}, \\
& = \sum_j \left\{ V_{ij}^N + \frac{\epsilon}{2} \partial \mathcal{V}_j \right\} \left\{ \epsilon \mathcal{M}_j + \epsilon^2 \mathcal{N}_j \right\}
\end{aligned}$$

Focusing on the second-order terms, we start by expanding the term involving $\partial \mathcal{V}_j$ and $\mathcal{M}_j$:

$$\frac{\epsilon^2}{2} \sum_j \partial \mathcal{V}_j \times \mathcal{M}_j = \left(\mathcal{V}^{N,N} \times_3 n + \mathcal{V}^{N,N} \times_2 n + \mathcal{V}^{N,K} \times_3 k + \mathcal{V}^{K,N} \times_2 k + 2V_t^{N,Z} z + 2V_t^{N,T} \right) \mathcal{M}$$

Expanding term by term, we get:

$$\begin{aligned}
\left(\mathcal{V}^{N,N} \times_3 n \right) \mathcal{M} &= \mathcal{V}^{N,N} \times_2 F^N \times_3 n \times_2 n + \mathcal{V}^{N,N} \times_2 F^K \times_3 k \times_2 n \\
&+ \left(\mathcal{V}^{N,N} \times_2 f_t^Z \right) n z + \left(\mathcal{V}^{N,N} \times_2 f_t^T \right) n.
\end{aligned}$$

$$\begin{aligned}
\left(\mathcal{V}^{N,N} \times_2 n \right) \mathcal{M} &= \mathcal{V}^{N,N} \times_3 F^N \times_3 n \times_2 n + \mathcal{V}^{N,N} \times_3 F^K \times_3 k \times_2 n \\
&+ \left(\mathcal{V}^{N,N} \times_3 f_t^Z \right) n z + \left(\mathcal{V}^{N,N} \times_3 f_t^T \right) n.
\end{aligned}$$

$$\begin{aligned}
\left(\mathcal{V}^{N,K} \times_3 k \right) \mathcal{M} &= \mathcal{V}^{N,K} \times_2 F^N \times_3 k \times_2 n + \mathcal{V}^{N,K} \times_2 F^K \times_3 k \times_2 k \\
&+ \left(\mathcal{V}^{N,K} \times_2 f_t^Z \right) k z + \left(\mathcal{V}^{N,K} \times_2 f_t^T \right) k.
\end{aligned}$$

$$\begin{aligned}
\left(\mathcal{V}^{K,N} \times_2 k \right) \mathcal{M} &= \mathcal{V}^{K,N} \times_3 F^N \times_3 n \times_2 k + \mathcal{V}^{K,N} \times_3 F^K \times_3 k \times_2 k \\
&+ \left(\mathcal{V}^{K,N} \times_3 f_t^Z \right) k z + \left(\mathcal{V}^{K,N} \times_3 f_t^T \right) k.
\end{aligned}$$

$$2V_t^{N,Z}{}_Z\mathcal{M} = 2 \left(V_t^{N,Z}F^N n z + V_t^{N,Z}F^K k z + V_t^{N,Z}f_t^Z z^2 + V_t^{N,Z}f_t^T z \right).$$

$$2V_t^{N,T}\mathcal{M} = 2 \left(V_t^{N,T}F^N n + V_t^{N,T}F^K k + V_t^{N,T}f_t^Z z + V_t^{N,T}f_t^T \right).$$

Now we work on the term involving V^N and $\mathcal{N}$:

$$\begin{aligned} \sum_j V_{ij}^N \epsilon^2 \mathcal{N}_j &= \epsilon^2 \left\{ \frac{1}{2} \left(\mathbb{G} \times_1 V^N \times_2 N \right) v_2 + \mathbb{G} \times_1 V^N \times_3 v_1 \times_2 n + \frac{1}{2} \left(\mathbb{H} \times_1 V^N \times_2 N \right) \times_3 v_1 \times_2 v_1 \right\}, \\ &= \epsilon^2 \left\{ \frac{1}{2} X v_2 + \mathbb{Y} \times_3 v_1 \times_2 n + \frac{1}{2} \mathbb{Z} \times_3 v_1 \times_2 v_1 \right\}, \end{aligned}$$

with the matrix X and the tensors $\mathbb{Y}$ and $\mathbb{Z}$ defined as

$$X \equiv \mathbb{G} \times_1 V^N \times_2 N, \quad \mathbb{Y} \equiv \mathbb{G} \times_1 V^N, \quad \mathbb{Z} = \mathbb{H} \times_1 V^N \times_2 N,$$

and we denote $\mathbb{Y}^*$ as the tensor resulting when transposing the last two coordinates of $\mathbb{Y}$.

Expanding each individual term, we get:

$$\begin{aligned} X v_2 &= \left(\mathcal{V}^{N,N} \times_1 X \right) \times_3 n \times_2 n + \left(\mathcal{V}^{N,K} \times_1 X \right) \times_3 k \times_2 n \\ &\quad + \left(\mathcal{V}^{K,N} \times_1 X \right) \times_3 n \times_2 k + \left(\mathcal{V}^{K,K} \times_1 X \right) \times_3 k \times_2 k \\ &\quad + 2X V_t^{N,Z} n z + 2X V_t^{K,Z} k z + 2X V_t^{N,T} n + 2X V_t^{K,T} k \\ &\quad + X \bar{v}_t + X v_t^{Z,Z} z^2 + X v_t^{T,T} + 2X v_t^{Z,T} z + X v_t^{\Sigma,Z} z + X v_t^{\Sigma,T}. \end{aligned}$$

For the second term, we get:

$$\begin{aligned} \mathbb{Y} \times_3 v_1 \times_2 n &= \mathbb{Y} \times_3 \left(V^N n + V^K k + v_t^Z z + v_t^T \right) \times_2 n \\ &= \left(\mathbb{Y} \times_3 V^N \right) \times_3 n \times_2 n + \left(\mathbb{Y} \times_3 V^K \right) \times_3 k \times_2 n + \left(\mathbb{Y} \times_3 v_t^Z \right) n z + \left(\mathbb{Y} \times_3 v_t^T \right) n \end{aligned}$$

For the last term, we use the results from the previous section to write:

$$\begin{aligned}
\mathbb{Z} \times_3 v_1 \times_2 v_1 &= \mathbb{Z} \times_3 V^N \times_2 V^N \times_3 n \times_2 n + \mathbb{Z} \times_3 V^K \times_2 V^N \times_3 k \times_2 n \\
&+ \mathbb{Z} \times_3 V^N \times_2 V^K \times_3 n \times_2 k + \mathbb{Z} \times_3 V^K \times_2 V^K \times_3 k \times_2 k \\
&+ 2 \left(\mathbb{Z} \times_2 V^N \times_3 v_t^Z \right) nz + 2 \left(\mathbb{Z} \times_2 V^N \times_3 v_t^T \right) n \\
&+ 2 \left(\mathbb{Z} \times_2 V^K \times_3 v_t^Z \right) kz + 2 \left(\mathbb{Z} \times_2 V^K \times_3 v_t^T \right) k \\
&+ \mathbb{Z} \times_3 v_t^Z \times_2 v_t^Z z^2 + \mathbb{Z} \times_3 v_t^T \times_2 v_t^T + 2\mathbb{Z} \times_3 v_t^Z \times_2 v_t^T z
\end{aligned}$$

Collecting all terms, we get:

$$\begin{aligned}
&\frac{1}{\epsilon^2} \sum_j \frac{\partial V_i}{\partial N_j} (M^*(V)N)_j \\
&= \frac{1}{2} \left\{ \mathcal{V}^{N,N} \times_2 F^N + \mathcal{V}^{N,N} \times_3 F^N + \mathcal{V}^{N,N} \times_1 X + \mathbb{Y}^* \times_2 V^N + \mathbb{Y} \times_3 V^N + \mathbb{Z} \times_3 V^N \times_2 V^N \right\} \times_3 n \times_2 n \\
&+ \frac{1}{2} \left\{ \mathcal{V}^{N,N} \times_3 F^K + \mathcal{V}^{N,K} \times_2 F^N + \mathcal{V}^{N,K} \times_1 X + \mathbb{Y} \times_3 V^K + \mathbb{Z} \times_3 V^K \times_2 V^N \right\} \times_3 k \times_2 n \\
&+ \frac{1}{2} \left\{ \mathcal{V}^{N,N} \times_2 F^K + \mathcal{V}^{K,N} \times_3 F^N + \mathcal{V}^{K,N} \times_1 X + \mathbb{Y}^* \times_2 V^K + \mathbb{Z} \times_3 V^N \times_2 V^K \right\} \times_3 n \times_2 k \\
&+ \frac{1}{2} \left\{ \mathcal{V}^{N,K} \times_2 F^K + \mathcal{V}^{K,N} \times_3 F^K + \mathcal{V}^{K,K} \times_1 X + \mathbb{Z} \times_3 V^K \times_2 V^K \right\} \times_3 k \times_2 k \\
&+ \left[\frac{1}{2} \mathcal{V}^{N,N} \times_2 f_t^Z + \frac{1}{2} \mathcal{V}^{N,N} \times_3 f_t^Z + V_t^{N,Z} F^N + X V_t^{N,Z} + \mathbb{Y} \times_3 v_t^Z + \mathbb{Z} \times_2 V^N \times_3 v_t^Z \right] nz \\
&+ \left[\frac{1}{2} \mathcal{V}^{N,N} \times_2 f_t^T + \frac{1}{2} \mathcal{V}^{N,N} \times_3 f_t^T + V_t^{N,T} F^N + X V_t^{N,T} + \mathbb{Y} \times_3 v_t^T + \mathbb{Z} \times_2 V^N \times_3 v_t^T \right] n \\
&+ \left[\frac{1}{2} \mathcal{V}^{N,K} \times_2 f_t^Z + \frac{1}{2} \mathcal{V}^{K,N} \times_3 f_t^Z + V_t^{N,Z} F^K + X V_t^{K,Z} + \mathbb{Z} \times_2 V^K \times_3 v_t^Z \right] kz \\
&+ \left[\frac{1}{2} \mathcal{V}^{N,K} \times_2 f_t^T + \frac{1}{2} \mathcal{V}^{K,N} \times_3 f_t^T + V_t^{N,T} F^K + X V_t^{K,T} + \mathbb{Z} \times_2 V^K \times_3 v_t^T \right] k \\
&+ \frac{1}{2} \left[2V_t^{N,Z} f_t^Z + X v_t^{Z,Z} + \mathbb{Z} \times_3 v_t^Z \times_2 v_t^Z \right] z^2 \\
&+ \frac{1}{2} \left[2V_t^{N,T} f_t^T + X v_t^{T,T} + \mathbb{Z} \times_3 v_t^T \times_2 v_t^T \right] \\
&+ \left[V_t^{N,Z} f_t^T + V_t^{N,T} f_t^Z + X v_t^{Z,T} + \mathbb{Z} \times_3 v_t^Z \times_2 v_t^T \right] z \\
&+ \frac{1}{2} X v_t^{\Sigma,Z} z + \frac{1}{2} X v_t^{\Sigma,T} \\
&+ \left[\frac{1}{2} X \bar{v}_t \right]
\end{aligned}$$

E.3.2 Capitalists

For capitalists, the continuation value from changes in the population distribution is given by:

$$\mu \sum_j \frac{\partial Q_i}{\partial N_j} \left(\sum_k m_{kj}(V) N_k - N_j \right)$$

Similarly to the workers case, (15), (16), and (18) imply that:

$$\frac{\partial Q_i}{\partial N_j} = Q_{ij}^N + \frac{\epsilon}{2} \left(\sum_s \mathcal{Q}_{ijs}^{N,N} n_s + \sum_j \mathcal{Q}_{ijs}^{N,N} n_j + \sum_s \mathcal{Q}_{ijs}^{N,K} k_s + \sum_j \mathcal{Q}_{ijs}^{K,N} k_j + 2Q_{ijt}^{N,Z} z + 2Q_{ijt}^{N,T} \right)$$

Following the derivations for the worker's case, we get that to the leading order:

$$\begin{aligned} & \sum_j \frac{\partial Q_i}{\partial N_j} (M^*(V)N)_j = \\ & \sum_j \left\{ Q_{ij}^N + \frac{\epsilon}{2} \left(\sum_s \mathcal{Q}_{ijs}^{N,N} n_s + \sum_j \mathcal{Q}_{ijs}^{N,N} n_j + \sum_s \mathcal{Q}_{ijs}^{N,K} k_s + \sum_j \mathcal{Q}_{ijs}^{K,N} k_j + 2Q_{ijt}^{N,Z} z + 2Q_{ijt}^{N,T} \right) \right\} \\ & \quad \times \left\{ \epsilon \mathcal{M}_j + \epsilon^2 \mathcal{N}_j \right\}, \\ & = \sum_j \left\{ Q_{ij}^N + \frac{\epsilon}{2} \partial \mathcal{Q}_j \right\} \left\{ \epsilon \mathcal{M}_j + \epsilon^2 \mathcal{N}_j \right\} \end{aligned}$$

First, the term $\partial \mathcal{Q}_j \times \mathcal{M}_j$ is analogous to the worker's case. Second, we get that:

$$\begin{aligned} \sum_j Q_{ij}^N \epsilon^2 \mathcal{N}_j &= \epsilon^2 \left\{ \frac{1}{2} \left(\mathbb{G} \times_1 Q^N \times_2 N \right) v_2 + \mathbb{G} \times_1 Q^N \times_3 v_1 \times_2 n + \frac{1}{2} \left(\mathbb{H} \times_1 Q^N \times_2 N \right) \times_3 v_1 \times_2 v_1 \right\}, \\ &= \epsilon^2 \left\{ \frac{1}{2} S v_2 + \mathbb{P} \times_3 v_1 \times_2 n + \frac{1}{2} \mathbb{R} \times_3 v_1 \times_2 v_1 \right\}, \end{aligned}$$

with the matrix S and the tensors $\mathbb{P}$ and $\mathbb{R}$ defined as

$$S \equiv \mathbb{G} \times_1 Q^N \times_2 N, \quad \mathbb{P} \equiv \mathbb{G} \times_1 Q^N, \quad \mathbb{R} = \mathbb{H} \times_1 Q^N \times_2 N.$$

We denote by $\mathbb{P}^*$ the tensor one obtains by transposing the last two coordinates of $\mathbb{P}$. Collecting all terms, the second-order term of the capitalist's continuation value from

changes in the population distribution is given by:

$$\begin{aligned}
& \frac{1}{\epsilon^2} \sum_j \frac{\partial Q_i}{\partial N_j} (M^*(V)N)_j \\
&= \frac{1}{2} \left\{ Q^{N,N} \times_2 F^N + Q^{N,N} \times_3 F^N + \mathcal{V}^{N,N} \times_1 S + \mathbb{P}^* \times_2 V^N + \mathbb{P} \times_3 V^N + \mathbb{R} \times_3 V^N \times_2 V^N \right\} \times_3 n \times_2 n \\
&+ \frac{1}{2} \left\{ Q^{N,N} \times_3 F^K + Q^{N,K} \times_2 F^N + \mathcal{V}^{N,K} \times_1 S + \mathbb{P} \times_3 V^K + \mathbb{R} \times_3 V^K \times_2 V^N \right\} \times_3 k \times_2 n \\
&+ \frac{1}{2} \left\{ Q^{N,N} \times_2 F^K + Q^{K,N} \times_3 F^N + \mathcal{V}^{K,N} \times_1 S + \mathbb{P}^* \times_2 V^K + \mathbb{R} \times_3 V^N \times_2 V^K \right\} \times_3 n \times_2 k \\
&+ \frac{1}{2} \left\{ Q^{N,K} \times_2 F^K + Q^{K,N} \times_3 F^K + \mathcal{V}^{K,K} \times_1 S + \mathbb{R} \times_3 V^K \times_2 V^K \right\} \times_3 k \times_2 k \\
&+ \left[\frac{1}{2} Q^{N,N} \times_2 f_t^Z + \frac{1}{2} Q^{N,N} \times_3 f_t^Z + Q_t^{N,Z} F^N + S V_t^{N,Z} + \mathbb{P} \times_3 v_t^Z + \mathbb{R} \times_2 V^N \times_3 v_t^Z \right] n z \\
&+ \left[\frac{1}{2} Q^{N,N} \times_2 f_t^T + \frac{1}{2} Q^{N,N} \times_3 f_t^T + Q_t^{N,T} F^N + S V_t^{N,T} + \mathbb{P} \times_3 v_t^T + \mathbb{R} \times_2 V^N \times_3 v_t^T \right] n \\
&+ \left[\frac{1}{2} Q^{N,K} \times_2 f_t^Z + \frac{1}{2} Q^{K,N} \times_3 f_t^Z + Q_t^{N,Z} F^K + S V_t^{K,Z} + \mathbb{R} \times_2 V^K \times_3 v_t^Z \right] k z \\
&+ \left[\frac{1}{2} Q^{N,K} \times_2 f_t^T + \frac{1}{2} Q^{K,N} \times_3 f_t^T + Q_t^{N,T} F^K + S V_t^{K,T} + \mathbb{R} \times_2 V^K \times_3 v_t^T \right] k \\
&+ \frac{1}{2} \left[2 Q_t^{N,Z} f_t^Z + S v_t^{Z,Z} + \mathbb{R} \times_3 v_t^Z \times_2 v_t^Z \right] z^2 \\
&+ \frac{1}{2} \left[2 Q_t^{N,T} f_t^T + S v_t^{T,T} + \mathbb{R} \times_3 v_t^T \times_2 v_t^T \right] \\
&+ \left[Q_t^{N,Z} f_t^T + Q_t^{N,T} f_t^Z + S v_t^{Z,T} + \mathbb{R} \times_3 v_t^Z \times_2 v_t^T \right] z \\
&+ \frac{1}{2} S v_t^{\Sigma,Z} z + \frac{1}{2} S v_t^{\Sigma,T} \\
&+ \frac{1}{2} [S \bar{v}_t]
\end{aligned}$$

E.4 Continuation Value from Changes in the Capital Distribution

E.4.1 Workers

Worker's continuation value from changes in the capital distribution is given by:

$$\sum_j \frac{\partial V_i}{\partial \mathcal{K}_j} \left(R_{jt} - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right)$$

We first work on approximating the law of motion for capital. Up to second-order we

get:

$$\begin{aligned}
\frac{dk_t}{dt} = & \mathbf{D}_{RK}k_t + \mathbf{D}_{RN}n_t + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbf{D}_{IQ}q_{1,t}, \\
& + \frac{\epsilon}{2} \left\{ \mathbb{D}_{RNN} \times_3 n_t \times_2 n_t + \mathbb{D}_{RNK} \times_3 k_t \times_2 n_t + \mathbb{D}_{RKN} \times_3 n_t \times_2 n_k + \mathbb{D}_{RKK} \times_3 k_t \times_2 k_t \right. \\
& + 2\mathbf{D}_{RNT}n_t(\sigma_t z + T_t) + 2\mathbf{D}_{RKT}k_t(\sigma_t z + T_t) + \mathbf{d}_{ITT}(\sigma_t z + T_t)^2 + \Sigma^2 \mathbf{d}_{IST}\zeta_t \\
& \left. + \mathbf{D}_{IQ}q_{2,t} + \mathbf{D}_{IQQ}(q_{1,t} \cdot q_{1,t}) \right\},
\end{aligned} \tag{51}$$

where:

$$\begin{aligned}
\mathbb{D}_{RNN} &= \psi(\psi - 1) \mathbf{diag}_3 \left(\frac{R_i^{ss}}{(N_i^{ss})^2} \right), \quad \mathbb{D}_{RKK} = \phi^2 \mathbf{diag}_3 (R_i^{ss}), \quad \mathbb{D}_{RNK} = \mathbb{D}_{RKN} = -\phi\psi \mathbf{diag}_3 \left(\frac{R_i^{ss}}{N_i^{ss}} \right) \\
\mathbf{D}_{RNT} &= \psi \mathbf{diag} \left(\chi_{i1} \frac{R_i^{ss}}{N_i^{ss}} \right), \quad \mathbf{D}_{RKT} = -\phi \mathbf{diag} (\chi_{i1} R_i^{ss}), \quad \mathbf{D}_{IQQ} = -\frac{1+\zeta}{\zeta} \mathbf{diag} \left((Q_i^{ss})^{-\frac{1}{\zeta}-2} \right), \\
\mathbf{d}_{ITT} &= \mathbf{vec} \left((\chi_{i1}^2 + \chi_{i2}) R_i^{ss} - \delta_{i2} \right), \quad \mathbf{d}_{IST} = -\frac{1}{2} \mathbb{1},
\end{aligned}$$

where $\mathbb{1}$ is a vector with ones in all of its entries. Now, turning to the value, we we have that:

$$\frac{\partial V_i}{\partial \mathcal{K}_j} = V_{ij}^K + \frac{\epsilon}{2} \underbrace{\left(\sum_s \mathcal{V}_{ijs}^{K,K} k_s + \sum_s \mathcal{V}_{isj}^{K,K} k_s + \sum_s \mathcal{V}_{isj}^{N,K} n_s + \sum_s \mathcal{V}_{ijs}^{K,N} n_s + 2V_{ijt}^{K,Z} z + 2V_{ijt}^{K,T} \right)}_{\equiv \partial_j V_i}.$$

Hence, we get:

$$\begin{aligned}
& \frac{1}{\epsilon} \sum_j \frac{\partial V_i}{\partial \mathcal{K}_j} \left(R_{jt} - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right) \\
&= \sum_j \left(V_{ij}^K + \frac{\epsilon}{2} \partial_j V_i \right) [\mathbb{D}_{RK}k + \mathbb{D}_{RN}n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ}q_1]_j \\
&+ \sum_j \left(V_{ij}^K + \frac{\epsilon}{2} \partial_j V_i \right) \left(\frac{\epsilon}{2} \left[\mathbb{D}_{RNN} \times_3 n_t \times_2 n_t + \mathbb{D}_{RNC} \times_3 k_t \times_2 n_t + \mathbb{D}_{RCN} \times_3 n_t \times_2 n_k + \mathbb{D}_{RKC} \times_3 k_t \right. \right. \\
&\quad \left. \left. + 2\mathbb{D}_{RNT}n_t(\sigma_t z + T_t) + 2\mathbb{D}_{RKT}k_t(\sigma_t z + T_t) + \mathbf{d}_{ITT}(\sigma_t z + T_t)^2 + \Sigma^2 \mathbf{d}_{IST}\zeta_t \right. \right. \\
&\quad \left. \left. + \mathbb{D}_{IQ}q_{2,t} + \mathbb{D}_{IQQ}(q_{1,t} \cdot q_{1,t}) \right]_j \right) \\
&= \sum_j V_{ij}^K [\mathbb{D}_{RK}k + \mathbb{D}_{RN}n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ}q_1]_j \\
&+ \frac{\epsilon}{2} \sum_j \partial_j V_i [\mathbb{D}_{RK}k + \mathbb{D}_{RN}n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ}q_1]_j \\
&+ \frac{\epsilon}{2} \sum_j V_{ij}^K \left[\mathbb{D}_{RNN} \times_3 n_t \times_2 n_t + \mathbb{D}_{RNC} \times_3 k_t \times_2 n_t + \mathbb{D}_{RCN} \times_3 n_t \times_2 n_k + \mathbb{D}_{RKC} \times_3 k_t \times_2 k_t \right. \\
&\quad \left. + 2\mathbb{D}_{RNT}n_t(\sigma_t z + T_t) + 2\mathbb{D}_{RKT}k_t(\sigma_t z + T_t) + \mathbf{d}_{ITT}(\sigma_t z + T_t)^2 + \Sigma^2 \mathbf{d}_{IST}\zeta_t \right. \\
&\quad \left. + \mathbb{D}_{IQ}q_{2,t} + \mathbb{D}_{IQQ}(q_{1,t} \cdot q_{1,t}) \right]_j,
\end{aligned}$$

where we drop the terms with an order higher than two. The first-order term is given by the first line:

$$\sum_j V_{ij}^K [\mathbb{D}_{RK}k + \mathbb{D}_{RN}n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ}q_1]_i = V^K \left\{ \mathbb{D}_{RK}k + \mathbb{D}_{RN}n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ}q_1 \right\}$$

Working on the second line, we get that

$$\begin{aligned}
& \sum_j \partial_j V_i [\mathbf{D}_{RK}k + \mathbf{D}_{RN}n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbf{D}_{IQ}q_1]_j \\
&= \sum_j \left(\sum_s \mathcal{V}_{ijs}^{K,K} k_s + \sum_s \mathcal{V}_{isj}^{K,K} k_s + \sum_s \mathcal{V}_{isj}^{N,K} n_s + \sum_s \mathcal{V}_{ijs}^{K,N} n_s + 2V_{ijt}^{K,Z} z + 2V_{ijt}^{K,T} \right) \\
&\quad \times (\mathbf{D}_{RK}k + \mathbf{D}_{RN}n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbf{D}_{IQ}q_1)_j \\
&= \left(\mathcal{V}^{K,K} \times_2 \mathbf{D}_{RK} \right) \times_3 k \times_2 k + \left(\mathcal{V}^{K,K} \times_3 \mathbf{D}_{RK} \right) \times_3 k \times_2 k + \left(\mathcal{V}^{N,K} \times_3 \mathbf{D}_{RK} \right) \times_3 k \times_2 n \\
&\quad + \left(\mathcal{V}^{K,N} \times_2 \mathbf{D}_{RK} \right) \times_3 n \times_2 k + 2V_t^{K,Z} \mathbf{D}_{RK} k z + 2V_t^{K,T} \mathbf{D}_{RK} k \\
&\quad + \left(\mathcal{V}^{K,K} \times_2 \mathbf{D}_{RN} \right) \times_3 k \times_2 n + \left(\mathcal{V}^{K,K} \times_3 \mathbf{D}_{RN} \right) \times_3 n \times_2 k + \left(\mathcal{V}^{N,K} \times_3 \mathbf{D}_{RN} \right) \times_3 n \times_2 n \\
&\quad + \left(\mathcal{V}^{K,N} \times_2 \mathbf{D}_{RN} \right) \times_3 n \times_2 n + 2V_t^{K,Z} \mathbf{D}_{RN} n z + 2V_t^{K,T} \mathbf{D}_{RN} n \\
&\quad + \left(\mathcal{V}^{K,K} \times_2 \mathbf{d}_{IT} \right) k(\sigma_t z + T_t) + \left(\mathcal{V}^{K,K} \times_3 \mathbf{d}_{IT} \right) k(\sigma_t z + T_t) + \left(\mathcal{V}^{N,K} \times_3 \mathbf{d}_{IT} \right) n(\sigma_t z + T_t) \\
&\quad + \left(\mathcal{V}^{K,N} \times_2 \mathbf{d}_{IT} \right) n(\sigma_t z + T_t) + 2V_t^{K,Z} \mathbf{d}_{IT} z(\sigma_t z + T_t) + 2V_t^{K,T} \mathbf{d}_{IT}(\sigma_t z + T_t) \\
&\quad + \left(\mathcal{V}^{K,K} \times_2 \mathbf{D}_{IQ} \right) \times_3 k \times_2 q_1 + \left(\mathcal{V}^{K,K} \times_3 \mathbf{D}_{IQ} \right) \times_3 q_1 \times_2 k + \left(\mathcal{V}^{N,K} \times_3 \mathbf{D}_{IQ} \right) \times_3 q_1 \times_2 n \\
&\quad + \left(\mathcal{V}^{K,N} \times_2 \mathbf{D}_{IQ} \right) \times_3 n \times_2 q_1 + 2V_t^{K,Z} \mathbf{D}_{IQ} q_1 z + 2V_t^{K,T} \mathbf{D}_{IQ} q_1.
\end{aligned}$$

We work on expanding the last two lines which involve the term q_1 . Define the matrices:

$$D^N \equiv \mathbf{D}_{IQ} Q^N, \quad D^K \equiv \mathbf{D}_{IQ} Q^K, \quad D_t^Z \equiv \mathbf{D}_{IQ} q_t^Z, \quad D_t^T \equiv \mathbf{D}_{IQ} q_t^T.$$

Using this notation, expanding each of the sub-terms gives:

$$\begin{aligned}
(\mathcal{V}^{K,K} \times_2 \mathbf{D}_{IQ}) \times_3 k \times_2 q_1 &= (\mathcal{V}^{K,K} \times_2 D^N) \times_3 k \times_2 n + (\mathcal{V}^{K,K} \times_2 D^K) \times_3 k \times_2 k \\
&\quad + (\mathcal{V}^{K,K} \times_2 D_t^Z) kz + (\mathcal{V}^{K,K} \times_2 D_t^T) k. \\
(\mathcal{V}^{K,K} \times_3 \mathbf{D}_{IQ}) \times_3 q_1 \times_2 k &= (\mathcal{V}^{K,K} \times_3 D^N) \times_3 n \times_2 k + (\mathcal{V}^{K,K} \times_3 D^K) \times_3 k \times_2 k \\
&\quad + (\mathcal{V}^{K,K} \times_3 D_t^Z) kz + (\mathcal{V}^{K,K} \times_3 D_t^T) k. \\
(\mathcal{V}^{N,K} \times_3 \mathbf{D}_{IQ}) \times_3 q_1 \times_2 n &= (\mathcal{V}^{N,K} \times_3 D^N) \times_3 n \times_2 n + (\mathcal{V}^{N,K} \times_3 D^K) \times_3 k \times_2 n \\
&\quad + (\mathcal{V}^{N,K} \times_3 D_t^Z) nz + (\mathcal{V}^{N,K} \times_3 D_t^T) n. \\
(\mathcal{V}^{K,N} \times_2 \mathbf{D}_{IQ}) \times_3 n \times_2 q_1 &= (\mathcal{V}^{K,N} \times_2 D^N) \times_3 n \times_2 n + (\mathcal{V}^{K,N} \times_2 D^K) \times_3 n \times_2 k \\
&\quad + (\mathcal{V}^{K,N} \times_2 D_t^Z) nz + (\mathcal{V}^{K,N} \times_2 D_t^T) n. \\
2V_t^{K,Z} \mathbf{D}_{IQ} q_1 z &= 2 \left(V_t^{K,Z} D^N nz + V_t^{K,Z} D^K kz + V_t^{K,Z} D_t^Z z^2 + V_t^{K,Z} D_t^T z \right) \\
2V_t^{K,T} \mathbf{D}_{IQ} q_1 &= 2 \left(V_t^{K,T} D^N n + V_t^{K,T} D^K k + V_t^{K,T} D_t^Z z + V_t^{K,T} D_t^T \right)
\end{aligned}$$

Finally, we work on the last line of the initial expansion.

$$\begin{aligned}
&\sum_j V_{ij}^K \left[\mathbb{D}_{RNN} \times_3 n \times_2 n + \mathbb{D}_{RNK} \times_3 k \times_2 n + \mathbb{D}_{RKN} \times_3 n \times_2 k + \mathbb{D}_{RKK} \times_3 k \times_2 k \right. \\
&\quad + 2\mathbb{D}_{RNT} n(\sigma_t z + T_t) + 2\mathbb{D}_{RKT} k(\sigma_t z + T_t) + \mathbf{d}_{ITT}(\sigma_t z + T_t)^2 + \Sigma^2 \mathbf{d}_{IST} \zeta_t \\
&\quad \left. + \mathbf{D}_{IQ} q_{2,t} + \mathbf{D}_{IQQ}(q_{1,t} \cdot q_{1,t}) \right]_j, \\
&= (\mathbb{D}_{RNN} \times_1 V^K) \times_3 n \times_2 n + (\mathbb{D}_{RNK} \times_1 V^K) \times_3 k \times_2 n + (\mathbb{D}_{RKN} \times_1 V^K) \times_3 n \times_2 k \\
&\quad + (\mathbb{D}_{RKK} \times_1 V^K) \times_3 k \times_2 k + 2V^K \mathbb{D}_{RNT} n_t(\sigma_t z + T_t) + 2V^K \mathbb{D}_{RKT} k_t(\sigma_t z + T_t) \\
&\quad + V^K \mathbf{d}_{ITT}(\sigma_t z + T_t)^2 + V^K \Sigma^2 \mathbf{d}_{IST} \zeta_t + V^K \mathbf{D}_{IQ} q_{2,t} + V^K \mathbf{D}_{IQQ}(q_{1,t} \cdot q_{1,t})
\end{aligned}$$

Then, the term involving second-order shocks can be expressed as:

$$V^K \mathbf{d}_{ITT}(\sigma_t z + T_t)^2 = V^K \mathbf{d}_{ITT}(\sigma_t z)^2 + V^K \mathbf{d}_{ITT}(T_t)^2 + 2V^K \mathbf{d}_{ITT} \sigma_t z T_t.$$

Define the matrix $A \equiv V^K D_{IQ}$. By expanding the term involving q_2 we obtain:

$$\begin{aligned}
Aq_2 = & \left(\mathcal{Q}^{N,N} \times_1 A \right) \times_3 n \times_2 n + \left(\mathcal{Q}^{N,K} \times_1 A \right) \times_3 k \times_2 n \\
& + \left(\mathcal{Q}^{K,N} \times_1 A \right) \times_3 n \times_2 k + \left(\mathcal{Q}^{K,K} \times_1 A \right) \times_3 k \times_2 k \\
& + 2AQ_t^{N,Z}nz + 2AQ_t^{N,T}n \\
& + 2AQ_t^{K,Z}kz + 2AQ_t^{K,T}k \\
& + A\bar{q}_t + Aq_t^{Z,Z}z^2 + Aq_t^{T,T} + 2Aq_t^{Z,T}z + Aq_t^\Sigma.
\end{aligned}$$

Now, define the matrix $B \equiv V^K D_{QQ}$. We expand the term involving $q_1 \cdot q_1$ as:

$$\begin{aligned}
B(q_1 \cdot q_1) = & \left(\mathbb{Q}^{N,N} \times_1 B \right) \times_3 n \times_2 n + \left(\mathbb{Q}^{N,K} \times_1 B \right) \times_3 k \times_2 n \\
& + \left(\mathbb{Q}^{K,N} \times_1 B \right) \times_3 n \times_2 k + \left(\mathbb{Q}^{K,K} \times_1 B \right) \times_3 k \times_2 k \\
& + B \left(q_t^Z \cdot q_t^Z \right) z^2 + B \left(q_t^T \cdot q_t^T \right) + 2B \left(q_t^Z \cdot q_t^T \right) z \\
& + 2B\mathbf{diag}(q_t^Z)Q^Nnz + 2B\mathbf{diag}(q_t^T)Q^Nn \\
& + 2B\mathbf{diag}(q_t^Z)Q^Kkz + 2B\mathbf{diag}(q_t^T)Q^Kk.
\end{aligned}$$

By collecting all terms, we construct the second-order approximation of the worker's continuation value from changes in capital distribution. Given that $\mathcal{V}^{K,K}$ is symmetric in the last two coordinates, we obtain:

$$\begin{aligned}
\sum_j \frac{\partial V_i}{\partial \mathcal{K}_j} \left(R_{jt} - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right) &= \epsilon V^K \left\{ \mathbf{D}_{RK} k + \mathbf{D}_{RN} n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbf{D}_{IQ} \left[Q^N n + Q^K k + q_t^Z z + q_t^T \right] \right\} \\
&= \frac{1}{2} \left\{ \mathcal{V}^{N,K} \times_3 \mathbf{D}_{RN} + \mathcal{V}^{K,N} \times_2 \mathbf{D}_{RN} + \mathcal{V}^{N,K} \times_3 D^N + \mathcal{V}^{K,N} \times_2 D^N + \mathbf{D}_{RNN} \times_1 V^K + \mathcal{Q}^{N,N} \times_1 A + \mathbb{Q}^{N,N} \times_1 B \right\} \times_3 n \times_2 n \\
&+ \frac{1}{2} \left\{ \mathcal{V}^{N,K} \times_3 \mathbf{D}_{RK} + \mathcal{V}^{K,K} \times_2 \mathbf{D}_{RN} + \mathcal{V}^{K,K} \times_2 D^N + \mathcal{V}^{N,K} \times_3 D^K + \mathbf{D}_{RNK} \times_1 V^K + \mathcal{Q}^{N,K} \times_1 A + \mathbb{Q}^{N,K} \times_1 B \right\} \times_3 k \times_2 n \\
&+ \frac{1}{2} \left\{ \mathcal{V}^{K,N} \times_2 \mathbf{D}_{RK} + \mathcal{V}^{K,K} \times_3 \mathbf{D}_{RN} + \mathcal{V}^{K,K} \times_3 D^N + \mathcal{V}^{K,N} \times_2 D^K + \mathbf{D}_{RKN} \times_1 V^K + \mathcal{Q}^{K,N} \times_1 A + \mathbb{Q}^{K,N} \times_1 B \right\} \times_3 n \times_2 k \\
&+ \frac{1}{2} \left\{ \mathcal{V}^{K,K} \times_2 \mathbf{D}_{RK} + \mathcal{V}^{K,K} \times_3 \mathbf{D}_{RK} + \mathcal{V}^{K,K} \times_2 D^K + \mathcal{V}^{K,K} \times_3 D^K + \mathbf{D}_{RKK} \times_1 V^K + \mathcal{Q}^{K,K} \times_1 A + \mathbb{Q}^{K,K} \times_1 B \right\} \times_3 k \times_2 k \\
&+ \left[V_t^{K,Z} \mathbf{D}_{RN} + \frac{1}{2} \left[\left(\mathcal{V}^{N,K} \times_3 \mathbf{d}_{IT} + \mathcal{V}^{K,N} \times_2 \mathbf{d}_{IT} \right) \sigma_t + \mathcal{V}^{N,K} \times_3 D_t^Z + \mathcal{V}^{K,N} \times_2 D_t^Z \right] + V_t^{K,Z} D^N + V^K \mathbf{D}_{RNT} \sigma_t + A Q_t^{N,Z} + B \mathbf{diag}(q_t^Z) \right] \\
&+ \left[V_t^{K,T} \mathbf{D}_{RN} + \frac{1}{2} \left[\left(\mathcal{V}^{N,K} \times_3 \mathbf{d}_{IT} + \mathcal{V}^{K,N} \times_2 \mathbf{d}_{IT} \right) T_t + \mathcal{V}^{N,K} \times_3 D_t^T + \mathcal{V}^{K,N} \times_2 D_t^T \right] + V_t^{K,T} D^N + V^K \mathbf{D}_{RNT} T_t + A Q_t^{N,T} + B \mathbf{diag}(q_t^T) \right] \\
&+ \left[V_t^{K,Z} \mathbf{D}_{RK} + \frac{1}{2} \left[\left(\mathcal{V}^{K,K} \times_2 \mathbf{d}_{IT} + \mathcal{V}^{K,K} \times_3 \mathbf{d}_{IT} \right) \sigma_t + \mathcal{V}^{K,K} \times_2 D_t^Z + \mathcal{V}^{K,K} \times_3 D_t^Z \right] + V_t^{K,Z} D^K + V^K \mathbf{D}_{RKT} \sigma_t + A Q_t^{K,Z} + B \mathbf{diag}(q_t^Z) \right] \\
&+ \left[V_t^{K,T} \mathbf{D}_{RK} + \frac{1}{2} \left[\left(\mathcal{V}^{K,K} \times_2 \mathbf{d}_{IT} + \mathcal{V}^{K,K} \times_3 \mathbf{d}_{IT} \right) T_t + \mathcal{V}^{K,K} \times_2 D_t^T + \mathcal{V}^{K,K} \times_3 D_t^T \right] + V_t^{K,T} D^K + V^K \mathbf{D}_{RKT} T_t + A Q_t^{K,T} + B \mathbf{diag}(q_t^T) \right] \\
&+ \frac{1}{2} \left[2V_t^{K,Z} \mathbf{d}_{IT} \sigma_t + 2V_t^{K,Z} D_t^Z + V^K \mathbf{d}_{ITT} \sigma_t^2 + A q_t^{Z,Z} + B \left(q_t^Z \cdot q_t^Z \right) \right] z^2 \\
&+ \frac{1}{2} \left[2V_t^{K,T} \mathbf{d}_{IT} T_t + 2V_t^{K,T} D_t^T + V^K \mathbf{d}_{ITT} (T_t)^2 + A q_t^{T,T} + B \left(q_t^T \cdot q_t^T \right) \right] \\
&+ \left[V_t^{K,Z} \mathbf{d}_{IT} T_t + V_t^{K,T} \mathbf{d}_{IT} \sigma_t + V_t^{K,Z} D_t^T + V_t^{K,T} D_t^Z + V^K \mathbf{d}_{ITT} \sigma_t T_t + A q_t^{Z,T} + B \left(q_t^Z \cdot q_t^T \right) \right] z \\
&+ \frac{1}{2} \left[V^K \Sigma^2 \mathbf{d}_{I\Sigma T} \zeta_t + A q_t^\Sigma \right] \\
&+ \frac{1}{2} [A \bar{q}_t]
\end{aligned}$$

E.4.2 Capitalists

Capitalists continuation value from changes in the capital distribution is:

$$\sum_j \frac{\partial Q_i}{\partial \mathcal{K}_j} \left(R_{jt} - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right).$$

The derivative of the capitalists' value with respect to capital is given by

$$\frac{\partial Q_i}{\partial \mathcal{K}_j} = Q_{ij}^K + \frac{\epsilon}{2} \underbrace{\left(\sum_s \mathcal{Q}_{ijs}^{K,K} k_s + \sum_s \mathcal{Q}_{isj}^{K,K} k_s + \sum_s \mathcal{Q}_{isj}^{N,K} n_s + \sum_s \mathcal{Q}_{ijs}^{K,N} n_s + 2Q_{ijt}^{K,Z} z + 2Q_{ijt}^{K,T} \right)}_{\equiv \partial_j Q_i}$$

Following the derivations for the worker's case, up to second order, we get that

$$\begin{aligned} & \frac{1}{\epsilon} \sum_j \frac{\partial Q_i}{\partial \mathcal{K}_j} \left(R_{jt} - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right) \\ &= \sum_j Q_{ij}^K [\mathbb{D}_{RK} k + \mathbb{D}_{RN} n + \mathbb{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ} q_1]_j \\ &+ \frac{\epsilon}{2} \sum_j \partial_j Q_i [\mathbb{D}_{RK} k + \mathbb{D}_{RN} n + \mathbb{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ} q_1]_j \\ &+ \frac{\epsilon}{2} \sum_j Q_{ij}^K \left[\mathbb{D}_{RNN} \times_3 n_t \times_2 n_t + \mathbb{D}_{RNC} \times_3 k_t \times_2 n_t + \mathbb{D}_{RCN} \times_3 n_t \times_2 n_k + \mathbb{D}_{RCK} \times_3 k_t \times_2 k_t \right. \\ &\quad \left. + 2\mathbb{D}_{RNT} n_t (\sigma_t z + T_t) + 2\mathbb{D}_{RKT} k_t (\sigma_t z + T_t) + \mathbb{d}_{ITT} (\sigma_t z + T_t)^2 + \Sigma^2 \mathbb{d}_{IST} \zeta_t \right. \\ &\quad \left. + \mathbb{D}_{IQ} q_{2,t} + \mathbb{D}_{IQQ} (q_{1,t} \cdot q_{1,t}) \right]_j. \end{aligned}$$

Defining the matrices

$$D \equiv Q^K \mathbb{D}_{IQ}, \quad E \equiv Q^K \mathbb{D}_{IQQ},$$

we can make the analogous derivations as in the worker's case to get that, up to the second order, the capitalist's continuation value from changes in the capital distribution is given by:

$$\begin{aligned}
& \sum_j \frac{\partial Q_i}{\partial \mathcal{K}_j} \left(R_{jt} - Q_{jt}^{-1/\zeta} - \Delta_{jt} - \frac{\Sigma_{jt}^2}{2} \right) = \epsilon Q^K \left\{ \mathbf{D}_{RK} k + \mathbf{D}_{RN} n + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbf{D}_{IQ} \left[Q^N n + Q^K k + q_t^Z z + q_t^T \right] \right\} \\
&= \frac{1}{2} \left\{ \mathcal{Q}^{N,K} \times_3 \mathbf{D}_{RN} + \mathcal{Q}^{K,N} \times_2 \mathbf{D}_{RN} + \mathcal{Q}^{N,K} \times_3 D^N + \mathcal{Q}^{K,N} \times_2 D^N + \mathbf{D}_{RNN} \times_1 Q^K + \mathcal{Q}^{N,N} \times_1 D + \mathcal{Q}^{N,N} \times_1 E \right\} \times_3 n \times_2 n \\
&+ \frac{1}{2} \left\{ \mathcal{Q}^{N,K} \times_3 \mathbf{D}_{RK} + \mathcal{Q}^{K,K} \times_2 \mathbf{D}_{RN} + \mathcal{Q}^{K,K} \times_2 D^N + \mathcal{Q}^{N,K} \times_3 D^K + \mathbf{D}_{RNK} \times_1 Q^K + \mathcal{Q}^{N,K} \times_1 D + \mathcal{Q}^{N,K} \times_1 E \right\} \times_3 k \times_2 n \\
&+ \frac{1}{2} \left\{ \mathcal{Q}^{K,N} \times_2 \mathbf{D}_{RK} + \mathcal{Q}^{K,K} \times_3 \mathbf{D}_{RN} + \mathcal{Q}^{K,K} \times_3 D^N + \mathcal{Q}^{K,N} \times_2 D^K + \mathbf{D}_{RKN} \times_1 Q^K + \mathcal{Q}^{K,N} \times_1 D + \mathcal{Q}^{K,N} \times_1 E \right\} \times_3 n \times_2 k \\
&+ \frac{1}{2} \left\{ \mathcal{Q}^{K,K} \times_2 \mathbf{D}_{RK} + \mathcal{Q}^{K,K} \times_3 \mathbf{D}_{RK} + \mathcal{Q}^{K,K} \times_2 D^K + \mathcal{Q}^{K,K} \times_3 D^K + \mathbf{D}_{RKK} \times_1 Q^K + \mathcal{Q}^{K,K} \times_1 D + \mathcal{Q}^{K,K} \times_1 E \right\} \times_3 k \times_2 k \\
&+ \left[Q_t^{K,Z} \mathbf{D}_{RN} + \frac{1}{2} \left[\left(\mathcal{Q}^{N,K} \times_3 \mathbf{d}_{IT} + \mathcal{Q}^{K,N} \times_2 \mathbf{d}_{IT} \right) \sigma_t + \mathcal{Q}^{N,K} \times_3 D_t^Z + \mathcal{Q}^{K,N} \times_2 D_t^Z \right] + Q_t^{K,Z} D^N + Q^K \mathbf{D}_{RNT} \sigma_t + D Q_t^{N,Z} + E \mathbf{diag}(q_t^Z) \right] z \\
&+ \left[Q_t^{K,T} \mathbf{D}_{RN} + \frac{1}{2} \left[\left(\mathcal{Q}^{N,K} \times_3 \mathbf{d}_{IT} + \mathcal{Q}^{K,N} \times_2 \mathbf{d}_{IT} \right) T_t + \mathcal{Q}^{N,K} \times_3 D_t^T + \mathcal{Q}^{K,N} \times_2 D_t^T \right] + Q_t^{K,T} D^N + Q^K \mathbf{D}_{RNT} T_t + D Q_t^{N,T} + E \mathbf{diag}(q_t^T) \right] z \\
&+ \left[Q_t^{K,Z} \mathbf{D}_{RK} + \frac{1}{2} \left[\left(\mathcal{Q}^{K,K} \times_2 \mathbf{d}_{IT} + \mathcal{Q}^{K,K} \times_3 \mathbf{d}_{IT} \right) \sigma_t + \mathcal{Q}^{K,K} \times_2 D_t^Z + \mathcal{Q}^{K,K} \times_3 D_t^Z \right] + Q_t^{K,Z} D^K + Q^K \mathbf{D}_{RKT} \sigma_t + D Q_t^{K,Z} + E \mathbf{diag}(q_t^Z) \right] z \\
&+ \left[Q_t^{K,T} \mathbf{D}_{RK} + \frac{1}{2} \left[\left(\mathcal{Q}^{K,K} \times_2 \mathbf{d}_{IT} + \mathcal{Q}^{K,K} \times_3 \mathbf{d}_{IT} \right) T_t + \mathcal{Q}^{K,K} \times_2 D_t^T + \mathcal{Q}^{K,K} \times_3 D_t^T \right] + Q_t^{K,T} D^K + Q^K \mathbf{D}_{RKT} T_t + D Q_t^{K,T} + E \mathbf{diag}(q_t^T) \right] z \\
&+ \frac{1}{2} \left[2 Q_t^{K,Z} \mathbf{d}_{IT} \sigma_t + 2 Q_t^{K,Z} D_t^Z + Q^K \mathbf{d}_{ITT} \sigma_t^2 + D q_t^{Z,Z} + E \left(q_t^Z \cdot q_t^Z \right) \right] z^2 \\
&+ \frac{1}{2} \left[2 Q_t^{K,T} \mathbf{d}_{IT} T_t + 2 Q_t^{K,T} D_t^T + Q^K \mathbf{d}_{ITT} (T_t)^2 + D q_t^{T,T} + E \left(q_t^T \cdot q_t^T \right) \right] z^2 \\
&+ \left[Q_t^{K,Z} \mathbf{d}_{IT} T_t + Q_t^{K,T} \mathbf{d}_{IT} \sigma_t + Q_t^{K,Z} D_t^T + Q_t^{K,T} D_t^Z + Q^K \mathbf{d}_{ITT} \sigma_t T_t + D q_t^{Z,T} + E \left(q_t^Z \cdot q_t^T \right) \right] z \\
&+ \frac{1}{2} \left[Q^K \Sigma^2 \mathbf{d}_{ITS} \zeta_t + D q_t^\Sigma \right] \\
&+ \frac{1}{2} [D \bar{q}_t]
\end{aligned}$$

F Continuation Value from Climate Shocks

Recall that we impose z to follow an AR(1) process with mean-reversion parameter θ and unit variance.

F.1 Workers

Omitting the first-order terms, we have the following:

$$\begin{aligned}\mathcal{A}(z)[V] &= \frac{\epsilon^2}{2} \mathcal{A}(z)[v_2] \\ &= \frac{\epsilon^2}{2} \left\{ \mathcal{A}(z)[2V_t^{N,Z}nz] + \mathcal{A}(z)[2V_t^{K,Z}kz] + \mathcal{A}(z)[v_t^{Z,Z}z^2] + \mathcal{A}(z)[2v_t^{Z,T}z] \right\} \\ &= -\epsilon^2\theta \left\{ V_t^{N,Z}nz + V_t^{K,Z}kz + v_t^{Z,Z}z^2 + v_t^{Z,T}z \right\} + \frac{\epsilon^2}{2}v_t^{Z,Z}.\end{aligned}$$

F.2 Capitalists

Similarly, ignoring the first-order terms, we have that:

$$\begin{aligned}\mathcal{A}(z)[Q] &= \frac{\epsilon^2}{2} \mathcal{A}(z)[q_2] \\ &= \frac{\epsilon^2}{2} \left\{ \mathcal{A}(z)[2Q_t^{N,Z}nz] + \mathcal{A}(z)[2Q_t^{K,Z}kz] + \mathcal{A}(z)[q_t^{Z,Z}z^2] + \mathcal{A}(z)[2q_t^{Z,T}z] \right\} \\ &= -\epsilon^2\theta \left\{ Q_t^{N,Z}nz + Q_t^{K,Z}kz + q_t^{Z,Z}z^2 + q_t^{Z,T}z \right\} + \frac{\epsilon^2}{2}q_t^{Z,Z}.\end{aligned}$$

G SAME

G.1 SAME in matrix notation

Due to the length of the equations, we omit the full expression of the linearized master equation. Applying the method of identifying coefficients, we get the following equations for the second-order terms:

G.1.1 Workers

Block 1: N & K

$$\begin{aligned}
\rho \mathfrak{V}^{N,N} &= \mathbb{D}_{UNN} + \mu\nu \left(\mathbb{V}^{N,N} \times_1 m + \mathbb{M} \times_3 V^N \times_2 V^N \right) + \mathbb{Y}^* \times_2 V^N + \mathbb{Y} \times_3 V^N + \mathbb{Z} \times_3 V^N \times_2 V^N \\
&+ \mathbb{D}_{RNN} \times_1 V^K + \mathbb{Q}^{N,N} \times_1 B \\
&+ \mathfrak{V}^{N,N} \times_1 (M + X) + \mathfrak{V}^{N,N} \times_2 F^N + \mathfrak{V}^{N,N} \times_3 F^N + \mathfrak{V}^{N,K} \times_3 (D_{RN} + D^N) + \mathfrak{V}^{K,N} \times_2 (D_{RN} + D^N) \\
&+ \mathfrak{Q}^{N,N} \times_1 A
\end{aligned}$$

$$\begin{aligned}
\rho \mathfrak{V}^{N,K} &= \mathbb{D}_{UNK} + \mu\nu \left(\mathbb{V}^{N,K} \times_1 m + \mathbb{M} \times_3 V^K \times_2 V^N \right) + \mathbb{Y} \times_3 V^K + \mathbb{Z} \times_3 V^K \times_2 V^N \\
&+ \mathbb{D}_{RNK} \times_1 V^K + \mathbb{Q}^{N,K} \times_1 B \\
&+ \mathfrak{V}^{N,N} \times_3 F^K + \mathfrak{V}^{N,K} \times_1 (M + X) + \mathfrak{V}^{N,K} \times_2 F^N + \mathfrak{V}^{N,K} \times_3 (D_{RK} + D^K) + \mathfrak{V}^{K,K} \times_2 (D_{RN} + D^N) \\
&+ \mathfrak{Q}^{N,K} \times_1 A
\end{aligned}$$

$$\begin{aligned}
\rho \mathfrak{V}^{K,N} &= \mathbb{D}_{UKN} + \mu\nu \left(\mathbb{V}^{K,N} \times_1 m + \mathbb{M} \times_3 V^N \times_2 V^K \right) + \mathbb{Y}^* \times_2 V^K + \mathbb{Z} \times_3 V^N \times_2 V^K \\
&+ \mathbb{D}_{RKN} \times_1 V^K + \mathbb{Q}^{K,N} \times_1 B \\
&+ \mathfrak{V}^{N,N} \times_2 F^K + \mathfrak{V}^{K,N} \times_1 (M + X) + \mathfrak{V}^{K,N} \times_3 F^N + \mathfrak{V}^{K,N} \times_2 (D_{RK} + D^K) + \mathfrak{V}^{K,K} \times_3 (D_{RN} + D^N) \\
&+ \mathfrak{Q}^{K,N} \times_1 A
\end{aligned}$$

$$\begin{aligned}
\rho \mathfrak{V}^{K,K} &= \mathbb{D}_{UKK} + \mu\nu \left(\mathbb{V}^{K,K} \times_1 m + \mathbb{M} \times_3 V^K \times_2 V^K \right) + \mathbb{Z} \times_3 V^K \times_2 V^K \\
&+ \mathbb{D}_{RKK} \times_1 V^K + \mathbb{Q}^{K,K} \times_1 B \\
&+ \mathfrak{V}^{K,K} \times_1 (M + X) + \mathfrak{V}^{K,K} \times_2 (D_{RK} + D^K) + \mathfrak{V}^{K,K} \times_3 (D_{RK} + D^K) + \mathfrak{V}^{N,K} \times_2 F^K + \mathfrak{V}^{K,N} \times_3 F^K \\
&+ \mathfrak{Q}^{K,K} \times_1 A
\end{aligned}$$

Block 2: N & σ_t^Z , and T_t

$$\begin{aligned}
& (\rho + \theta) \mathbf{V}_t^{N,Z} - \frac{\mathbf{V}_t^{N,Z}}{\partial t} \\
&= \mathbf{D}_{UNT} \sigma_t + \mu \nu \left(m \mathbf{diag}(v_t^Z) V^N + \mathbb{M} \times_2 V^N \times_3 v_t^Z \right) + \mathbb{Y} \times_3 v_t^Z + \mathbb{Z} \times_2 V^N \times_3 v_t^Z \\
&+ V^K \mathbf{D}_{RNT} \sigma_t + B \mathbf{diag}(q_t^Z) Q^N \\
&+ \frac{1}{2} \left[\mathbf{V}^{N,N} \times_2 f_t^Z + \mathbf{V}^{N,N} \times_3 f_t^Z + \mathbf{V}^{N,K} \times_3 D_t^Z + \mathbf{V}^{K,N} \times_2 D_t^Z + \left(\mathbf{V}^{N,K} \times_3 \mathbf{d}_{IT} + \mathbf{V}^{K,N} \times_2 \mathbf{d}_{IT} \right) \right] \\
&+ (M + X) \mathbf{V}_t^{N,Z} + \mathbf{V}_t^{N,Z} F^N + A \mathbf{Q}_t^{N,Z} + \mathbf{V}_t^{K,Z} (\mathbf{D}_{RN} + D^N)
\end{aligned}$$

$$\begin{aligned}
& \rho \mathbf{V}_t^{N,T} - \frac{\mathbf{V}_t^{N,T}}{\partial t} \\
&= \mathbf{D}_{UNT} T_t + \mu \nu \left(m \mathbf{diag}(v_t^T) V^N + \mathbb{M} \times_2 V^N \times_3 v_t^T \right) + \mathbb{Y} \times_3 v_t^T + \mathbb{Z} \times_2 V^N \times_3 v_t^T \\
&+ V^K \mathbf{D}_{RNT} T_t + B \mathbf{diag}(q_t^T) Q^N \\
&+ \frac{1}{2} \left[\mathbf{V}^{N,N} \times_2 f_t^T + \mathbf{V}^{N,N} \times_3 f_t^T + \mathbf{V}^{N,K} \times_3 D_t^T + \mathbf{V}^{K,N} \times_2 D_t^T + \left(\mathbf{V}^{N,K} \times_3 \mathbf{d}_{IT} + \mathbf{V}^{K,N} \times_2 \mathbf{d}_{IT} \right) \right] \\
&+ (M + X) \mathbf{V}_t^{N,T} + \mathbf{V}_t^{N,T} F^N + A \mathbf{Q}_t^{N,T} + \mathbf{V}_t^{K,T} (\mathbf{D}_{RN} + D^N)
\end{aligned}$$

Block 3: K & σ_t^Z , and T_t

$$\begin{aligned}
& (\rho + \theta) \mathbf{V}_t^{K,Z} - \frac{\partial \mathbf{V}_t^{K,Z}}{\partial t} \\
&= \mathbf{D}_{UKT} \sigma_t + \mu \nu \left(m \mathbf{diag}(v_t^Z) V^K + \mathbb{M} \times_2 V^K \times_3 v_t^Z \right) + \mathbb{Z} \times_2 V^K \times_3 v_t^Z + V^K \mathbf{D}_{RKT} \sigma_t + B \mathbf{diag}(q_t^Z) Q^K \\
&+ \frac{1}{2} \left[\mathbf{V}^{N,K} \times_2 f_t^Z + \mathbf{V}^{K,N} \times_3 f_t^Z + \mathbf{V}^{K,K} \times_2 D_t^Z + \mathbf{V}^{K,K} \times_3 D_t^Z + \left(\mathbf{V}^{K,K} \times_2 \mathbf{d}_{IT} + \mathbf{V}^{K,K} \times_3 \mathbf{d}_{IT} \right) \sigma_t \right] \\
&+ (M + X) \mathbf{V}_t^{K,Z} + \mathbf{V}_t^{N,Z} F^K + A \mathbf{Q}_t^{K,Z} + \mathbf{V}_t^{K,Z} (\mathbf{D}_{RK} + D^K)
\end{aligned}$$

$$\begin{aligned}
& \rho \mathbf{V}_t^{K,T} - \frac{\partial \mathbf{V}_t^{K,T}}{\partial t} \\
&= \mathbf{D}_{UKT} T_t + \mu \nu \left(m \mathbf{diag}(v_t^T) V^K + \mathbb{M} \times_2 V^K \times_3 v_t^T \right) + \mathbb{Z} \times_2 V^K \times_3 v_t^T + V^K \mathbf{D}_{RKT} T_t + B \mathbf{diag}(q_t^T) Q^K \\
&+ \frac{1}{2} \left[\mathbf{V}^{N,K} \times_2 f_t^T + \mathbf{V}^{K,N} \times_3 f_t^T + \mathbf{V}^{K,K} \times_2 D_t^T + \mathbf{V}^{K,K} \times_3 D_t^T + \left(\mathbf{V}^{K,K} \times_2 \mathbf{d}_{IT} + \mathbf{V}^{K,K} \times_3 \mathbf{d}_{IT} \right) T_t \right] \\
&+ (M + X) \mathbf{V}_t^{K,T} + \mathbf{V}_t^{N,T} F^K + A \mathbf{Q}_t^{K,T} + \mathbf{V}_t^{K,T} (\mathbf{D}_{RK} + D^K)
\end{aligned}$$

Block 4: $(\sigma_t^Z)^2$, $(T_t)^2$, and $(\sigma_t^Z)T_t$

$$\begin{aligned}
(\rho + 2\theta) \mathbf{v}_t^{Z,Z} - \frac{\partial \mathbf{v}_t^{Z,Z}}{\partial t} &= \mathbf{d}_{UTT} \sigma_t^2 + \mu\nu \left(m(v_t^Z \cdot v_t^Z) + \mathbb{M} \times_3 v_t^Z \times_2 v_t^Z \right) + \mathbb{Z} \times_3 v_t^Z \times_2 v_t^Z + B \left(q_t^Z \cdot q_t^Z \right) + V^K \mathbf{d}_{ITT} \sigma_t^2 \\
&+ 2 \left[\mathbf{V}_t^{N,Z} f_t^Z + \mathbf{V}_t^{K,Z} \left(D_t^Z + \mathbf{d}_{IT} \sigma_t \right) \right] \\
&+ (M + X) \mathbf{v}_t^{Z,Z} + A \mathbf{q}_t^{Z,Z}
\end{aligned}$$

$$\begin{aligned}
\rho \mathbf{v}_t^{T,T} - \frac{\partial \mathbf{v}_t^{T,T}}{\partial t} &= \mathbf{d}_{UTT} (T_t)^2 + \mu\nu \left(m(v_t^T \cdot v_t^T) + \mathbb{M} \times_3 v_t^T \times_2 v_t^T \right) + \mathbb{Z} \times_3 v_t^T \times_2 v_t^T + B \left(q_t^T \cdot q_t^T \right) + V^K \mathbf{d}_{ITT} (T_t)^2 \\
&+ 2 \left[\mathbf{V}_t^{N,T} f_t^T + \mathbf{V}_t^{K,T} \left(D_t^T + \mathbf{d}_{IT} T_t \right) \right] \\
&+ (M + X) \mathbf{v}_t^{T,T} + A \mathbf{q}_t^{T,T}
\end{aligned}$$

$$\begin{aligned}
(\rho + \theta) \mathbf{v}_t^{Z,T} - \frac{\partial \mathbf{v}_t^{Z,T}}{\partial t} &= \mathbf{d}_{UTT} \sigma_t T_t + \mu\nu \left(m(v_t^Z \cdot v_t^T) + \mathbb{M} \times_3 v_t^Z \times_2 v_t^T \right) + \mathbb{Z} \times_3 v_t^Z \times_2 v_t^T + B \left(q_t^Z \cdot q_t^T \right) + V^K \mathbf{d}_{ITT} \sigma_t T_t \\
&+ \mathbf{V}_t^{N,Z} f_t^T + \mathbf{V}_t^{N,T} f_t^Z + \mathbf{V}_t^{K,Z} \left(D_t^T + \mathbf{d}_{IT} T_t \right) + \mathbf{V}_t^{K,T} \left(D_t^Z + \mathbf{d}_{IT} \sigma_t \right) \\
&+ (M + X) \mathbf{v}_t^{Z,T} + A \mathbf{q}_t^{Z,T}
\end{aligned}$$

Block 6: $\Sigma_i^2 \zeta_t$

$$\rho \mathbf{v}_t^\Sigma - \frac{\partial \mathbf{v}_t^\Sigma}{\partial t} = V^K \Sigma^2 \mathbf{d}_{\Sigma T} \zeta_t + (M + X) \mathbf{v}_t^\Sigma + A \mathbf{q}_t^\Sigma$$

Block 7: Constant

$$\rho \bar{\mathbf{v}}_t - \frac{\partial \bar{\mathbf{v}}_t}{\partial t} = \mathbf{v}_t^{Z,Z} + (M + X) \bar{\mathbf{v}}_t + A \bar{\mathbf{q}}_t$$

G.1.2 Capitalists

Block 1: N & K

$$\begin{aligned}
\rho \mathcal{Q}^{N,N} &= \mathbb{D}_{QNN} + \mathbb{Q}^{N,N} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQN} \times_1 Q^N + \mathbb{P}^* \times_2 V^N + \mathbb{P} \times_3 V^N + \mathbb{R} \times_3 V^N \times_2 V^N \\
&+ \mathbb{D}_{RNN} \times_1 Q^K + \mathbb{Q}^{N,N} \times_1 E \\
&+ \mathcal{Q}^{N,N} \times_1 D + \mathcal{Q}^{N,N} \times_2 F^N + \mathcal{Q}^{N,N} \times_3 F^N + \mathcal{Q}^{N,K} \times_3 (\mathbb{D}_{RN} + D^N) + \mathcal{Q}^{K,N} \times_2 (\mathbb{D}_{RN} + D^N) \\
&+ \mathcal{V}^{N,N} \times_1 S
\end{aligned}$$

$$\begin{aligned}
\rho \mathcal{Q}^{N,K} &= \mathbb{D}_{QKN} + \mathbb{Q}^{N,K} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQN} \times_1 Q^K + \mathbb{P} \times_3 V^K + \mathbb{R} \times_3 V^K \times_2 V^N \\
&+ \mathbb{D}_{RKN} \times_1 Q^K + \mathbb{Q}^{N,K} \times_1 E \\
&+ \mathcal{Q}^{N,N} \times_3 F^K + \mathcal{Q}^{N,K} \times_1 D + \mathcal{Q}^{N,K} \times_2 F^N + \mathcal{Q}^{N,K} \times_3 (\mathbb{D}_{RK} + D^K) + \mathcal{Q}^{K,K} \times_2 (\mathbb{D}_{RN} + D^N) \\
&+ \mathcal{V}^{N,K} \times_1 S
\end{aligned}$$

$$\begin{aligned}
\rho \mathcal{Q}^{K,N} &= \mathbb{D}_{QNK} + \mathbb{Q}^{K,N} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQK} \times_1 Q^N + \mathbb{P}^* \times_2 V^K + \mathbb{R} \times_3 V^N \times_2 V^K \\
&+ \mathbb{D}_{RKN} \times_1 Q^K + \mathbb{Q}^{K,N} \times_1 E \\
&+ \mathcal{Q}^{N,N} \times_2 F^K + \mathcal{Q}^{K,N} \times_1 D + \mathcal{Q}^{K,N} \times_3 F^N + \mathcal{Q}^{K,N} \times_2 (\mathbb{D}_{RK} + D^K) + \mathcal{Q}^{K,K} \times_3 (\mathbb{D}_{RN} + D^N) \\
&+ \mathcal{V}^{K,N} \times_1 S
\end{aligned}$$

$$\begin{aligned}
\rho \mathcal{Q}^{K,K} &= \mathbb{D}_{QKK} + \mathbb{Q}^{K,K} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQK} \times_1 Q^K + \mathbb{R} \times_3 V^K \times_2 V^K \\
&+ \mathbb{D}_{RKK} \times_1 Q^K + \mathbb{Q}^{K,K} \times_1 E \\
&+ \mathcal{Q}^{K,K} \times_1 D + \mathcal{Q}^{K,K} \times_2 (\mathbb{D}_{RK} + D^K) + \mathcal{Q}^{K,K} \times_3 (\mathbb{D}_{RK} + D^K) + \mathcal{Q}^{N,K} \times_2 F^K + \mathcal{Q}^{K,N} \times_3 F^K \\
&+ \mathcal{V}^{K,K} \times_1 S
\end{aligned}$$

Block 2: N & σ_t^Z , and T_t

$$\begin{aligned}
& (\rho + \theta) \mathcal{Q}_t^{N,Z} - \frac{\mathcal{Q}_t^{N,Z}}{\partial t} \\
&= \mathbb{D}_{QQQ} \mathbf{diag}(q_t^Z) Q^N + \left(\mathbb{D}_{QNT} + \mathbb{D}_{QQT} Q^N \right) \sigma_t + \mathbb{D}_{QQN} \times_1 q_t^Z + \mathbb{P} \times_3 v_t^Z + \mathbb{R} \times_2 V^N \times_3 v_t^Z \\
&+ Q^K \mathbb{D}_{RNT} \sigma_t + E \mathbf{diag}(q_t^Z) Q^N \\
&+ \frac{1}{2} \left[\mathcal{Q}^{N,N} \times_2 f_t^Z + \mathcal{Q}^{N,N} \times_3 f_t^Z + \mathcal{Q}^{N,K} \times_3 D_t^Z + \mathcal{Q}^{K,N} \times_2 D_t^Z + \left(\mathcal{Q}^{N,K} \times_3 \mathbf{d}_{IT} + \mathcal{Q}^{K,N} \times_2 \mathbf{d}_{IT} \right) \sigma_t \right] \\
&+ D \mathcal{Q}_t^{N,Z} + \mathcal{Q}_t^{N,Z} F^N + S \mathbf{V}_t^{N,Z} + \mathcal{Q}_t^{K,Z} (\mathbb{D}_{RN} + D^N)
\end{aligned}$$

$$\begin{aligned}
& \rho \mathcal{Q}_t^{N,T} - \frac{\mathcal{Q}_t^{N,T}}{\partial t} \\
&= \mathbb{D}_{QQQ} \mathbf{diag}(q_t^T) Q^N + \left(\mathbb{D}_{QNT} + \mathbb{D}_{QQT} Q^N \right) T_t + \mathbb{D}_{QQN} \times_1 q_t^T + \mathbb{P} \times_3 v_t^T + \mathbb{R} \times_2 V^N \times_3 v_t^T \\
&+ Q^K \mathbb{D}_{RNT} T_t + E \mathbf{diag}(q_t^T) Q^N \\
&+ \frac{1}{2} \left[\mathcal{Q}^{N,N} \times_2 f_t^T + \mathcal{Q}^{N,N} \times_3 f_t^T + \mathcal{Q}^{N,K} \times_3 D_t^T + \mathcal{Q}^{K,N} \times_2 D_t^T + \left(\mathcal{Q}^{N,K} \times_3 \mathbf{d}_{IT} + \mathcal{Q}^{K,N} \times_2 \mathbf{d}_{IT} \right) T_t \right] \\
&+ D \mathcal{Q}_t^{N,T} + \mathcal{Q}_t^{N,T} F^N + S \mathbf{V}_t^{N,T} + \mathcal{Q}_t^{K,T} (\mathbb{D}_{RN} + D^N)
\end{aligned}$$

Block 3: K & σ_t^Z , and T_t

$$\begin{aligned}
& (\rho + \theta) \mathcal{Q}_t^{K,Z} - \frac{\partial \mathcal{Q}_t^{K,Z}}{\partial t} \\
&= \mathbb{D}_{QQQ} \mathbf{diag}(q_t^Z) Q^K + (\mathbb{D}_{QKT} + \mathbb{D}_{QQT} Q^K) \sigma_t + \mathbb{D}_{QQK} \times_1 q_t^Z + \mathbb{R} \times_2 V^K \times_3 v_t^Z \\
&+ Q^K \mathbb{D}_{RKT} \sigma_t + E \mathbf{diag}(q_t^Z) Q^K \\
&+ \frac{1}{2} \left[\mathcal{Q}^{N,K} \times_2 f_t^Z + \mathcal{Q}^{K,N} \times_3 f_t^Z + \mathcal{Q}^{K,K} \times_2 D_t^Z + \mathcal{Q}^{K,K} \times_3 D_t^Z + \left(\mathcal{Q}^{K,K} \times_2 \mathbf{d}_{IT} + \mathcal{Q}^{K,K} \times_3 \mathbf{d}_{IT} \right) \sigma_t \right] \\
&+ D \mathcal{Q}_t^{K,Z} + \mathcal{Q}_t^{N,Z} F^K + S \mathbf{V}_t^{K,Z} + \mathcal{Q}_t^{K,Z} (\mathbb{D}_{RK} + D^K)
\end{aligned}$$

$$\begin{aligned}
& \rho \mathcal{Q}_t^{K,T} - \frac{\partial \mathcal{Q}_t^{K,T}}{\partial t} \\
&= \mathbb{D}_{QQQ} \mathbf{diag}(q_t^T) Q^K + \left(\mathbb{D}_{QK\mathcal{T}} + \mathbb{D}_{QQT} Q^K \right) T_t + \mathbb{D}_{QQ\mathcal{K}} \times_1 q_t^T + \mathbb{R} \times_2 V^K \times_3 v_t^T \\
&+ Q^K \mathbb{D}_{RKT} T_t + E \mathbf{diag}(q_t^T) Q^K \\
&+ \frac{1}{2} \left[\mathcal{Q}_t^{N,K} \times_2 f_t^T + \mathcal{Q}_t^{K,N} \times_3 f_t^T + \mathcal{Q}_t^{K,K} \times_2 D_t^T + \mathcal{Q}_t^{K,K} \times_3 D_t^T + \left(\mathcal{Q}_t^{K,K} \times_2 \mathbf{d}_{IT} + \mathcal{Q}_t^{K,K} \times_3 \mathbf{d}_{IT} \right) T_t \right] \\
&+ D \mathcal{Q}_t^{K,T} + \mathcal{Q}_t^{N,T} F^K + S \mathcal{V}_t^{K,T} + \mathcal{Q}_t^{K,T} (\mathbb{D}_{RK} + D^K)
\end{aligned}$$

Block 4: $(\sigma_t \mathbf{z})^2$, $(T_t)^2$, and $(\sigma_t \mathbf{z}) T_t$

$$\begin{aligned}
& (\rho + 2\theta) q_t^{Z,Z} - \frac{\partial q_t^{Z,Z}}{\partial t} \\
&= \mathbb{D}_{QQQ} (q_t^Z \cdot q_t^Z) + 2\mathbb{D}_{QQT} q_t^Z \sigma_t + \mathbf{d}_{QTT} \sigma_t^2 + \mathbb{R} \times_3 v_t^Z \times_2 v_t^Z + E \left(q_t^Z \cdot q_t^Z \right) + Q^K \mathbf{d}_{ITT} \sigma_t^2 \\
&+ 2 \left[\mathcal{Q}_t^{N,Z} f_t^Z + \mathcal{Q}_t^{K,Z} \left(D_t^Z + \mathbf{d}_{IT} \sigma_t \right) \right] \\
&+ D q_t^{Z,Z} + S \mathcal{V}_t^{Z,Z}
\end{aligned}$$

$$\begin{aligned}
& \rho q_t^{T,T} - \frac{\partial q_t^{T,T}}{\partial t} \\
&= \mathbb{D}_{QQQ} (q_t^T \cdot q_t^T) + 2\mathbb{D}_{QQT} q_t^T T_t + \mathbf{d}_{QTT} (T_t)^2 + \mathbb{R} \times_3 v_t^T \times_2 v_t^T + E \left(q_t^T \cdot q_t^T \right) + Q^K \mathbf{d}_{ITT} (T_t)^2 \\
&+ 2 \left[\mathcal{Q}_t^{N,T} f_t^T + \mathcal{Q}_t^{K,T} \left(D_t^T + \mathbf{d}_{IT} T_t \right) \right] \\
&+ D q_t^{T,T} + S \mathcal{V}_t^{T,T}
\end{aligned}$$

$$\begin{aligned}
& (\rho + \theta) q_t^{Z,T} - \frac{\partial q_t^{Z,T}}{\partial t} \\
&= \mathbb{D}_{QQQ} q_t^Z q_t^T + \mathbb{D}_{QQT} (q_t^Z T_t + q_t^T \sigma_t) + \mathbf{d}_{QTT} \sigma_t T_t + \mathbb{R} \times_3 v_t^Z \times_2 v_t^T + E \left(q_t^Z \cdot q_t^T \right) + Q^K \mathbf{d}_{ITT} \sigma_t T_t \\
&+ \mathcal{Q}_t^{N,Z} f_t^T + \mathcal{Q}_t^{N,T} f_t^Z + \mathcal{Q}_t^{K,Z} \left(D_t^T + \mathbf{d}_{IT} T_t \right) + \mathcal{Q}_t^{K,T} \left(D_t^Z + \mathbf{d}_{IT} \sigma_t \right) \\
&+ D q_t^{Z,T} + S \mathcal{V}_t^{Z,T}
\end{aligned}$$

Block 6: $\Sigma_i^2 \zeta_t$

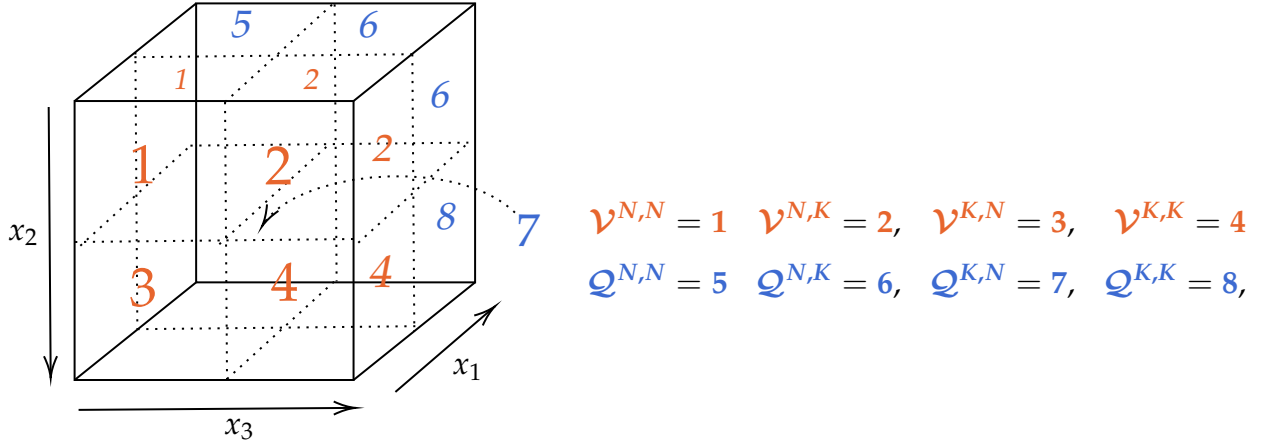
$$\rho \mathbf{q}_t^\Sigma - \frac{\partial \mathbf{q}_t^\Sigma}{\partial t} = \Sigma^2 \mathbf{d}_{Q\Sigma T \zeta_t} + Q^K \Sigma^2 \mathbf{d}_{I\Sigma T \zeta_t} + D \mathbf{q}_t^\Sigma + S \mathbf{v}_t^\Sigma$$

Block 7: Constant

$$\rho \bar{\mathbf{q}}_t - \frac{\partial \bar{\mathbf{q}}_t}{\partial t} = \mathbf{q}_t^{Z,Z} + D \bar{\mathbf{q}}_t + S \bar{\mathbf{v}}_t$$

G.2 Deterministic SAME

We stack the deterministic tensors as follows:



which we will represent as:

$$\mathcal{T} \equiv \underbrace{\begin{bmatrix} \mathbf{v}^{N,N} & \mathbf{v}^{N,K} \\ \mathbf{v}^{K,N} & \mathbf{v}^{K,K} \end{bmatrix}}_{\text{"front panel"}} \odot \underbrace{\begin{bmatrix} \mathbf{Q}^{N,N} & \mathbf{Q}^{N,K} \\ \mathbf{Q}^{K,N} & \mathbf{Q}^{K,K} \end{bmatrix}}_{\text{"back panel"}}$$

Define the tensor $\mathbb{T}_0$ as:

$$\mathbb{T}_0 \equiv \begin{bmatrix} \mathbb{V}_0^{N,N} & \mathbb{V}_0^{N,K} \\ \mathbb{V}_0^{K,N} & \mathbb{V}_0^{K,K} \end{bmatrix} \odot \begin{bmatrix} \mathbb{Q}_0^{N,N} & \mathbb{Q}_0^{N,K} \\ \mathbb{Q}_0^{K,N} & \mathbb{Q}_0^{K,K} \end{bmatrix},$$

with $\mathbb{V}_0^{N,N}$, $\mathbb{V}_0^{N,K}$, $\mathbb{V}_0^{K,N}$, $\mathbb{V}_0^{K,K}$, $\mathbb{Q}_0^{N,N}$, $\mathbb{Q}_0^{N,K}$, $\mathbb{Q}_0^{K,N}$ and $\mathbb{Q}_0^{K,K}$ defined as:

$$\begin{aligned}
\mathbb{V}_0^{N,N} &\equiv \mathbb{D}_{UNN} + \mu\nu \left(\mathbb{V}^{N,N} \times_1 m + \mathbb{M} \times_3 V^N \times_2 V^N \right) + \mathbb{Y}^* \times_2 V^N + \mathbb{Y} \times_3 V^N + \mathbb{Z} \times_3 V^N \times_2 V^N + \mathbb{D}_{RNN} \times_1 V^K \\
&\quad + \mathbb{Q}^{N,N} \times_1 B \\
\mathbb{V}_0^{N,K} &\equiv \mathbb{D}_{UNK} + \mu\nu \left(\mathbb{V}^{N,K} \times_1 m + \mathbb{M} \times_3 V^K \times_2 V^N \right) + \mathbb{Y} \times_3 V^K + \mathbb{Z} \times_3 V^K \times_2 V^N + \mathbb{D}_{RNK} \times_1 V^K + \mathbb{Q}^{N,K} \times_1 B \\
\mathbb{V}_0^{K,N} &\equiv \mathbb{D}_{UKN} + \mu\nu \left(\mathbb{V}^{K,N} \times_1 m + \mathbb{M} \times_3 V^N \times_2 V^K \right) + \mathbb{Y}^* \times_2 V^K + \mathbb{Z} \times_3 V^N \times_2 V^K + \mathbb{D}_{RKN} \times_1 V^K + \mathbb{Q}^{K,N} \times_1 B \\
\mathbb{V}_0^{K,K} &\equiv \mathbb{D}_{UKK} + \mu\nu \left(\mathbb{V}^{K,K} \times_1 m + \mathbb{M} \times_3 V^K \times_2 V^K \right) + \mathbb{Z} \times_3 V^K \times_2 V^K + \mathbb{D}_{RKK} \times_1 V^K + \mathbb{Q}^{K,K} \times_1 B
\end{aligned}$$

$$\begin{aligned}
\mathbb{V}_0^{N,N} &\equiv \mathbb{D}_{QNN} + \mathbb{Q}^{N,N} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQN} \times_1 Q^N + \mathbb{P}^* \times_2 V^N + \mathbb{P} \times_3 V^N + \mathbb{R} \times_3 V^N \times_2 V^N + \mathbb{D}_{RNN} \times_1 Q^K + \mathbb{Q}^{N,N} \times_1 E \\
\mathbb{V}_0^{N,K} &\equiv \mathbb{D}_{QNK} + \mathbb{Q}^{N,K} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQN} \times_1 Q^K + \mathbb{P} \times_3 V^K + \mathbb{R} \times_3 V^K \times_2 V^N + \mathbb{D}_{RNK} \times_1 Q^K + \mathbb{Q}^{N,K} \times_1 E \\
\mathbb{V}_0^{K,N} &\equiv \mathbb{D}_{QKN} + \mathbb{Q}^{K,N} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQK} \times_1 Q^N + \mathbb{P}^* \times_2 V^K + \mathbb{R} \times_3 V^N \times_2 V^K + \mathbb{D}_{RKN} \times_1 Q^K + \mathbb{Q}^{K,N} \times_1 E \\
\mathbb{V}_0^{K,K} &\equiv \mathbb{D}_{QKK} + \mathbb{Q}^{K,K} \times_1 \mathbb{D}_{QQQ} + \mathbb{D}_{QQK} \times_1 Q^K + \mathbb{R} \times_3 V^K \times_2 V^K + \mathbb{D}_{RKK} \times_1 Q^K + \mathbb{Q}^{K,K} \times_1 E
\end{aligned}$$

Recall from the FAME notation that we define the matrix Φ as:

$$\Phi \equiv \begin{pmatrix} M + X & A \\ S & D \end{pmatrix}.$$

Moreover, define the matrix Θ as:

$$\Theta \equiv \begin{pmatrix} F^N & F^K \\ \mathbb{D}_{RN} + D^N & \mathbb{D}_{RK} + D^K \end{pmatrix}.$$

With this notation, the deterministic SAME equations take the form:

$$\begin{aligned}
\rho \mathbf{v}^{N,N} &= \mathbb{V}_0^{N,N} \\
&+ \mathbf{v}^{N,N} \times_1 (M + X) + \mathbf{q}^{N,N} \times_1 A \\
&+ \mathbf{v}^{N,N} \times_2 F^N + \mathbf{v}^{K,N} \times_2 (D_{RN} + D^N) \\
&+ \mathbf{v}^{N,N} \times_3 F^N + \mathbf{v}^{N,K} \times_3 (D_{RN} + D^N) \\
\rho \mathbf{q}^{N,N} &= \mathbb{Q}_0^{N,N} \\
&+ \mathbf{q}^{N,N} \times_1 D + \mathbf{v}^{N,N} \times_1 S \\
&+ \mathbf{q}^{N,N} \times_2 F^N + \mathbf{q}^{K,N} \times_2 (D_{RN} + D^N) \\
&+ \mathbf{q}^{N,N} \times_3 F^N + \mathbf{q}^{N,K} \times_3 (D_{RN} + D^N)
\end{aligned}$$

$$\begin{aligned}
\rho \mathbf{v}^{N,K} &= \mathbb{V}_0^{N,K} \\
&+ \mathbf{v}^{N,K} \times_1 (M + X) + \mathbf{q}^{N,K} \times_1 A \\
&+ \mathbf{v}^{N,K} \times_2 F^N + \mathbf{v}^{K,K} \times_2 (D_{RN} + D^N) \\
&+ \mathbf{v}^{N,N} \times_3 F^K + \mathbf{v}^{N,K} \times_3 (D_{RK} + D^K) \\
\rho \mathbf{q}^{N,K} &= \mathbb{V}_0^{N,K} \\
&+ \mathbf{q}^{N,K} \times_1 D + \mathbf{v}^{N,K} \times_1 S \\
&+ \mathbf{q}^{N,K} \times_2 F^N + \mathbf{q}^{K,K} \times_2 (D_{RN} + D^N) \\
&+ \mathbf{q}^{N,N} \times_3 F^K + \mathbf{q}^{N,K} \times_3 (D_{RK} + D^K)
\end{aligned}$$

$$\begin{aligned}
\rho \mathbf{v}^{K,N} &= \mathbb{V}_0^{K,N} \\
&+ \mathbf{v}^{K,N} \times_1 (M + X) + \mathbf{q}^{K,N} \times_1 A \\
&+ \mathbf{v}^{N,N} \times_2 F^K + \mathbf{v}^{K,N} \times_2 (D_{RK} + D^K) \\
&+ \mathbf{v}^{K,N} \times_3 F^N + \mathbf{v}^{K,K} \times_3 (D_{RN} + D^N) \\
\rho \mathbf{q}^{K,N} &= \mathbb{V}_0^{K,N} \\
&+ \mathbf{q}^{K,N} \times_1 D + \mathbf{v}^{K,N} \times_1 S \\
&+ \mathbf{q}^{N,N} \times_2 F^K + \mathbf{q}^{K,N} \times_2 (D_{RK} + D^K) \\
&+ \mathbf{q}^{K,N} \times_3 F^N + \mathbf{q}^{K,K} \times_3 (D_{RN} + D^N)
\end{aligned}$$

$$\begin{aligned}
\rho \mathbf{v}^{K,K} &= \mathbb{V}_0^{K,K} \\
&+ \mathbf{v}^{K,K} \times_1 (M + X) + \mathbf{q}^{K,K} \times_1 A \\
&+ \mathbf{v}^{K,K} \times_2 (D_{RK} + D^K) + \mathbf{v}^{N,K} \times_2 F^K \\
&+ \mathbf{v}^{K,K} \times_3 (D_{RK} + D^K) + \mathbf{v}^{K,N} \times_3 F^K \\
\rho \mathbf{q}^{K,K} &= \mathbb{V}_0^{K,K} \\
&+ \mathbf{q}^{K,K} \times_1 D + \mathbf{v}^{K,K} \times_1 S \\
&+ \mathbf{q}^{K,K} \times_2 (D_{RK} + D^K) + \mathbf{q}^{N,K} \times_2 F^K \\
&+ \mathbf{q}^{K,K} \times_3 (D_{RK} + D^K) + \mathbf{q}^{K,N} \times_3 F^K
\end{aligned}$$

which can be written compactly as a tensor equation for $\mathcal{T}$ as:

$$\rho \mathcal{T} = \mathbb{T}_0 + \mathcal{T} \times_1 \Phi + \mathcal{T} \times_2 \Theta + \mathcal{T} \times_3 \Theta \quad (52)$$

G.3 Cross SAME: $\sigma_t^Z, T_t, \mathbf{N}$ & $\mathbf{K}$

Define the matrices:

$$\begin{aligned}
\mathbf{V}_{0t}^{N,Z} &= \mathbf{D}_{UNT}\sigma_t + \mu\nu \left(m \mathbf{diag}(v_t^Z) V^N + \mathbb{M} \times_2 V^N \times_3 v_t^Z \right) + \mathbb{Y} \times_3 v_t^Z + \mathbb{Z} \times_2 V^N \times_3 v_t^Z \\
&\quad + V^K \mathbf{D}_{RNT}\sigma_t + B \mathbf{diag}(q_t^Z) Q^N \\
&\quad + \frac{1}{2} \left[\mathbf{v}^{N,N} \times_2 f_t^Z + \mathbf{v}^{N,N} \times_3 f_t^Z + \mathbf{v}^{N,K} \times_3 D_t^Z + \mathbf{v}^{K,N} \times_2 D_t^Z + \left(\mathbf{v}^{N,K} \times_3 \mathbf{d}_{IT} + \mathbf{v}^{K,N} \times_2 \mathbf{d}_{IT} \right) \sigma_t \right] \\
\mathbf{V}_{0t}^{K,Z} &= \mathbf{D}_{UKT}\sigma_t + \mu\nu \left(m \mathbf{diag}(v_t^Z) V^K + \mathbb{M} \times_2 V^K \times_3 v_t^Z \right) + \mathbb{Z} \times_2 V^K \times_3 v_t^Z + V^K \mathbf{D}_{RKT}\sigma_t + B \mathbf{diag}(q_t^Z) Q^K \\
&\quad + \frac{1}{2} \left[\mathbf{v}^{N,K} \times_2 f_t^Z + \mathbf{v}^{K,N} \times_3 f_t^Z + \mathbf{v}^{K,K} \times_2 D_t^Z + \mathbf{v}^{K,K} \times_3 D_t^Z + \left(\mathbf{v}^{K,K} \times_2 \mathbf{d}_{IT} + \mathbf{v}^{K,K} \times_3 \mathbf{d}_{IT} \right) \sigma_t \right] \\
\mathbf{Q}_{0t}^{N,Z} &= \mathbf{D}_{QQQ} \mathbf{diag}(q_t^Z) Q^N + \left(\mathbf{D}_{QNT} + \mathbf{D}_{QQT} Q^N \right) \sigma_t + \mathbb{D}_{QQN} \times_1 q_t^Z + \mathbb{P} \times_3 v_t^Z + \mathbb{R} \times_2 V^N \times_3 v_t^Z \\
&\quad + Q^K \mathbf{D}_{RNT}\sigma_t + E \mathbf{diag}(q_t^Z) Q^N \\
&\quad + \frac{1}{2} \left[\mathbf{q}^{N,N} \times_2 f_t^Z + \mathbf{q}^{N,N} \times_3 f_t^Z + \mathbf{q}^{N,K} \times_3 D_t^Z + \mathbf{q}^{K,N} \times_2 D_t^Z + \left(\mathbf{q}^{N,K} \times_3 \mathbf{d}_{IT} + \mathbf{q}^{K,N} \times_2 \mathbf{d}_{IT} \right) \sigma_t \right] \\
\mathbf{Q}_{0t}^{K,Z} &= \mathbf{D}_{QQQ} \mathbf{diag}(q_t^Z) Q^K + \left(\mathbf{D}_{QKT} + \mathbf{D}_{QQT} Q^K \right) \sigma_t + \mathbb{D}_{QKK} \times_1 q_t^Z + \mathbb{R} \times_2 V^K \times_3 v_t^Z \\
&\quad + Q^K \mathbf{D}_{RKT}\sigma_t + E \mathbf{diag}(q_t^Z) Q^K \\
&\quad + \frac{1}{2} \left[\mathbf{q}^{N,K} \times_2 f_t^Z + \mathbf{q}^{K,N} \times_3 f_t^Z + \mathbf{q}^{K,K} \times_2 D_t^Z + \mathbf{q}^{K,K} \times_3 D_t^Z + \left(\mathbf{q}^{K,K} \times_2 \mathbf{d}_{IT} + \mathbf{q}^{K,K} \times_3 \mathbf{d}_{IT} \right) \sigma_t \right]
\end{aligned}$$

Similarly, define the matrices:

$$\begin{aligned}
\mathbf{V}_{0t}^{N,T} &= \mathbf{D}_{UNT} T_t + \mu\nu \left(m \mathbf{diag}(v_t^T) V^N + \mathbb{M} \times_2 V^N \times_3 v_t^T \right) + \mathbb{Y} \times_3 v_t^T + \mathbb{Z} \times_2 V^N \times_3 v_t^T \\
&\quad + V^K \mathbf{D}_{RNT} T_t + B \mathbf{diag}(q_t^T) Q^N \\
&\quad + \frac{1}{2} \left[\mathbf{V}^{N,N} \times_2 f_t^T + \mathbf{V}^{N,N} \times_3 f_t^T + \mathbf{V}^{N,K} \times_3 D_t^T + \mathbf{V}^{K,N} \times_2 D_t^T + \left(\mathbf{V}^{N,K} \times_3 \mathbf{d}_{IT} + \mathbf{V}^{K,N} \times_2 \mathbf{d}_{IT} \right) T_t \right] \\
\mathbf{V}_{0t}^{K,T} &= \mathbf{D}_{UKT} T_t + \mu\nu \left(m \mathbf{diag}(v_t^T) V^K + \mathbb{M} \times_2 V^K \times_3 v_t^T \right) + \mathbb{Z} \times_2 V^K \times_3 v_t^T + V^K \mathbf{D}_{RKT} T_t + B \mathbf{diag}(q_t^T) Q^K \\
&\quad + \frac{1}{2} \left[\mathbf{V}^{N,K} \times_2 f_t^T + \mathbf{V}^{K,N} \times_3 f_t^T + \mathbf{V}^{K,K} \times_2 D_t^T + \mathbf{V}^{K,K} \times_3 D_t^T + \left(\mathbf{V}^{K,K} \times_2 \mathbf{d}_{IT} + \mathbf{V}^{K,K} \times_3 \mathbf{d}_{IT} \right) T_t \right] \\
\mathbf{Q}_{0t}^{N,T} &= \mathbf{D}_{QQQ} \mathbf{diag}(q_t^T) Q^N + \left(\mathbf{D}_{QNT} + \mathbf{D}_{QQT} Q^N \right) T_t + \mathbb{D}_{QQN} \times_1 q_t^T + \mathbb{P} \times_3 v_t^T + \mathbb{R} \times_2 V^N \times_3 v_t^T \\
&\quad + Q^K \mathbf{D}_{RNT} T_t + E \mathbf{diag}(q_t^T) Q^N \\
&\quad + \frac{1}{2} \left[\mathbf{Q}^{N,N} \times_2 f_t^T + \mathbf{Q}^{N,N} \times_3 f_t^T + \mathbf{Q}^{N,K} \times_3 D_t^T + \mathbf{Q}^{K,N} \times_2 D_t^T + \left(\mathbf{Q}^{N,K} \times_3 \mathbf{d}_{IT} + \mathbf{Q}^{K,N} \times_2 \mathbf{d}_{IT} \right) T_t \right] \\
\mathbf{Q}_{0t}^{K,T} &= \mathbf{D}_{QQQ} \mathbf{diag}(q_t^T) Q^K + \left(\mathbf{D}_{QKT} + \mathbf{D}_{QQT} Q^K \right) T_t + \mathbb{D}_{QQK} \times_1 q_t^T + \mathbb{R} \times_2 V^K \times_3 v_t^T \\
&\quad + Q^K \mathbf{D}_{RKT} T_t + E \mathbf{diag}(q_t^T) Q^K \\
&\quad + \frac{1}{2} \left[\mathbf{Q}^{N,K} \times_2 f_t^T + \mathbf{Q}^{K,N} \times_3 f_t^T + \mathbf{Q}^{K,K} \times_2 D_t^T + \mathbf{Q}^{K,K} \times_3 D_t^T + \left(\mathbf{Q}^{K,K} \times_2 \mathbf{d}_{IT} + \mathbf{Q}^{K,K} \times_3 \mathbf{d}_{IT} \right) T_t \right].
\end{aligned}$$

Then, $\mathbf{V}_t^{N,Z}$, $\mathbf{V}_t^{K,Z}$, $\mathbf{Q}_t^{N,Z}$, $\mathbf{Q}_t^{K,Z}$, $\mathbf{V}_t^{N,T}$, $\mathbf{V}_t^{K,T}$, $\mathbf{Q}_t^{N,T}$, and $\mathbf{Q}_t^{K,T}$ satisfy:

$$\begin{aligned}
(\rho + \theta) \begin{pmatrix} \mathbf{V}_t^{N,Z} & \mathbf{V}_t^{K,Z} \\ \mathbf{Q}_t^{N,Z} & \mathbf{Q}_t^{K,Z} \end{pmatrix} &= \begin{pmatrix} \mathbf{V}_{0t}^{N,Z} & \mathbf{V}_{0t}^{K,Z} \\ \mathbf{Q}_{0t}^{N,Z} & \mathbf{Q}_{0t}^{K,Z} \end{pmatrix} + \Phi \begin{pmatrix} \mathbf{V}_t^{N,Z} & \mathbf{V}_t^{K,Z} \\ \mathbf{Q}_t^{N,Z} & \mathbf{Q}_t^{K,Z} \end{pmatrix} + \begin{pmatrix} \mathbf{V}_t^{N,Z} & \mathbf{V}_t^{K,Z} \\ \mathbf{Q}_t^{N,Z} & \mathbf{Q}_t^{K,Z} \end{pmatrix} \Theta \\
&\quad + \begin{pmatrix} \mathbf{V}_t^{N,Z} & \mathbf{V}_t^{K,Z} \\ \mathbf{Q}_t^{N,Z} & \mathbf{Q}_t^{K,Z} \end{pmatrix} \mathcal{D}_T^* \tag{53} \\
\rho \begin{pmatrix} \mathbf{V}_t^{N,T} & \mathbf{V}_t^{K,T} \\ \mathbf{Q}_t^{N,T} & \mathbf{Q}_t^{K,T} \end{pmatrix} &= \begin{pmatrix} \mathbf{V}_{0t}^{N,T} & \mathbf{V}_{0t}^{K,T} \\ \mathbf{Q}_{0t}^{N,T} & \mathbf{Q}_{0t}^{K,T} \end{pmatrix} + \Phi \begin{pmatrix} \mathbf{V}_t^{N,T} & \mathbf{V}_t^{K,T} \\ \mathbf{Q}_t^{N,T} & \mathbf{Q}_t^{K,T} \end{pmatrix} + \begin{pmatrix} \mathbf{V}_t^{N,T} & \mathbf{V}_t^{K,T} \\ \mathbf{Q}_t^{N,T} & \mathbf{Q}_t^{K,T} \end{pmatrix} \Theta \\
&\quad + \begin{pmatrix} \mathbf{V}_t^{N,T} & \mathbf{V}_t^{K,T} \\ \mathbf{Q}_t^{N,T} & \mathbf{Q}_t^{K,T} \end{pmatrix} \mathcal{D}_T^*
\end{aligned}$$

where $\mathcal{D}_T^*$ is the matrix that discretizes the time derivative.

G.4 Stochastic SAME: $(\sigma_t z)^2, T_t^2, (\sigma_t z)T_t$

Define the vectors

$$\begin{aligned}
\mathbf{v}_{0t}^{Z,Z} &= \mathbf{d}_{UTT}\sigma_t^2 + \mu\nu \left(m(v_t^Z \cdot v_t^Z) + \mathbb{M} \times_3 v_t^Z \times_2 v_t^Z \right) + \mathbb{Z} \times_3 v_t^Z \times_2 v_t^Z + B \left(q_t^Z \cdot q_t^Z \right) + V^K \mathbf{d}_{ITT}\sigma_t^2 \\
&\quad + 2 \left[\mathbf{V}_t^{N,Z} f_t^Z + \mathbf{V}_t^{K,Z} \left(D_t^Z + \mathbf{d}_{IT}\sigma_t \right) \right] \\
\mathbf{v}_{0t}^{T,T} &= \mathbf{d}_{UTT}(T_t)^2 + \mu\nu \left(m(v_t^T \cdot v_t^T) + \mathbb{M} \times_3 v_t^T \times_2 v_t^T \right) + \mathbb{Z} \times_3 v_t^T \times_2 v_t^T + B \left(q_t^T \cdot q_t^T \right) + V^K \mathbf{d}_{ITT}(T_t)^2 \\
&\quad + 2 \left[\mathbf{V}_t^{N,T} f_t^T + \mathbf{V}_t^{K,T} \left(D_t^T + \mathbf{d}_{IT}T_t \right) \right] \\
\mathbf{v}_{0t}^{Z,T} &= \mathbf{d}_{UTT}\sigma_t T_t + \mu\nu \left(m(v_t^Z \cdot v_t^T) + \mathbb{M} \times_3 v_t^Z \times_2 v_t^T \right) + \mathbb{Z} \times_3 v_t^Z \times_2 v_t^T + B \left(q_t^Z \cdot q_t^T \right) + V^K \mathbf{d}_{ITT}\sigma_t T_t \\
&\quad + \mathbf{V}_t^{N,Z} f_t^T + \mathbf{V}_t^{N,T} f_t^Z + \mathbf{V}_t^{K,Z} \left(D_t^T + \mathbf{d}_{IT}T_t \right) + \mathbf{V}_t^{K,T} \left(D_t^Z + \mathbf{d}_{IT}\sigma_t \right) \\
\mathbf{v}_{0t}^\Sigma &= V^K \Sigma^2 \mathbf{d}_{IST}\zeta_t, \\
\mathbf{q}_{0t}^{Z,Z} &= \mathbf{D}_{QQQ} \left(q_t^Z \cdot q_t^Z \right) + 2\mathbf{D}_{QQT}q_t^Z \sigma_t + \mathbf{d}_{QTT}\sigma_t^2 + \mathbb{R} \times_3 v_t^Z \times_2 v_t^Z + E \left(q_t^Z \cdot q_t^Z \right) + Q^K \mathbf{d}_{ITT}\sigma_t^2 \\
&\quad + 2 \left[\mathbf{Q}_t^{N,Z} f_t^Z + \mathbf{Q}_t^{K,Z} \left(D_t^Z + \mathbf{d}_{IT}\sigma_t \right) \right] \\
\mathbf{q}_{0t}^{T,T} &= \mathbf{D}_{QQQ}(q_t^T \cdot q_t^T) + 2\mathbf{D}_{QQT}q_t^T T_t + \mathbf{d}_{QTT}(T_t)^2 + \mathbb{R} \times_3 v_t^T \times_2 v_t^T + E \left(q_t^T \cdot q_t^T \right) + Q^K \mathbf{d}_{ITT}(T_t)^2 \\
&\quad + 2 \left[\mathbf{Q}_t^{N,T} f_t^T + \mathbf{Q}_t^{K,T} \left(D_t^T + \mathbf{d}_{IT}T_t \right) \right] \\
\mathbf{q}_{0t}^{Z,T} &= \mathbf{D}_{QQQ}(q_t^Z \cdot q_t^T) + \mathbf{D}_{QQT}(q_t^Z T_t + q_t^T \sigma_t) + \mathbf{d}_{QTT}\sigma_t T_t + \mathbb{R} \times_3 v_t^Z \times_2 v_t^T + E \left(q_t^Z \cdot q_t^T \right) + Q^K \mathbf{d}_{ITT}\sigma_t T_t \\
&\quad + \mathbf{Q}_t^{N,Z} f_t^T + \mathbf{Q}_t^{N,T} f_t^Z + \mathbf{Q}_t^{K,Z} \left(D_t^T + \mathbf{d}_{IT}T_t \right) + \mathbf{Q}_t^{K,T} \left(D_t^Z + \mathbf{d}_{IT}\sigma_t \right) \\
\mathbf{q}_{0t}^\Sigma &= (\Sigma^2 \mathbf{d}_{QST} + Q^K \Sigma^2 \mathbf{d}_{IST})\zeta_t.
\end{aligned}$$

Then, $v_t^{Z,Z}, q_t^{Z,Z}, v_t^{T,T}, q_t^{T,T}, v_t^{Z,T}, q_t^{Z,T}, v_t^{\Sigma,Z}, v_t^{\Sigma,T}, q_t^{\Sigma,Z}$ and $v_t^{\Sigma,T}$ satisfy:

$$\begin{aligned}
(\rho + 2\theta) \begin{pmatrix} v_t^{Z,Z} \\ q_t^{Z,Z} \end{pmatrix} &= \begin{pmatrix} v_{0t}^{Z,Z} \\ q_{0t}^{Z,Z} \end{pmatrix} + \Phi \begin{pmatrix} v_t^{Z,Z} \\ q_t^{Z,Z} \end{pmatrix} + \begin{pmatrix} v_t^{Z,Z} \\ q_t^{Z,Z} \end{pmatrix} \mathcal{D}_T^* \\
\rho \begin{pmatrix} v_t^{T,T} \\ q_t^{T,T} \end{pmatrix} &= \begin{pmatrix} v_{0t}^{T,T} \\ q_{0t}^{T,T} \end{pmatrix} + \Phi \begin{pmatrix} v_t^{T,T} \\ q_t^{T,T} \end{pmatrix} + \begin{pmatrix} v_t^{T,T} \\ q_t^{T,T} \end{pmatrix} \mathcal{D}_T^* \\
(\rho + \theta) \begin{pmatrix} v_t^{Z,T} \\ q_t^{Z,T} \end{pmatrix} &= \begin{pmatrix} v_{0t}^{Z,T} \\ q_{0t}^{Z,T} \end{pmatrix} + \Phi \begin{pmatrix} v_t^{Z,T} \\ q_t^{Z,T} \end{pmatrix} + \begin{pmatrix} v_t^{Z,T} \\ q_t^{Z,T} \end{pmatrix} \mathcal{D}_T^* \\
\rho \begin{pmatrix} v_t^{\Sigma} \\ q_t^{\Sigma} \end{pmatrix} &= \begin{pmatrix} v_{0t}^{\Sigma} \\ q_{0t}^{\Sigma} \end{pmatrix} + \Phi \begin{pmatrix} v_t^{\Sigma} \\ q_t^{\Sigma} \end{pmatrix} + \begin{pmatrix} v_t^{\Sigma} \\ q_t^{\Sigma} \end{pmatrix} \mathcal{D}_T^*
\end{aligned}$$

G.5 Constant SAME

Given $v_t^{Z,Z}$ and $q_t^{Z,Z}$, the vectors $\bar{v}_t$ and $\bar{q}_t$ satisfy:

$$\rho \begin{pmatrix} \bar{v}_t \\ \bar{q}_t \end{pmatrix} = \begin{pmatrix} v_t^{Z,Z} \\ q_t^{Z,Z} \end{pmatrix} + \Phi \begin{pmatrix} \bar{v}_t \\ \bar{q}_t \end{pmatrix} + \begin{pmatrix} \bar{v}_t \\ \bar{q}_t \end{pmatrix} \mathcal{D}_T^*$$

G.6 Linearized KFEs

On the one hand, the law of motion for the population is given by (50). Noticing that $\frac{dn_{it}}{dt} = \frac{1}{\epsilon} \frac{dN_{it}}{dt}$, $\mathbb{G} \times_2 N = G$, and defining the tensor $\mathbb{I} \equiv \mathbb{H} \times_2 N$, we obtain the following expression:

$$\frac{dn_t}{dt} = M^* n_t + G v_{1,t} + \frac{\epsilon}{2} \{ G v_{2,t} + 2\mathbb{G} \times_3 v_{1,t} \times_2 n_t + \mathbb{I} \times_3 v_{1,t} \times_2 v_{1,t} \} \quad (54)$$

On the other hand, the law of motion for capital is given by (51). Just for convenience, we will write the same expression here:

$$\begin{aligned} \frac{dk_t}{dt} = & \mathbb{D}_{RK}k_t + \mathbb{D}_{RN}n_t + \mathbf{d}_{IT}(\sigma_t z + T_t) + \mathbb{D}_{IQ}q_{1,t}, \\ & + \frac{\epsilon}{2} \left\{ \mathbb{D}_{RNN} \times_3 n_t \times_2 n_t + \mathbb{D}_{RNK} \times_3 k_t \times_2 n_t + \mathbb{D}_{RKN} \times_3 n_t \times_2 k_t + \mathbb{D}_{RKK} \times_3 k_t \times_2 k_t \right. \\ & + 2\mathbb{D}_{RNT}n_t(\sigma_t z + T_t) + 2\mathbb{D}_{RKT}k_t(\sigma_t z + T_t) + \mathbf{d}_{ITT}(\sigma_t z + T_t)^2 + \Sigma^2 \mathbf{d}_{IST}(\sigma_t z + T_t) \\ & \left. + \mathbb{D}_{IQ}q_{2,t} + \mathbb{D}_{IQQ}(q_{1,t} \cdot q_{1,t}) \right\}, \end{aligned}$$

G.7 Approximation of Prices

G.7.1 Wages

From equation (23) we have that:

$$\begin{aligned} w_{it} &= (1 - \alpha)\alpha^\alpha \Xi^{-\alpha} (1 - x)^{\omega\alpha} Z_i L_i^{\omega\alpha} e^{\chi_{it}} (K_{it}^{1-\omega-\omega} N_{it}^{\omega-1})^\alpha, \\ &= w_{i,0} e^{\chi_{it}} (K_{it}^{1-\omega-\omega} N_{it}^{\omega-1})^\alpha. \end{aligned}$$

Therefore, up to second order:

$$\begin{aligned} \frac{w_t - w^{ss}}{\epsilon} = & \mathbb{D}_{wN}n + \mathbb{D}_{wK}k + \mathbf{d}_{wT}(\sigma_t z + T_t) \\ & + \frac{\epsilon}{2} \{ \mathbb{D}_{wNN} \times_3 n \times_2 n + \mathbb{D}_{wNK} \times_3 k \times_2 n + \mathbb{D}_{wKN} \times_3 n \times_2 k + \mathbb{D}_{wKK} \times_3 k \times_2 k \} \\ & + \epsilon \{ \mathbb{D}_{wNT}n(\sigma_t z + T_t) + \mathbb{D}_{wKT}k(\sigma_t z + T_t) \} \\ & + \frac{\epsilon}{2} \mathbf{d}_{wTT}(\sigma_t z + T_t)^2. \end{aligned}$$

where $\mathbb{D}_{wNN}$, $\mathbb{D}_{wKK}$, $\mathbb{D}_{wNK}$, $\mathbb{D}_{wKN}$, $\mathbb{D}_{wNT}$, $\mathbb{D}_{wKT}$, and $\mathbf{d}_{wTT}$ are given by

$$\begin{aligned} \mathbb{D}_{wNN} &= \alpha(1 - \omega) [\alpha(1 - \omega) + 1] \mathbf{diag}_3 \left(\frac{w_i^{ss}}{(N_i^{ss})^2} \right) & \mathbb{D}_{wNT} &= -\alpha(1 - \omega) \mathbf{diag} \left(\chi_{i1} \frac{w_i^{ss}}{N_i^{ss}} \right) \\ \mathbb{D}_{wKK} &= \alpha(1 - \omega - \omega) [\alpha(1 - \omega - \omega) - 1] \mathbf{diag}_3 (w_i^{ss}) & \mathbb{D}_{wKT} &= \alpha(1 - \omega - \omega) \mathbf{diag} (\chi_{i1} w_i^{ss}) \\ \mathbb{D}_{wNK} &= \mathbb{D}_{wKN} = -\alpha^2(1 - \omega)(1 - \omega - \omega) \mathbf{diag}_3 \left(\frac{w_i^{ss}}{N_i^{ss}} \right) & \mathbf{d}_{wTT} &= \mathbf{vec} \left(w_i^{ss} (\chi_{i1}^2 + \chi_{i2}) \right). \end{aligned}$$

The matrices D_{wN} , D_{wK} and the vector d_{wT} are defined as in Section D.7.

G.7.2 Capital's Rental Rate

The return to capital is given by equation (25):

$$R_{it} = R_{0i} e^{\chi_{it}} K_{it}^{-\phi} N_{it}^{\psi}$$

Hence, up to a second-order:

$$\begin{aligned} \frac{R_t - R^{ss}}{\epsilon} &= D_{RN}n + D_{RK}k + d_{RT}(\sigma_t z + T_t) \\ &+ \frac{\epsilon}{2} \{ \mathbb{D}_{RNN} \times_3 n \times_2 n + \mathbb{D}_{RNK} \times_3 k \times_2 n + \mathbb{D}_{RKN} \times_3 n \times_2 k + \mathbb{D}_{RKK} \times_3 k \times_2 k \} \\ &+ \epsilon \{ D_{RNT}n(\sigma_t z + T_t) + D_{RKT}k(\sigma_t z + T_t) \} + \frac{\epsilon}{2} d_{RTT}(\sigma_t z + T_t)^2, \end{aligned}$$

where D_{RNT} , D_{RKT} and d_{RT} are defined in Section D.1.2. Moreover, $\mathbb{D}_{RNN}$, $\mathbb{D}_{RNK}$, $\mathbb{D}_{RKN}$, $\mathbb{D}_{RKK}$, D_{RNT} and D_{RKT} are defined as in Section E.1.2. Finally, define the vector d_{RTT} as:

$$d_{RTT} = \mathbf{vec} \left(R_i^{ss} (\chi_{i1}^2 + \chi_{i2}) \right).$$

G.7.3 Investment

Investment is given by

$$I_{it} = \left(R_{it} - Q_{it}^{-1/\zeta} \right) K_{it}.$$

Hence, up to a second-order:

$$\begin{aligned} \frac{I_t - I^{ss}}{\epsilon} &= D_{IN}n + D_{IK}k + D_{IQ}q_1 + d_{IT}(\sigma_t z + T_t) \\ &+ \frac{\epsilon}{2} \{ \mathbb{D}_{INN} \times_3 n \times_2 n + \mathbb{D}_{INK} \times_3 k \times_2 n + \mathbb{D}_{IKN} \times_3 n \times_2 k + \mathbb{D}_{IKK} \times_3 k \times_2 k \}, \\ &+ \epsilon \{ D_{INT}n(\sigma_t z + T_t) + D_{IKT}k(\sigma_t z + T_t) \}, \\ &+ \frac{\epsilon}{2} \left\{ \mathbb{D}_{IKQ} \times_3 q_1 \times_2 k + \mathbb{D}_{IQK} \times_3 k \times_2 q_1 + D_{IQ}q_2 + \mathbb{D}_{IQQ} \times_3 q_1 \times_2 q_1 + d_{ITT}(\sigma_t z + T_t)^2 \right\} \end{aligned}$$

where:

$$\begin{aligned}
\mathbb{D}_{INN} &= \psi(\psi - 1) \mathbf{diag}_3 \left(K_i^{ss} R_i^{ss} / (N_i^{ss})^2 \right), & \mathbb{D}_{IKN} = \mathbb{D}_{INK} &= \psi(1 - \phi) \mathbf{diag}_3 (R_i^{ss} K_i^{ss} / N_i^{ss}) \\
\mathbb{D}_{IKK} &= \phi(\phi - 2) \mathbf{diag}_3 (R_i^{ss} K_i^{ss}) + \mathbf{diag}_3 (I_i^{ss}) & \mathbb{D}_{IKQ} = \mathbb{D}_{IQK} &= \frac{1}{\zeta} \mathbf{diag}_3 \left((Q_i^{ss})^{-\frac{1}{\zeta} - 1} K_i^{ss} \right) \\
\mathbb{D}_{IQQ} &= - \left[\frac{1}{\zeta} + 1 \right] \frac{1}{\zeta} \mathbf{diag}_3 \left((Q_i^{ss})^{-\frac{1}{\zeta} - 2} K_i^{ss} \right) & \mathbf{d}_{ITT} &= \mathbf{vec} \left((\chi_{i1}^2 + \chi_{i2}) R_i^{ss} K_i^{ss} \right) \\
\mathbb{D}_{INT} &= \psi \mathbf{diag} (\chi_{i1} K_i^{ss} R_i^{ss} / N_i^{ss}), & \mathbb{D}_{IKT} &= (1 - \phi) \mathbf{diag} (\chi_{i1} K_i^{ss} R_i^{ss})
\end{aligned}$$

H Consumption equivalent welfare

H.1 Workers

For any period τ , we compute consumption equivalent welfare change between the steady-state SS and any counterfactual IRF path as $\omega_{i\tau}$, such that

$$\mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \left\{ u \left((1 + \omega_{i\tau}) A_{i(t)}^{ss} C_{i(t)}^{ss} \right) + \varepsilon_{i(t)t}^{ss} \right\} dt = \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \left\{ u(A_{i(t),t}^{IRF} C_{i(t),t}^{IRF}) + \varepsilon_{i(t)t}^{IRF} \right\} dt$$

where ε_{it} denotes the preference shock for the current (chosen) location. We use general CRRA utility $u(C) = \frac{C^{1-\gamma}-1}{1-\gamma}$. We denote by:

$$\mathcal{E}_{i\tau}^{ss} = \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \varepsilon_{i(t)t}^{ss} dt$$

the present value of preference shocks.

Then:

$$\begin{aligned}
& \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \left\{ u \left((1 + \omega_{i\tau}) A_{i(t)}^{ss} C_{i(t)}^{ss} \right) + \varepsilon_{i(t)t}^{ss} \right\} dt \\
&= \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \left\{ \frac{\left((1 + \omega_{i\tau}) A_{i(t)}^{ss} C_{i(t)}^{ss} \right)^{1-\gamma} - 1}{1-\gamma} + \varepsilon_{i(t)t}^{ss} \right\} dt \\
&= (1 + \omega_{i\tau})^{1-\gamma} \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \frac{\left(A_{i(t)}^{ss} C_{i(t)}^{ss} \right)^{1-\gamma}}{1-\gamma} dt + \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \left\{ -\frac{1}{1-\gamma} + \varepsilon_{i(t)t}^{ss} \right\} dt \\
&= (1 + \omega_{i\tau})^{1-\gamma} \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} \left\{ \frac{\left(A_{i(t)}^{ss} C_{i(t)}^{ss} \right)^{1-\gamma} - 1}{1-\gamma} dt + \varepsilon_{i(t)t}^{ss} + \frac{1}{1-\gamma} - \varepsilon_{i(t)t}^{ss} \right\} dt - \frac{1}{\rho(1-\gamma)} + \mathcal{E}_{i\tau}^{ss} \\
&= (1 + \omega_{i\tau})^{1-\gamma} \left\{ V_{i\tau}^{ss} + \frac{1}{\rho(1-\gamma)} - \mathcal{E}_{i\tau}^{ss} \right\} - \frac{1}{\rho(1-\gamma)} + \mathcal{E}_{i\tau}^{ss}
\end{aligned}$$

Then:

$$(1 + \omega_{i\tau})^{1-\gamma} = \frac{V_{i\tau}^{IRF} + \frac{1}{\rho(1-\gamma)} - \mathcal{E}_{i\tau}^{ss}}{V_{i\tau}^{ss} + \frac{1}{\rho(1-\gamma)} - \mathcal{E}_{i\tau}^{ss}} = 1 + \frac{dV_{i\tau}^{IRF}}{V_{i\tau}^{ss} + \frac{1}{\rho(1-\gamma)} - \mathcal{E}_{i\tau}^{ss}},$$

To compute $\mathcal{E}_{i,\tau}^{ss}$, we use the identify:

$$V_{i\tau}^{ss} = V_{i\tau}^{C,ss} + \mathcal{E}_{i,\tau}^{ss},$$

where $V_{i\tau}^{C,ss}$ is the value excluding taste shocks and is defined by:

$$V_{i\tau}^{C,ss} = \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} u \left(A_{i(t)}^{ss} C_{i(t)}^{ss} \right) dt.$$

$V_{i\tau}^{C,ss}$ satisfies the Bellman equation:

$$\rho V_{it}^{C,ss} = u \left(A_i^{ss} C_i^{ss} \right) + \mu \sum_j m_{ij}^{ss} V_{jt}^{C,ss}$$

Hence, we can easily calculate $V_{it}^{C,ss}$ (as we were doing previously). Subtituting into the

welfare formula:

$$\begin{aligned}(1 + \omega_{i\tau})^{1-\gamma} &= 1 + \frac{dV_{i\tau}^{IRF}}{V_{i\tau}^{ss} + \frac{1}{\rho(1-\gamma)} + V_{i\tau}^{C,SS} - V_{i\tau}^{ss}} \\ &= 1 + \frac{dV_{i\tau}^{IRF}}{V_{i\tau}^{C,SS} + \frac{1}{\rho(1-\gamma)}}\end{aligned}$$

which implies:

$$\omega_{i\tau} = \left(1 + \frac{dV_{i\tau}^{IRF}}{V_{i\tau}^{C,SS} + \frac{1}{\rho(1-\gamma)}} \right)^{\frac{1}{1-\gamma}} - 1$$

When shocks are small, to a first order

$$\omega_{i\tau} \approx \frac{1}{1-\gamma} \frac{dV_{i\tau}}{V_{i\tau}^{C,SS} + \frac{1}{\rho(1-\gamma)}} = \frac{\rho dV_{i\tau}}{1 + \rho(1-\gamma)V_{i\tau}^{C,SS}}. \quad (55)$$

Crucially, we need V^C in the denominator.

H.2 Capitalists

We define the capitalist's consumption equivalent welfare analogously as in the workers case. For any period τ , we compute consumption equivalent welfare change between the steady-state SS and any counterfactual IRF path as $\omega_{i\tau}^{cap}$, such that

$$\mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} u^{cap}((1 + \omega_{i\tau}^{cap})C_{i(t)}^{ss}) dt = \mathbb{E}_\tau \int_\tau^\infty e^{-\rho(t-\tau)} u^{cap}(C_{i(t),t}^{IRF}) dt.$$

Using $u^{cap}(c) = \frac{c^{1-\zeta}-1}{1-\zeta}$, we obtain:

$$\omega_{i\tau}^{cap} = \left[\frac{\rho(1-\zeta)\mathcal{P}_{i\tau}^{IRF} + 1}{\rho(1-\zeta)\mathcal{P}_{i\tau}^{ss} + 1} \right]^{\frac{1}{1-\zeta}} - 1,$$

where

$$\mathcal{P}_{i\tau}^{SS} \equiv \int_{\tau}^{\infty} e^{-\rho(t-\tau)} u^{cap}(C_{i(t)}^{SS}) dt,$$

$$\mathcal{P}_{i\tau}^{IRF} \equiv \int_{\tau}^{\infty} e^{-\rho(t-\tau)} u^{cap}(C_{i(t),t}^{IRF}) dt.$$

Hence, for small shocks, we get:

$$\omega_{i\tau}^{cap} \approx \frac{\rho d\mathcal{P}_{i\tau}^{IRF}}{\rho(1-\zeta)\mathcal{P}_{i\tau}^{SS} + 1}.$$

Using the definition of $\mathcal{P}_{it}$ from (28) we get that:

$$\omega_{i\tau}^{cap} \approx \frac{1}{1-\zeta} \frac{dQ_{i\tau}^{IRF}}{Q_{i\tau}^{SS}} + \frac{dK_{i\tau}}{K_{i\tau}^{SS}}.$$

I Additional Figures

Figure 6: Effect of Local and Aggregate Risk on Workers

